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Visualization Interface Tool

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The Visualization Interface Tool is a Python API that provides an interface between PyAnsys libraries and different plotting backends.

The Visualization Interface Tool offers these main features:

- Serves as an interface between PyAnsys and other plotting libraries (although only PyVista is supported currently).
- Provides out-of-the box picking, viewing, and measuring functionalities.
- Supplies an extensible class for adding custom functionalities.

Getting started Learn how to install the Visualization Interface Tool in user mode and quickly begin using it.

Getting started User guide Understand key concepts for implementing the Visualization Interface Tool in your workflow.

User guide API reference Understand how to use Python to interact programmatically with the Visualization Interface Tool.

API reference Examples Explore examples that show how to use the Visualization Interface Tool to perform many different types of operations.

Examples Contribute Learn how to contribute to the Visualization Interface Tool codebase or documentation.

Contribute

CHAPTER

ONE

GETTING STARTED

This section describes how to install the Visualization Interface Tool in user mode and quickly begin using it. If you are interested in contributing to the Visualization Interface Tool, see *Contribute* for information on installing in developer mode.

1.1 Installation

To use pip to install the Visualization Interface Tool, run this command:

```
pip install ansys-tools-visualization-interface
```

Alternatively, to install the latest version from this library's GitHub repository, run these commands:

```
git clone https://github.com/ansys/ansys-tools-visualization-interface
cd ansys-tools-visualization-interface
pip install .
```

1.1.1 Quick start

The following examples show how to use the Visualization Interface Tool to visualize a mesh file.

This code uses only a PyVista mesh:

```
from ansys.tools.visualization_interface import Plotter

my_mesh = my_custom_object.get_mesh()

# Create a Visualization Interface Tool object
pl = Plotter()
pl.plot(my_mesh)

# Plot the result
pl.show()
```

This code uses objects from a PyAnsys library:

```
from ansys.tools.visualization_interface import Plotter, MeshObjectPlot

my_custom_object = MyObject()
my_mesh = my_custom_object.get_mesh()

mesh_object = MeshObjectPlot(my_custom_object, my_mesh)
```

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```
# Create a Visualization Interface Tool object
pl = Plotter()
pl.plot(mesh_object)

# Plot the result
pl.show()
```

CHAPTER

TWO

USER GUIDE

This section explains key concepts for implementing the Visualization Interface Tool in your workflow. You can use the Visualization Interface Tool in your examples as well as integrate this library into your own code. For information on how to migrate from PyVista to the Ansys Visualization Interface Tool, see *Migration*.

2.1 Default plotter usage

The Visualization Interface Tool provides a default plotter that can be used out of the box, using the PyVista backend. This default plotter provides common functionalities so that you do not need to create a custom plotter.

2.1.1 Use with PyVista meshes

You can use the default plotter to plot simple PyVista meshes. This code shows how to use it to visualize a simple PyVista mesh:

```
## Usage example with pyvista meshes ##

import pyvista as pv
from ansys.tools.visualization_interface import Plotter

# Create a pyvista mesh
mesh = pv.Cube()

# Create a plotter
pl = Plotter()

# Add the mesh to the plotter
pl.plot(mesh)

# Show the plotter
pl.show()
```

2.1.2 Use with PyAnsys custom objects

You can also use the default plotter to visualize PyAnsys custom objects. The only requirement is that the custom object must have a method that returns a PyVista mesh a method that exposes a `name` or `id` attribute of your object. To expose a custom object, you use a *`MeshObjectPlot`* instance. This class relates PyVista meshes with any object.

The following code shows how to use the default plotter to visualize a PyAnsys custom object:

```
## Usage example with PyAnsys custom objects ##

from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface import MeshObjectPlot


# Create a custom object for this example
class CustomObject:
    def __init__(self):
        self.name = "CustomObject"
        self.mesh = pv.Cube()

    def get_mesh(self):
        return self.mesh

    def name(self):
        return self.name


custom_object = CustomObject()

# Create a MeshObjectPlot instance
mesh_object = MeshObjectPlot(custom_object, custom_object.get_mesh())

# Create a plotter
pl = Plotter()

# Add the MeshObjectPlot instance to the plotter
pl.plot(mesh_object)

# Show the plotter
pl.show()
```

2.2 Customize your own plotter

The Visualization Interface Tool provides a base class, `PlotterInterface`, for customizing certain functions of the plotter. This class provides a set of methods that can be overridden so that you can adapt the plotter to the specific need of your PyAnsys library.

The first thing you must do is to create a class that inherits from the `PlotterInterface` class. After that, see these main use cases for customizing the plotter:

- The most common use case is to customize the way that the objects you represent are shown in the plotter. To this end, you can override the `plot` and `plot_iter` methods. These methods are called every time a new object is added to the plotter. The default implementation of this method is to add a PyVista mesh or a *MeshObjectPlot* instance to the plotter. You can override this method to add your own meshes or objects to the plotter in a manner that fits the way that you want to represent the meshes.
- Another use case is the need to have custom button functionalities for your library. For example, you may want buttons for hiding or showing certain objects. To add custom buttons to the plotter, you use the implementable interface provided by the `PlotterWidget` class.

Some practical examples of how to use the `PlotterInterface` class are included in some PyAnsys libraries, such as PyAnsys Geometry.

For comprehensive migration information with code examples, see *Migration*.

2.3 Customizing the picker and hover callbacks

The Visualization Interface Tool provides a base class, `AbstractPicker`, for customizing the picker and hover callbacks of the plotter. This class provides a set of methods that can be overridden so that you can adapt the picker and hover functionalities to the specific need of your PyAnsys library.

The first thing you must do is to create a class that inherits from the `AbstractPicker` class. After that, see these main use cases for customizing the picker and hover callbacks:

- You may want to change the way that objects are picked in the plotter. To do this, you can override the `pick_select_object` and `pick_unselect_object` methods. These methods are called when an object is selected or unselected, respectively.
- Similarly, you may want to change the way that objects are hovered over in the plotter. To do this, you can override the `hover_select_object` and `hover_unselect_object` methods. These methods are called when an object is hovered over or unhovered, respectively.

A practical example of how to use the `AbstractPicker` class is included in *Create custom picker*.

CHAPTER

THREE

MIGRATION

This section helps you migrate from PyVista plotters to the Ansys Tools Visualization Interface plotters. It consists of two major topics:

- *Code migration*
- *Documentation configuration migration*

3.1 Code migration

This topic explains how to migrate from PyVista plotters to the new Ansys Tools Visualization Interface plotters. Because cases vary greatly, it provides a few examples that cover the most common scenarios.

3.1.1 Replace PyVista plotter code with Ansys Tools Visualization Interface plotter code

If you only need to plot simple PyVista meshes, you can directly replace your PyVista plotter code with the Ansys Tools Visualization Interface plotter code. On top of common PyVista functionalities, the Ansys Tools Visualization Interface plotter provides additional interactivity such as view buttons and mesh slicing.

The following code shows how to do the plotter code replacement:

- PyVista code:

```
import pyvista as pv

# Create a PyVista mesh
mesh = pv.Cube()

# Create a plotter
pl = pv.Plotter()

# Add the mesh to the plotter
pl.add_mesh(mesh)

# Show the plotter
pl.show()
```

- Ansys Tools Visualization Interface code:

```
import pyvista as pv
from ansys.tools.visualization_interface import Plotter

```

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```
# Create a PyVista mesh
mesh = pv.Cube()

# Create a plotter
pl = Plotter()

# Add the mesh to the plotter
pl.plot(mesh)

# Show the plotter
pl.show()
```

3.1.2 Convert your custom meshes to objects usable by the Ansys Tools Visualization Interface plotter

Your custom object must have a method that returns a PyVista mesh and a method that exposes a `name` or `id` attribute of your object:

```
class CustomObject:
    def __init__(self):
        self.name = "CustomObject"
        self.mesh = pv.Cube(center=(1, 1, 0))

    def get_mesh(self):
        return self.mesh

    def name(self):
        return self.name
```

You then need to create a *MeshObjectPlot* instance that relates the PyVista mesh with your custom object:

```
from ansys.tools.visualization_interface import MeshObjectPlot

custom_object = CustomObject()
mesh_object_plot = MeshObjectPlot(
    custom_object=custom_object,
    mesh=custom_object.get_mesh(),
)
```

With this, you can use the Ansys Tools Visualization Interface plotter to visualize your custom object. It enables interactivity such as picking and hovering.

3.1.3 Customize the PyVista backend

You can customize the backend of the Ansys Tools Visualization Interface plotter to enable or turn off certain functionalities. The following code shows how to enable picking:

```
from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.backends import PyVistaBackend

backend = PyVistaBackend(allow_picking=True)
```

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```
# Create a plotter
pl = Plotter(backend=backend)

# Add the MeshObjectPlot instance to the plotter
pl.plot(mesh_object_plot)

# Show the plotter
pl.show()
```

If you want to customize the backend even more, you can create your own backend by inheriting from the `PyVistaBackendInterface` class and implementing the required methods:

```
@abstractmethod
def plot_iter(self, plottable_object: Any, name_filter: str = None, **plotting_options):
    """Plot one or more compatible objects to the plotter.

    Parameters
    ----------
    plottable_object : Any
        One or more objects plot.
    name_filter : str, default: None.
        Regular expression with the desired name or names to include in the plotter.
    **plotting_options : dict, default: None
        Keyword arguments. For allowable keyword arguments, see the
        :meth:`Plotter.add_mesh <pyvista.Plotter.add_mesh>` method.

    """
    pass


@abstractmethod
def plot(self, plottable_object: Any, name_filter: str = None, **plotting_options):
    """Plot a single object to the plotter.

    Parameters
    ----------
    plottable_object : Any
        Object to plot.
    name_filter : str
        Regular expression with the desired name or names to include in the plotter.
    **plotting_options : dict, default: None
        Keyword arguments. For allowable keyword arguments, see the
        :meth:`Plotter.add_mesh <pyvista.Plotter.add_mesh>` method.

    """
    pass
```

The rest of the methods are implemented for you. This ensures that while you can customize what you need for plotting, the rest of the functionalities still work as expected. For more information, see the backend documentation. If you need to even go further, you can create your own plotter by inheriting from the `BaseBackend` class and implementing the required methods, although this may break existing features.

3.1.4 Customize the picker or hover behavior

You can customize the picker of the Ansys Tools Visualization Interface plotter to decide what happens when you pick or hover over an object. For example, if you want to print the name of the picked object, you can do it as described in the *Create custom picker* example.

3.1.5 Use the PyVista Qt backend

You can use the PyVista Qt backend with the Ansys Tools Visualization Interface plotter. To do this, you must set the PyVista backend to Qt before creating the plotter:

```
cube = pv.Cube()
pv_backend = PyVistaBackend(use_qt=True, show_qt=True)
pl = Plotter(backend=pv_backend)
pl.plot(cube)
pl.backend.enable_widgets()
pv_backend.scene.show()
```

You can then integrate the plotter into a PyQt or PySide app by disabling the `show_qt` parameter. For more information about this, see the PyVista documentation.

3.2 Documentation configuration migration

This topic explains how to migrate from the PyVista documentation configuration to the new Ansys Tools Visualization Interface documentation configuration.

1. Add environment variables for documentation:

```
os.environ["PYANSYS_VISUALIZER_DOC_MODE"] = "true"
os.environ["PYANSYS_VISUALIZER_HTML_BACKEND"] = "true"
```

2. Use PyVista DynamicScraper:

```
from pyvista.plotting.utilities.sphinx_gallery import DynamicScraper

sphinx_gallery_conf = {
    "image_scrapers": (DynamicScraper()),
}
```

3. Add PyVista viewer directive to extensions:

```
extensions = ["pyvista.ext.viewer_directive"]
```

4. Make sure you are executing the notebook cells:

```
nbsphinx_execute = "always"
```

For Plotly, in `conf.py`, do the following:

1. Add environment variables for documentation:

```
os.environ["PYANSYS_VISUALIZER_DOC_MODE"] = "true"
```

2. Add plotly configuration

```
import plotly.io as pio

pio.renderers.default = "sphinx_gallery"
```

3. Import and add scraper

```
from plotly.io._sg_scraper import plotly_sg_scraper

sphinx_gallery_conf = {
    "image_scrapers": (DynamicScraper(), "matplotlib", plotly_sg_scraper),
}
```

4. **[IMPORTANT]** The `pl.show()` must be the last line of code in the cell, or else it won't show.

CHAPTER

FOUR

API REFERENCE

This section describes ansys-tools-visualization-interface endpoints, their capabilities, and how to interact with them programmatically.

4.1 The `ansys.tools.visualization_interface` library

4.1.1 Summary

Subpackages

backends	Provides interfaces.
types	Provides custom types.
utils	Provides the Utils package.

Submodules

plotter	Module for the Plotter class.

Attributes

__version__

Constants

USE_TRAME	
DOCUMENTATION_BUILD	Whether the documentation is being built or not.
TESTING_MODE	Whether the library is being built or not, used to avoid showing plots while testing.
USE_HTML_BACKEND	Whether the library is being built or not, used to avoid showing plots while testing.

The `backends` package

Summary

Subpackages

plotly	Plotly initialization.
pyvista	Provides interfaces.
usd	USD backend initialization.

The plotly package

Summary

Subpackages

widgets	Widgets module init.

Submodules

plotly_dash	Module for dash plotly.
plotly_interface	Plotly backend interface for visualization.

The widgets package

Summary

Submodules

button_manager	Module for button management.
dropdown_manager	Module for dropdown management in Plotly figures.

The button_manager.py module

Summary

Classes

ButtonManager	Class to manage buttons in a Plotly figure.

ButtonManager

class ansys.tools.visualization_interface.backends.plotly.widgets.button_manager.**ButtonManager**(*fig: plotly.grap*

Class to manage buttons in a Plotly figure.

This class allows adding buttons to a Plotly figure for various functionalities such as toggling visibility of traces, resetting the view, and custom actions.

Parameters

fig

[go.Figure] The Plotly figure to which buttons will be added.

Overview

Methods

add_button	Add a button to the Plotly figure.
show_hide_bbox_dict	Generate dictionary for showing/hiding coordinate system elements.
update_layout	Update the figure layout with all controls as buttons in a single row.
args_xy_view_button	Get camera configuration for XY plane view (top-down view).
args_xz_view_button	Get camera configuration for XZ plane view (front view).
args_yz_view_button	Get camera configuration for YZ plane view (side view).
args_iso_view_button	Get camera configuration for isometric view (3D perspective).
add_measurement_toggle_button	Get configuration for measurement toggle button.
args_projection_toggle_button	Get configuration for projection toggle button.
args_theme_toggle_button	Get configuration for theme toggle button.

Import detail

```
from ansys.tools.visualization_interface.backends.plotly.widgets.button_manager import␣
↪ButtonManager
```

Method detail

ButtonManager.**add_button**(*label: str*, *x: float*, *y: float*, *xanchor: str = 'left'*, *yanchor: str = 'bottom'*, *method: str = 'restyle'*, *args: List[Any] = None*, *args2: List[Any] = None*) → None

Add a button to the Plotly figure.

Parameters

label
[str] The text to display on the button.

x
[float] X position of the button (0-1).

y
[float] Y position of the button (0-1).

xanchor
[str, optional] X anchor point for the button, by default “left”.

yanchor
[str, optional] Y anchor point for the button, by default “bottom”.

method
[str, optional] The method to call when the button is clicked. Options include: ‘restyle’, ‘relayout’, ‘update’, ‘animate’, by default ‘restyle’.

args
[List[Any], optional] Arguments to pass to the method when the button is clicked, by default None.

args2
[List[Any], optional] Secondary arguments for toggle functionality, by default None.

ButtonManager.**show_hide_bbox_dict**(*toggle: bool = True*)

Generate dictionary for showing/hiding coordinate system elements.

Parameters

toggle
[bool, optional] Whether to show (True) or hide (False) the coordinate system, by default True.

Returns

dict
Dictionary with coordinate system visibility settings.

ButtonManager.**update_layout**() → None

Update the figure layout with all controls as buttons in a single row.

This method builds buttons using the configuration methods and any additional buttons that were added via add_button().

ButtonManager.**args_xy_view_button**(*label: str = 'XY View'*, *x: float = 0.02*, *y: float = 1.02*) → dict

Get camera configuration for XY plane view (top-down view).

Parameters

label
[str, optional] The text to display on the button, by default "XY View".

x
[float, optional] X position of the button (0-1), by default 0.02.

y
[float, optional] Y position of the button (0-1), by default 1.02.

Returns

dict
Camera configuration for XY plane view.

ButtonManager.**args_xz_view_button**(*label: str = 'XZ View'*, *x: float = 0.02*, *y: float = 1.02*) → dict

Get camera configuration for XZ plane view (front view).

Parameters

label
[str, optional] The text to display on the button, by default "XZ View".

x
[float, optional] X position of the button (0-1), by default 0.02.

y
[float, optional] Y position of the button (0-1), by default 1.02.

Returns

dict
Camera configuration for XZ plane view.

ButtonManager.**args_yz_view_button**(*label: str = 'YZ View'*, *x: float = 0.02*, *y: float = 1.02*) → dict

Get camera configuration for YZ plane view (side view).

Parameters

label
[str, optional] The text to display on the button, by default "YZ View".

x
[float, optional] X position of the button (0-1), by default 0.02.

y
[float, optional] Y position of the button (0-1), by default 1.02.

Returns

dict
Camera configuration for YZ plane view.

ButtonManager.**args_iso_view_button**(*label: str = 'ISO View'*, *x: float = 0.02*, *y: float = 1.02*) → dict

Get camera configuration for isometric view (3D perspective).

Parameters

label
[str, optional] The text to display on the button, by default "ISO View".

x
[float, optional] X position of the button (0-1), by default 0.02.

y
[float, optional] Y position of the button (0-1), by default 1.02.

Returns

dict
Camera configuration for isometric view.

ButtonManager.**add_measurement_toggle_button**(*label: str = 'Toggle Measurement'*, *x: float = 0.02*, *y: float = 0.87*) → Tuple[dict, dict]

Get configuration for measurement toggle button.

Parameters

label
[str, optional] The text to display on the button, by default "Toggle Measurement".

x
[float, optional] X position of the button (0-1), by default 0.02.

y
[float, optional] Y position of the button (0-1), by default 0.87.

Returns

Tuple[dict, dict]
Tuple containing (enable_measurement_config, disable_measurement_config).

ButtonManager.**args_projection_toggle_button**() → Tuple[dict, dict]

Get configuration for projection toggle button.

Parameters

label
[str, optional] The text to display on the button, by default "Toggle Projection".

x
[float, optional] X position of the button (0-1), by default 0.14.

y
[float, optional] Y position of the button (0-1), by default 1.02.

Returns

Tuple[dict, dict]
Tuple containing (perspective_projection_config, orthographic_projection_config).

ButtonManager.**args_theme_toggle_button**(*label: str = 'Toggle Theme'*, *x: float = 0.32*, *y: float = 1.02*) → Tuple[dict, dict]

Get configuration for theme toggle button.

Parameters

label
[str, optional] The text to display on the button, by default "Toggle Theme".

x
[float, optional] X position of the button (0-1), by default 0.22.

y
[float, optional] Y position of the button (0-1), by default 1.02.

Returns

Tuple[dict, dict]
Tuple containing (light_theme_config, dark_theme_config).

Description

Module for button management.

The dropdown_manager.py module

Summary

Classes

DashDropdownManager	Class to manage dropdown menus in a Plotly figure.

DashDropdownManager

class ansys.tools.visualization_interface.backends.plotly.widgets.dropdown_manager.**DashDropdownManager**(*f*

Class to manage dropdown menus in a Plotly figure.

This class allows adding dropdown menus to a Plotly figure for controlling mesh visibility and other properties.

Parameters

fig
[go.Figure] The Plotly figure to which dropdowns will be added.

Overview

Methods

add_mesh_name	Add a mesh name to track for dropdown functionality.
get_mesh_names	Get the list of tracked mesh names.
get_visibility_args_for_meshes	Get visibility arguments for showing only specified meshes.
clear	Clear all tracked mesh names.

Import detail

```
from ansys.tools.visualization_interface.backends.plotly.widgets.dropdown_manager import␣
↪DashDropdownManager
```

Method detail

DashDropdownManager.**add_mesh_name**(*name: str*) → None

Add a mesh name to track for dropdown functionality.

Parameters

name
[str] The name of the mesh to track.

DashDropdownManager.**get_mesh_names**() → List[str]

Get the list of tracked mesh names.

Returns

List[str]
List of mesh names.

DashDropdownManager.**get_visibility_args_for_meshes**(*visible_mesh_names: List[str]*) → Dict[str, Any]

Get visibility arguments for showing only specified meshes.

Parameters

visible_mesh_names
[List[str]] List of mesh names that should be visible.

Returns

Dict[str, Any]
Arguments for restyle method to set mesh visibility.

DashDropdownManager.**clear**() → None

Clear all tracked mesh names.

Description

Module for dropdown management in Plotly figures.

Description

Widgets module init.

The plotly_dash.py module

Summary

Classes

PlotlyDashBackend	Plotly Dash interface for visualization.

PlotlyDashBackend

class ansys.tools.visualization_interface.backends.plotly.plotly_dash.**PlotlyDashBackend**(*app: dash.Dash = None*)

Bases: *ansys.tools.visualization_interface.backends.plotly.plotly_interface.PlotlyBackend*

Plotly Dash interface for visualization.

Overview

Methods

plot	Plot a single object using Plotly and track mesh names for dropdown.
create_dash_layout	Create the Dash layout with optional dropdown for mesh visibility.
show	Render the Plotly scene.

Properties

dropdown_manager	Get the dropdown manager for this backend.

Import detail

```
from ansys.tools.visualization_interface.backends.plotly.plotly_dash import␣
↪PlotlyDashBackend
```

Property detail

property PlotlyDashBackend.**dropdown_manager**: *ansys.tools.visualization_interface.backends.plotly.widgets.dropdown_manager.DashDropdownManager*

Get the dropdown manager for this backend.

Returns

DashDropdownManager
The dropdown manager instance.

Method detail

PlotlyDashBackend.**plot**(*plottable_object*, *name: str = None*, ***plotting_options*) → None

Plot a single object using Plotly and track mesh names for dropdown.

Parameters

plottable_object
[Any] The object to plot.

name
[str, optional] Name of the mesh for labeling in Plotly.

plotting_options
[dict] Additional plotting options.

PlotlyDashBackend.**create_dash_layout**() → dash.html.Div

Create the Dash layout with optional dropdown for mesh visibility.

Returns

html.Div
The Dash layout component.

PlotlyDashBackend.**show**(*plottable_object=None*, *screenshot: str = None*, ***kwargs*) → plotly.graph_objects.Figure | None

Render the Plotly scene.

Parameters

plottable_object
[Any, optional] Object to show, by default None.

screenshot
[str, optional] Path to save a screenshot, by default None.

kwargs
[dict] Additional options the selected backend accepts.

Returns

Union[go.Figure, None]
The figure of the plot if in doc building environment. Else, None.

Description

Module for dash plotly.

The plotly_interface.py module

Summary

Classes

PlotlyBackend	Plotly interface for visualization.

PlotlyBackend

class ansys.tools.visualization_interface.backends.plotly.plotly_interface.**PlotlyBackend**

Bases: ansys.tools.visualization_interface.backends._base.BaseBackend

Plotly interface for visualization.

Overview

Methods

plot_iter	Plot multiple objects using Plotly.
plot	Plot a single object using Plotly.
set_trace_visibility	Set the visibility of all traces with the given name.
filter_traces	Show only traces whose names are in the provided list, hiding all others.
show	Render the Plotly scene.
add_points	Add point markers to the scene.
add_lines	Add line segments to the scene.
add_planes	Add a plane to the scene.
add_text	Add text to the scene.
add_labels	Add labels at 3D point locations.
clear	Clear all traces from the figure.

Properties

layout	Get the current layout of the Plotly figure.

Import detail

```
from ansys.tools.visualization_interface.backends.plotly.plotly_interface import␣
↪PlotlyBackend
```

Property detail

property PlotlyBackend.**layout: Any**

Get the current layout of the Plotly figure.

Returns

Any
The current layout of the Plotly figure.

Method detail

PlotlyBackend.**plot_iter**(*plotting_list: Iterable[Any]*, ***kwargs*) → None

Plot multiple objects using Plotly.

Parameters

plotting_list
[Iterable[Any]] An iterable of objects to plot.

kwargs
[dict] Additional plotting options.

PlotlyBackend.**plot**(*plottable_object: pyvista.PolyData | pyvista.MultiBlock | ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot | plotly.graph_objects.Mesh3d*, *name: str | None = None*, ***plotting_options*) → None

Plot a single object using Plotly.

Parameters

plottable_object
[Union[pv.PolyData, pv.MultiBlock, MeshObjectPlot, go.Mesh3d]] The object to plot. Can be a PyVista PolyData, MultiBlock, a MeshObjectPlot, or a Plotly Mesh3d.

name
[str, optional] Name of the mesh for labeling in Plotly. Overrides the name from MeshObjectPlot if provided.

plotting_options
[dict] Additional plotting options.

PlotlyBackend.**set_trace_visibility**(*name: str*, *visible: bool*) → None

Set the visibility of all traces with the given name.

Parameters

name
[str] Name of the trace(s) to update.

visible
[bool] Whether to make the trace(s) visible.

PlotlyBackend.**filter_traces**(*names: List[str]*) → None

Show only traces whose names are in the provided list, hiding all others.

Parameters

names
[List[str]] Names of traces to show. All other traces will be hidden.

PlotlyBackend.**show**(*plottable_object=None*, *screenshot: str | None = None*, ***kwargs*) → plotly.graph_objects.Figure | None

Render the Plotly scene.

Parameters

plottable_object
[Any, optional] Object to show, by default None.

screenshot
[str, optional] Path to save a screenshot, by default None.

kwargs
[dict] Additional options the selected backend accepts.

Returns

Union[go.Figure, None]
The figure of the plot if in doc building environment. Else, None.

PlotlyBackend.**add_points**(*points: List | Any*, *color: str = 'red'*, *size: float = 10.0*, ***kwargs*) → Any

Add point markers to the scene.

Parameters

points
[Union[List, Any]] Points to add. Expected format: [[x1, y1, z1], [x2, y2, z2], ...] or Nx3 array.

color
[str, default: "red"] Color of the points.

size
[float, default: 10.0] Size of the point markers.

****kwargs**
[dict] Additional keyword arguments passed to Plotly's Scatter3d.

Returns

go.Scatter3d
Plotly Scatter3d trace representing the added points.

PlotlyBackend.**add_lines**(*points: List | Any*, *connections: List | Any | None = None*, *color: str = 'white'*, *width: float = 1.0*, ***kwargs*) → Any

Add line segments to the scene.

Parameters

points
[Union[List, Any]] Points defining the lines. Expected format: [[x1, y1, z1], [x2, y2, z2], ...] or Nx3 array.

connections
[Optional[Union[List, Any]], default: None] Line connectivity. If None, connects points sequentially. Expected format: [[start_idx1, end_idx1], [start_idx2, end_idx2], ...] or Mx2 array.

color
[str, default: "white"] Color of the lines.

width
[float, default: 1.0] Width of the lines.

****kwargs**
[dict] Additional keyword arguments passed to Plotly's Scatter3d.

Returns

Union[go.Scatter3d, List[go.Scatter3d]]
Plotly Scatter3d trace(s) representing the added lines.

PlotlyBackend.**add_planes**(*center: Tuple[float, float, float] = (0.0, 0.0, 0.0)*, *normal: Tuple[float, float, float] = (0.0, 0.0, 1.0)*, *i_size: float = 1.0*, *j_size: float = 1.0*, ***kwargs*) → Any

Add a plane to the scene.

Parameters

center
[Tuple[float, float, float], default: (0.0, 0.0, 0.0)] Center point of the plane (x, y, z).

normal
[Tuple[float, float, float], default: (0.0, 0.0, 1.0)] Normal vector of the plane (x, y, z).

i_size
[float, default: 1.0] Size of the plane in the i direction.

j_size
[float, default: 1.0] Size of the plane in the j direction.

****kwargs**
[dict] Additional keyword arguments passed to Plotly's Mesh3d (e.g., color, opacity).

Returns

go.Mesh3d
Plotly Mesh3d trace representing the added plane.

PlotlyBackend.**add_text**(*text: str*, *position: Tuple[float, float] | Tuple[float, float, float] | str*, *font_size: int = 12*, *color: str = 'white'*, ***kwargs*) → Any

Add text to the scene.

Parameters

text
[str] Text string to display.

position
[Union[Tuple[float, float], Tuple[float, float, float], str]] Position for the text as 2D screen coordinates (x, y). Values should be between 0 and 1 for normalized coordinates, or pixel values for absolute positioning.

font_size
[int, default: 12] Font size for the text.

color
[str, default: "white"] Color of the text.

****kwargs**
[dict] Additional keyword arguments passed to Plotly's annotation.

Returns

dict
Plotly annotation representing the added text.

PlotlyBackend.**add_labels**(*points: List | Any*, *labels: List[str]*, *font_size: int = 12*, *point_size: float = 5.0*, ***kwargs*) → Any

Add labels at 3D point locations.

Parameters

points
[Union[List, Any]] Points where labels should be placed.

labels
[List[str]] List of label strings to display at each point.

font_size
[int, default: 12] Font size for the labels.

point_size
[float, default: 5.0] Size of the point markers shown with labels.

****kwargs**
[dict] Additional keyword arguments.

Returns

Any
Plotly trace representing the labels.

PlotlyBackend.**clear**() → None

Clear all traces from the figure.

Description

Plotly backend interface for visualization.

Description

Plotly initialization.

The `pyvista` package

Summary

Subpackages

widgets	Provides widgets for the Visualization Interface Tool plotter.

Submodules

animation	Animation support for PyVista backend.
picker	Module for managing picking and hovering of objects in a PyVista plotter.
pyvista	Provides a wrapper to aid in plotting.
pyvista_interface	Provides plotting for various PyAnsys objects.
trame_local	Provides trame visualizer interface for visualization.
trame_remote	Module for trame websocket client functions.
trame_service	Trame service module.

The `widgets` package

Summary

Submodules

button	Provides for implementing buttons in PyAnsys.
dark_mode	Provides the dark mode button widget for the PyAnsys plotter.
displace_arrows	Provides the displacement arrows widget for the PyVista plotter.
dynamic_tree_menu	Provides a dynamic tree menu widget with clickable buttons.
hide_buttons	Provides the hide buttons widget for the PyAnsys plotter.
measure	Provides the measure widget for the PyAnsys plotter.
mesh_slider	Provides the measure widget for the PyAnsys plotter.
next_button	Provides the next button widget for animation control.
parallel_projection	Provides the parallel projection button widget.
pick_rotation_center	Provides the measure widget for the PyAnsys plotter.
play_pause_button	Provides the play/pause button widget for animation control.
previous_button	Provides the previous button widget for animation control.
ruler	Provides the ruler widget for the Visualization Interface Tool plotter.
save_gif_button	Provides the save GIF button widget for animation control.
screenshot	Provides the screenshot widget for the Visualization Interface Tool plotter.
stop_button	Provides the stop button widget for animation control.
tree_menu_toggle	Provides a button to toggle the tree menu visibility.
view_button	Provides the view button widget for changing the camera view.
widget	Provides the abstract implementation of plotter widgets.

The button.py module

Summary

Classes

Button	Provides the abstract class for implementing buttons in PyAnsys.

Button

class ansys.tools.visualization_interface.backends.pyvista.widgets.button.**Button**(*plotter: pyvista.Plotter*, *button_config: tuple*, *dark_mode: bool = False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Provides the abstract class for implementing buttons in PyAnsys.

Parameters

plotter
[Plotter] Plotter to draw the buttons on.

button_config
[tuple] Tuple containing the position and the path to the icon of the button.

dark_mode
[bool, optional] Whether to activate the dark mode or not.

Notes

This class wraps the PyVista add_checkbox_button_widget() method.

Overview

Abstract methods

callback	Get the functionality of the button, which is implemented by subclasses.

Methods

update	Assign the image that represents the button.

Attributes

button_config

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.button import Button
```

Attribute detail

Button.**button_config**

Method detail

abstractmethod Button.**callback**(*state: bool*) → None

Get the functionality of the button, which is implemented by subclasses.

Parameters

state
[bool] Whether the button is active.

Button.**update**() → None

Assign the image that represents the button.

Description

Provides for implementing buttons in PyAnsys.

The dark_mode.py module

Summary

Classes

DarkModeButton	Provides the dark mode widget for the Visualization Interface Tool Plotter class.

DarkModeButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.dark_mode.**DarkModeButton**(*plotter: ansys.tools.visual dark_mode: bool = False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Provides the dark mode widget for the Visualization Interface Tool Plotter class.

Parameters

plotter_helper
[PlotterHelper] Plotter to add the dark mode widget to.

dark_mode
[bool, optional] Whether to activate the dark mode or not.

Overview

Methods

callback	Remove or add the dark mode widget actor upon click.
update	Define the dark mode widget button parameters.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.dark_mode import␣
↪DarkModeButton
```

Method detail

DarkModeButton.**callback**(*state: bool*) → None

Remove or add the dark mode widget actor upon click.

Parameters

state

[bool] Whether the state of the button, which is inherited from PyVista, is active.

DarkModeButton.**update**() → None

Define the dark mode widget button parameters.

Description

Provides the dark mode button widget for the PyAnsys plotter.

The displace_arrows.py module

Summary

Classes

DisplacementArrow	Defines the arrow to draw and what it is to do.

Enums

CameraPanDirection	Provides an enum with the available movement directions of the camera.

DisplacementArrow

class ansys.tools.visualization_interface.backends.pyvista.widgets.displace_arrows.**DisplacementArrow**(*plo pyv di- rec tio Ca er- a- Pa rec tio da bo = Fa

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.button.Button*

Defines the arrow to draw and what it is to do.

Parameters

plotter
[Plotter] Plotter to draw the buttons on.

direction
[CameraPanDirection] Direction that the camera is to move.

dark_mode
[bool, optional] Whether to activate the dark mode or not.

Overview

Methods

callback	Move the camera in the direction defined by the button.

Attributes

direction

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.displace_arrows import␣
↪DisplacementArrow
```

Attribute detail

DisplacementArrow.**direction**

Method detail

DisplacementArrow.**callback**(*state: bool*) → None

Move the camera in the direction defined by the button.

Parameters

state
[bool] Whether the state of the button, which is inherited from PyVista, is active. However, this parameter is unused by this `callback` method.

CameraPanDirection

class ansys.tools.visualization_interface.backends.pyvista.widgets.displace_arrows.**CameraPanDirection**

Bases: enum.Enum

Provides an enum with the available movement directions of the camera.

Overview

Attributes

XUP
XDOWN
YUP
YDOWN
ZUP
ZDOWN

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.displace_arrows import␣
↪CameraPanDirection
```

Attribute detail

CameraPanDirection.**XUP** = (0, 'upxarrow', (5, 230))

CameraPanDirection.**XDOWN** = (1, 'downarrow', (5, 190))

CameraPanDirection.**YUP** = (2, 'upyarrow', (35, 230))

CameraPanDirection.**YDOWN** = (3, 'downarrow', (35, 190))

CameraPanDirection.**ZUP** = (4, 'upzarrow', (65, 230))

CameraPanDirection.**ZDOWN** = (5, 'downarrow', (65, 190))

Description

Provides the displacement arrows widget for the PyVista plotter.

The dynamic_tree_menu.py module

Summary

Classes

DynamicTreeMenuWidget	Provides a dynamic tree menu with clickable buttons for each object.

DynamicTreeMenuWidget

class ansys.tools.visualization_interface.backends.pyvista.widgets.dynamic_tree_menu.**DynamicTreeMenuWid**

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Provides a dynamic tree menu with clickable buttons for each object.

This widget displays a hierarchical tree view of all plotted objects on the right side of the screen. Each object has a clickable checkbox button to toggle visibility.

Parameters

plotter
[Plotter] The plotter to add the tree menu widget to.

position
[tuple, optional] The (x, y) position of the top-left corner of the menu. Default is (0.65, 0.95).

button_size
[int, optional] Size of the checkbox buttons in pixels. Default is 20.

spacing
[int, optional] Vertical spacing between items in pixels. Default is 30.

font_size
[int, optional] Font size for the object labels. Default is 12.

dark_mode
[bool, optional] Whether to use dark mode colors. Default is False. In light mode (False): text is black In dark mode (True): text is white

Notes

Each object gets a checkbox button that toggles visibility of the object and its subtree. The menu updates dynamically as objects are added or removed.

Overview

Methods

refresh	Refresh the menu display.
update	Update the widget's appearance for current dark mode setting.
show_menu	Show the menu by enabling all text actors and buttons.
hide_menu	Hide the menu by disabling all text actors and buttons.
callback	Placeholder callback (not used for this widget).

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.dynamic_tree_menu␣
↪import DynamicTreeMenuWidget
```

Method detail

DynamicTreeMenuWidget.**refresh**()

Refresh the menu display.

DynamicTreeMenuWidget.**update**()

Update the widget's appearance for current dark mode setting.

DynamicTreeMenuWidget.**show_menu**()

Show the menu by enabling all text actors and buttons.

DynamicTreeMenuWidget.**hide_menu**()

Hide the menu by disabling all text actors and buttons.

DynamicTreeMenuWidget.**callback**(*state: bool*) → None

Placeholder callback (not used for this widget).

Description

Provides a dynamic tree menu widget with clickable buttons.

The hide_buttons.py module

Summary

Classes

HideButton	Provides the hide widget for the Visualization Interface Tool Plotter class.

HideButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.hide_buttons.**HideButton**(*plotter: ansys.tools.visualiz dark_mode: bool = False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Provides the hide widget for the Visualization Interface Tool Plotter class.

Parameters

plotter_helper
[PlotterHelper] Plotter to add the hide widget to.

dark_mode
[bool, optional] Whether to activate the dark mode or not.

Overview

Methods

callback	Remove or add the hide widget actor upon click.
update	Define the hide widget button parameters.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.hide_buttons import␣
↪HideButton
```

Method detail

HideButton.**callback**(*state: bool*) → None

Remove or add the hide widget actor upon click.

Parameters

state
[bool] Whether the state of the button, which is inherited from PyVista, is active.

HideButton.**update**() → None

Define the hide widget button parameters.

Description

Provides the hide buttons widget for the PyAnsys plotter.

The `measure.py` module

Summary

Classes

MeasureWidget	Provides the measure widget for the Visualization Interface Tool `Plotter` class.

MeasureWidget

class ansys.tools.visualization_interface.backends.pyvista.widgets.measure.**MeasureWidget**(*plotter_helper: ansys.tools.visualizat dark_mode: bool = False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Provides the measure widget for the Visualization Interface Tool `Plotter` class.

Parameters

plotter_helper
[`PlotterHelper`] Plotter to add the measure widget to.

dark_mode
[bool, `optional`] Whether to activate the dark mode or not.

Overview

Methods

callback	Remove or add the measurement widget actor upon click.
update	Define the measurement widget button parameters.

Attributes

plotter_helper

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.measure import␣
↪MeasureWidget
```

Attribute detail

`MeasureWidget.`**`plotter_helper`**

Method detail

`MeasureWidget.`**`callback`**(*state: bool*) → None

Remove or add the measurement widget actor upon click.

Parameters

state
[bool] Whether the state of the button, which is inherited from PyVista, is active.

`MeasureWidget.`**`update`**() → None

Define the measurement widget button parameters.

Description

Provides the measure widget for the PyAnsys plotter.

The `mesh_slider.py` module

Summary

Classes

MeshSliderWidget	Provides the mesh slider widget for the Visualization Interface Tool `Plotter` class.

MeshSliderWidget

class ansys.tools.visualization_interface.backends.pyvista.widgets.mesh_slider.**MeshSliderWidget**(*plotter_he*
an-
sys.tools.v
dark_mod
bool
=
False)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Provides the mesh slider widget for the Visualization Interface Tool `Plotter` class.

Parameters

plotter_helper
[`PlotterHelper`] Plotter to add the mesh slider widget to.

dark_mode
[bool, `optional`] Whether to activate the dark mode or not.

Overview

Methods

callback	Remove or add the mesh slider widget actor upon click.
update	Define the mesh slider widget button parameters.

Attributes

plotter_helper

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.mesh_slider import␣
↪MeshSliderWidget
```

Attribute detail

MeshSliderWidget.**plotter_helper**

Method detail

MeshSliderWidget.**callback**(*state: bool*) → None

Remove or add the mesh slider widget actor upon click.

Parameters

state

[bool] Whether the state of the button, which is inherited from PyVista, is active.

MeshSliderWidget.**update**() → None

Define the mesh slider widget button parameters.

Description

Provides the measure widget for the PyAnsys plotter.

The `next_button.py` module

Summary

Classes

NextButton	Provides next frame control for animations.

Enums

NextButtonConfig	Provides configuration for the next button.

NextButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.next_button.**NextButton**(*plotter: pyvista.Plotter*, *animation*, *dark_mode: bool = False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.button.Button*

Provides next frame control for animations.

This button steps forward one frame.

Parameters

plotter
[Plotter] Plotter to draw the button on.

animation
[Animation] Animation instance to control.

dark_mode
[bool, optional] Whether to activate the dark mode or not.

Overview

Methods

callback	Step to next frame.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.next_button import␣
↪NextButton
```

Method detail

NextButton.**callback**(*state: bool*) → None

Step to next frame.

Parameters

state
[bool] Whether the button is active.

NextButtonConfig

class ansys.tools.visualization_interface.backends.pyvista.widgets.next_button.**NextButtonConfig**

Bases: enum.Enum

Provides configuration for the next button.

Overview

Attributes

NEXT	

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.next_button import␣
↪NextButtonConfig
```

Attribute detail

NextButtonConfig.**NEXT = (0, 'next', (160, 10))**

Description

Provides the next button widget for animation control.

The parallel_projection.py module

Summary

Classes

ParallelProjectionButton	Toggle parallel projection for the camera.

ParallelProjectionButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.parallel_projection.**ParallelProjectio**

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Toggle parallel projection for the camera.

Parameters

plotter
[Plotter] Plotter to add the widget to.

dark_mode
[bool, optional] Whether dark mode is active.

Overview

Methods

callback	Toggle parallel projection.
update	Update the button appearance.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.parallel_projection␣
↪import ParallelProjectionButton
```

Method detail

ParallelProjectionButton.**callback**(*state: bool*) → None

Toggle parallel projection.

Parameters

state
[bool] Button state from PyVista.

ParallelProjectionButton.**update**() → None

Update the button appearance.

Description

Provides the parallel projection button widget.

The pick_rotation_center.py module

Summary

Classes

PickRotCenterButton	Provides the pick rotation center widget for the Visualization Interface Tool Plotter class.

PickRotCenterButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.pick_rotation_center.**PickRotCenterBu**

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Provides the pick rotation center widget for the Visualization Interface Tool Plotter class.

Parameters

plotter_helper
[PlotterHelper] Plotter to add the pick rotation center widget to.

dark_mode
[bool, optional] Whether to activate the dark mode or not.

Overview

Methods

callback	Remove or add the pick rotation center widget actor upon click.
update	Define the measurement widget button parameters.

Attributes

plotter_helper

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.pick_rotation_center␣
↪import PickRotCenterButton
```

Attribute detail

PickRotCenterButton.**plotter_helper**

Method detail

PickRotCenterButton.**callback**(*state: bool*) → None

Remove or add the pick rotation center widget actor upon click.

Parameters

state
[bool] Whether the state of the button, which is inherited from PyVista, is active.

PickRotCenterButton.**update**() → None

Define the measurement widget button parameters.

Description

Provides the measure widget for the PyAnsys plotter.

The play_pause_button.py module

Summary

Classes

PlayPauseButton	Provides play/pause toggle control for animations.

Enums

PlayPauseButtonConfig	Provides configuration for the play/pause button.

PlayPauseButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.play_pause_button.**PlayPauseButton**(*plo*
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Fa

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.button.Button*

Provides play/pause toggle control for animations.

This button switches between play and pause icons dynamically based on the animation state.

Parameters

plotter
[Plotter] Plotter to draw the button on.

animation
[Animation] Animation instance to control.

dark_mode
[bool, optional] Whether to activate the dark mode or not.

Overview

Methods

callback	Toggle between play and pause.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.play_pause_button␣
↪import PlayPauseButton
```

Method detail

PlayPauseButton.**callback**(*state: bool*) → None

Toggle between play and pause.

Parameters

state
[bool] Whether the button is active (playing).

PlayPauseButtonConfig

class ansys.tools.visualization_interface.backends.pyvista.widgets.play_pause_button.**PlayPauseButtonConfig**

Bases: enum.Enum

Provides configuration for the play/pause button.

Overview

Attributes

PLAY_PAUSE	

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.play_pause_button␣
↪import PlayPauseButtonConfig
```

Attribute detail

PlayPauseButtonConfig.**PLAY_PAUSE = (0, 'play', (10, 10))**

Description

Provides the play/pause button widget for animation control.

The previous_button.py module

Summary

Classes

PreviousButton	Provides previous frame control for animations.

Enums

PreviousButtonConfig	Provides configuration for the previous button.

PreviousButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.previous_button.**PreviousButton**(*plotter*: *pyvista.an-i-ma-tion*, *dark_m*: *bool* = *False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.button.Button*

Provides previous frame control for animations.

This button steps backward one frame.

Parameters

plotter
[Plotter] Plotter to draw the button on.

animation
[Animation] Animation instance to control.

dark_mode
[bool, optional] Whether to activate the dark mode or not.

Overview

Methods

callback	Step to previous frame.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.previous_button import␣
↪PreviousButton
```

Method detail

PreviousButton.**callback**(*state: bool*) → None

Step to previous frame.

Parameters

state
[bool] Whether the button is active.

PreviousButtonConfig

class ansys.tools.visualization_interface.backends.pyvista.widgets.previous_button.**PreviousButtonConfig**

Bases: enum.Enum

Provides configuration for the previous button.

Overview

Attributes

PREVIOUS

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.previous_button import␣
↪PreviousButtonConfig
```

Attribute detail

PreviousButtonConfig.**PREVIOUS = (0, 'previous', (110, 10))**

Description

Provides the previous button widget for animation control.

The `ruler.py` module

Summary

Classes

Ruler	Provides the ruler widget for the Visualization Interface Tool `Plotter` class.

Ruler

class ansys.tools.visualization_interface.backends.pyvista.widgets.ruler.**Ruler**(*plotter: pyvista.Plotter*, *dark_mode: bool = False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Provides the ruler widget for the Visualization Interface Tool `Plotter` class.

Parameters

plotter
[`Plotter`] Provides the plotter to add the ruler widget to.

dark_mode
[bool, `optional`] Whether to activate the dark mode or not.

Overview

Methods

callback	Remove or add the ruler widget actor upon click.
update	Define the configuration and representation of the ruler widget button.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.ruler import Ruler
```

Method detail

Ruler.**callback**(*state: bool*) → None

Remove or add the ruler widget actor upon click.

Parameters

state
[bool] Whether the state of the button, which is inherited from PyVista, is `True`.

Notes

This method provides a callback function for the ruler widet. It is called every time the ruler widget is clicked.

Ruler.**update**() → None

Define the configuration and representation of the ruler widget button.

Description

Provides the ruler widget for the Visualization Interface Tool plotter.

The `save_gif_button.py` module

Summary

Classes

SaveGifButton	Provides GIF export control for animations.

Enums

SaveGifButtonConfig	Provides configuration for the save GIF button.

SaveGifButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.save_gif_button.**SaveGifButton**(*plotter: pyvista.I an- i- ma- tion, dark_mo bool = False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.button.Button*

Provides GIF export control for animations.

This button exports the animation as a timestamped GIF file.

Parameters

plotter
[`Plotter`] Plotter to draw the button on.

animation
[Animation] Animation instance to control.

dark_mode
[bool, optional] Whether to activate the dark mode or not.

Overview

Methods

callback	Export animation as GIF file.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.save_gif_button import␣
↪SaveGifButton
```

Method detail

SaveGifButton.**callback**(*state: bool*) → None

Export animation as GIF file.

Parameters

state
[bool] Whether the button is active.

SaveGifButtonConfig

class ansys.tools.visualization_interface.backends.pyvista.widgets.save_gif_button.**SaveGifButtonConfig**

Bases: enum.Enum

Provides configuration for the save GIF button.

Overview

Attributes

SAVE_GIF

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.save_gif_button import␣
↪SaveGifButtonConfig
```

Attribute detail

SaveGifButtonConfig.**SAVE_GIF = (0, 'save', (210, 10))**

Description

Provides the save GIF button widget for animation control.

The screenshot.py module

Summary

Classes

ScreenshotButton	Provides the screenshot widget for the Visualization Interface Tool `Plotter` class.

ScreenshotButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.screenshot.**ScreenshotButton**(*plotter: pyvista.Plo dark_mode bool = False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Provides the screenshot widget for the Visualization Interface Tool `Plotter` class.

Parameters

plotter
[`Plotter`] Provides the plotter to add the screenshot widget to.

dark_mode
[bool, `optional`] Whether to activate the dark mode or not.

Overview

Methods

callback	Remove or add the screenshot widget actor upon click.
update	Define the configuration and representation of the screenshot widget button.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.screenshot import␣
↪ScreenshotButton
```

Method detail

ScreenshotButton.**callback**(*state: bool*) → None

Remove or add the screenshot widget actor upon click.

Parameters

state
[bool] Whether the state of the button, which is inherited from PyVista, is `True`.

Notes

This method provides a callback function for the screenshot widget. It is called every time the screenshot widget is clicked.

ScreenshotButton.**update**() → None

Define the configuration and representation of the screenshot widget button.

Description

Provides the screenshot widget for the Visualization Interface Tool plotter.

The stop_button.py module

Summary

Classes

StopButton	Provides stop control for animations.

Enums

StopButtonConfig	Provides configuration for the stop button.

StopButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.stop_button.**StopButton**(*plotter: pyvista.Plotter*, *animation*, *dark_mode: bool = False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.button.Button*

Provides stop control for animations.

This button stops the animation and resets to the first frame.

Parameters

plotter
[Plotter] Plotter to draw the button on.

animation
[Animation] Animation instance to control.

dark_mode
[bool, optional] Whether to activate the dark mode or not.

Overview

Methods

callback	Stop animation and reset to first frame.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.stop_button import␣
↪StopButton
```

Method detail

StopButton.**callback**(*state: bool*) → None

Stop animation and reset to first frame.

Parameters

state
[bool] Whether the button is active.

StopButtonConfig

class
ansys.tools.visualization_interface.backends.pyvista.widgets.stop_button.**StopButtonConfig**

Bases: enum.Enum

Provides configuration for the stop button.

Overview

Attributes

STOP

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.stop_button import␣
↪StopButtonConfig
```

Attribute detail

StopButtonConfig.**STOP = (0, 'stop', (60, 10))**

Description

Provides the stop button widget for animation control.

The tree_menu_toggle.py module

Summary

Classes

TreeMenuToggleButton	Provides a button to show/hide the tree menu widget.

TreeMenuToggleButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.tree_menu_toggle.**TreeMenuToggleButton**

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*

Provides a button to show/hide the tree menu widget.

This button toggles the visibility of the dynamic tree menu panel.

Parameters

plotter
[Plotter] Plotter to draw the button on.

dark_mode
[bool, optional] Whether to activate dark mode. Default is False.

tree_menu
[DynamicTreeMenuWidget, optional] The tree menu widget to toggle. If None, will search for it in plotter widgets.

Overview

Methods

update	Assign the image that represents the button.
callback	Toggle the tree menu visibility.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.tree_menu_toggle␣
↪import TreeMenuToggleButton
```

Method detail

TreeMenuToggleButton.**update**() → None

Assign the image that represents the button.

TreeMenuToggleButton.**callback**(*state: bool*) → None

Toggle the tree menu visibility.

Parameters

state
[bool] The state of the button (True = show menu, False = hide menu).

Description

Provides a button to toggle the tree menu visibility.

The `view_button.py` module

Summary

Classes

ViewButton	Provides for changing the view.

Enums

ViewDirection	Provides an enum with the available views.

ViewButton

class ansys.tools.visualization_interface.backends.pyvista.widgets.view_button.**ViewButton**(*plotter: pyvista.Plotter*, *direction: tuple*, *dark_mode: bool = False*)

Bases: *ansys.tools.visualization_interface.backends.pyvista.widgets.button.Button*

Provides for changing the view.

Parameters

plotter
[`Plotter`] Plotter to draw the buttons on.

direction
[`ViewDirection`] Direction of the view.

dark_mode
[bool, `optional`] Whether to activate the dark mode or not.

Overview

Methods

callback	Change the view depending on button interaction.

Attributes

direction

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.view_button import␣
↪ViewButton
```

Attribute detail

ViewButton.**direction**

Method detail

ViewButton.**callback**(*state: bool*) → None

Change the view depending on button interaction.

Parameters

state
[bool] Whether the state of the button, which is inherited from PyVista, is `True`.

Raises

`NotImplementedError`
Raised if the specified direction is not implemented.

ViewDirection

class
ansys.tools.visualization_interface.backends.pyvista.widgets.view_button.**ViewDirection**

Bases: `enum.Enum`

Provides an enum with the available views.

Overview

Attributes

XYPLUS
XYMINUS
XZPLUS
XZMINUS
YZPLUS
YZMINUS
ISOMETRIC

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.view_button import␣
↪ViewDirection
```

Attribute detail

ViewDirection.**XYPLUS = (0, '+xy', (5, 280))**

ViewDirection.**XYMINUS = (1, '-xy', (5, 311))**

ViewDirection.**XZPLUS = (2, '+xz', (5, 342))**

ViewDirection.**XZMINUS = (3, '-xz', (5, 373))**

ViewDirection.**YZPLUS = (4, '+yz', (5, 404))**

ViewDirection.**YZMINUS = (5, '-yz', (5, 435))**

ViewDirection.**ISOMETRIC = (6, 'isometric', (5, 466))**

Description

Provides the view button widget for changing the camera view.

The `widget.py` module

Summary

Classes

PlotterWidget	Provides an abstract class for plotter widgets.

PlotterWidget

class ansys.tools.visualization_interface.backends.pyvista.widgets.widget.**PlotterWidget**(*plotter: pyvista.Plotter*)

Bases: abc.ABC

Provides an abstract class for plotter widgets.

Parameters

plotter
[Plotter] Plotter instance to add the widget to.

Notes

These widgets are intended to be used with PyVista plotter objects. More specifically, the way in which this abstraction has been built ensures that these widgets can be easily integrated with the Visualization Interface Tool's widgets.

Overview

Abstract methods

callback	General callback function for PlotterWidget objects.
update	General update function for PlotterWidget objects.

Properties

plotter	Plotter object that the widget is assigned to.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.widgets.widget import␣
↪PlotterWidget
```

Property detail

property PlotterWidget.**plotter:** pyvista.Plotter

Plotter object that the widget is assigned to.

Method detail

abstractmethod PlotterWidget.**callback**(*state*) → None

General callback function for `PlotterWidget` objects.

abstractmethod PlotterWidget.**update**() → None

General update function for `PlotterWidget` objects.

Description

Provides the abstract implementation of plotter widgets.

Description

Provides widgets for the Visualization Interface Tool plotter.

The `animation.py` module

Summary

Classes

FrameSequence	Abstract interface for frame data sources.
InMemoryFrameSequence	Frame sequence with all frames pre-loaded in memory.
Animation	Animation controller for PyVista visualizations.

Enums

AnimationState	Animation playback states.

Constants

DARK_MODE_THRESHOLD

FrameSequence

class ansys.tools.visualization_interface.backends.pyvista.animation.**FrameSequence**

Bases: abc.ABC

Abstract interface for frame data sources.

This class provides an abstraction for different frame storage strategies, allowing for in-memory, lazy-loaded, or computed frame sequences.

Overview

Abstract methods

get_frame	Retrieve frame at given index.
__len__	Return total number of frames.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.animation import FrameSequence
```

Method detail

abstractmethod FrameSequence.**get_frame**(*index: int*) → Any

Retrieve frame at given index.

Parameters

index
[int] Frame index to retrieve.

Returns

Any
Frame data (typically a PyVista mesh or MeshObjectPlot).

abstractmethod FrameSequence.**__len__**() → int

Return total number of frames.

Returns

int
Total frame count.

InMemoryFrameSequence

class ansys.tools.visualization_interface.backends.pyvista.animation.**InMemoryFrameSequence**(*frames: List[Any]*)

Bases: FrameSequence

Frame sequence with all frames pre-loaded in memory.

This is the simplest strategy, suitable for small to medium datasets where all frames can fit in memory.

Parameters

frames
[List[Any]] List of frame objects (meshes or mesh objects).

Overview

Methods

get_frame	Retrieve frame at given index.

Special methods

__len__	Return total number of frames.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.animation import␣
↪InMemoryFrameSequence
```

Method detail

InMemoryFrameSequence.**get_frame**(*index: int*) → Any

Retrieve frame at given index.

InMemoryFrameSequence.**__len__**() → int

Return total number of frames.

Animation

class ansys.tools.visualization_interface.backends.pyvista.animation.**Animation**(*plotter: pyvista.Plotter*, *frames: FrameSequence | List[Any]*, *fps: int = 30*, *loop: bool = False*, *scalar_bar_args: dict | None = None*, ***plot_kwargs*)

Animation controller for PyVista visualizations.

This class manages animation playback, providing play/pause/stop controls, frame stepping, timeline scrubbing, and export capabilities.

Parameters

plotter
[pv.Plotter] PyVista plotter instance to animate.

frames
[FrameSequence or List[Any]] Frame sequence or list of frame objects to animate.

fps
[int, optional] Frames per second for playback (default: 30).

loop
[bool, optional] Whether to loop animation continuously (default: False).

scalar_bar_args
[dict, optional] Scalar bar and rendering arguments to apply. Supports: - `clim` : tuple - Fixed color scale (min, max) for all frames - `title` : str - Scalar bar title - `color` : str - Scalar bar text color - Other parameters accepted by PyVista's `add_mesh` method

Examples

Create and play a simple animation:

```
>>> from ansys.tools.visualization_interface import Plotter
>>> plotter = Plotter(backend='pyvista')
>>> frames = [mesh1, mesh2, mesh3]
>>> animation = plotter.animate(frames, fps=30, loop=True)
>>> animation.play()
```

Overview

Methods

play	Start or resume animation playback.
pause	Pause animation playback.
stop	Stop animation and reset to first frame.
step_forward	Advance one frame forward.
step_backward	Rewind one frame backward.
seek	Jump to specific frame.
save	Export animation to video file.
show	Display animation with the plotter.

Properties

state	Current animation state.
current_frame	Current frame index.
total_frames	Total number of frames.
fps	Frames per second.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.animation import Animation
```

Property detail

property Animation.**state:** *AnimationState*
Current animation state.

property Animation.**current_frame:** int
Current frame index.

property Animation.**total_frames:** int
Total number of frames.

property Animation.**fps**: int

Frames per second.

Method detail

Animation.**play**()

Start or resume animation playback.

Animation.**pause**()

Pause animation playback.

Animation.**stop**()

Stop animation and reset to first frame.

Animation.**step_forward**()

Advance one frame forward.

Animation.**step_backward**()

Rewind one frame backward.

Animation.**seek**(*frame_index: int*)

Jump to specific frame.

Parameters

frame_index
[int] Target frame index.

Animation.**save**(*filename: str | pathlib.Path*, *quality: int = 5*, *framerate: int | None = None*, *close_plotter: bool = False*)

Export animation to video file.

Parameters

filename
[str or Path] Output filename (.mp4, .gif, .avi).

quality
[int, optional] Video quality (1-10, higher is better). Default is 5.

framerate
[int, optional] Output framerate. If None, uses animation fps.

close_plotter
[bool, optional] If True, closes plotter after saving. Default is False.

Animation.**show**(*show_controls: bool = True*, ***kwargs*)

Display animation with the plotter.

Parameters

show_controls
[bool, optional] If True, shows interactive controls. Default is True.

****kwargs**
Additional arguments passed to plotter.show().

AnimationState

class ansys.tools.visualization_interface.backends.pyvista.animation.**AnimationState**

Bases: enum.Enum

Animation playback states.

Overview

Attributes

STOPPED
PLAYING
PAUSED

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.animation import AnimationState
```

Attribute detail

AnimationState.**STOPPED** = **'stopped'**

AnimationState.**PLAYING** = **'playing'**

AnimationState.**PAUSED** = **'paused'**

Description

Animation support for PyVista backend.

Module detail

animation.**DARK_MODE_THRESHOLD** = **120**

The picker.py module

Summary

Classes

AbstractPicker	Abstract base class for pickers.
Picker	Class to manage picking and hovering of objects in the plotter.

AbstractPicker

class ansys.tools.visualization_interface.backends.pyvista.picker.**AbstractPicker**(*plotter_backend: an-sys.tools.visualization_interfa **kwargs*)

Bases: abc.ABC

Abstract base class for pickers.

Overview

Abstract methods

pick_select_object	Determine actions to take when an object is selected.
pick_unselect_object	Determine actions to take when an object is unselected.
hover_select_object	Determine actions to take when an object is hovered over.
hover_unselect_object	Determine actions to take when an object is unhovered.

Properties

picked_dict	Return the dictionary of picked objects.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.picker import AbstractPicker
```

Property detail

property AbstractPicker.**picked_dict:** dict

Abstractmethod

Return the dictionary of picked objects.

Method detail

abstractmethod AbstractPicker.**pick_select_object**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPl | ansys.tools.visualization_interface.types.edge_plot.EdgePlot, *pt:* *numpy.ndarray*) → None

Determine actions to take when an object is selected.

abstractmethod AbstractPicker.**pick_unselect_object**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjec | ansys.tools.visualization_interface.types.edge_plot.EdgePlot) → None

Determine actions to take when an object is unselected.

abstractmethod AbstractPicker.**hover_select_object**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectI | ansys.tools.visualization_interface.types.edge_plot.EdgePlot, *pt:* *numpy.ndarray*) → None

Determine actions to take when an object is hovered over.

abstractmethod AbstractPicker.**hover_unselect_object**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObje | ansys.tools.visualization_interface.types.edge_plot.EdgePlot) → None

Determine actions to take when an object is unhovered.

Picker

class ansys.tools.visualization_interface.backends.pyvista.picker.**Picker**(*plotter_backend: ansys.tools.visualization_interface.backen*, *plot_picked_names: bool = True*)

Bases: `AbstractPicker`

Class to manage picking and hovering of objects in the plotter.

This class is responsible for managing the selection and deselection of objects in the plotter, both through direct picking and hovering. It keeps track of the currently selected and hovered objects, and provides methods to select and unselect them.

Parameters

plotter_backend
[`Plotter`] The plotter instance to which this picker is attached.

plot_picked_names
[bool, `optional`] Whether to display the names of picked objects in the plotter. Defaults to True.

Overview

Methods

pick_select_object	Add actor to picked list and add label if required.
pick_unselect_object	Remove actor from picked list and remove label if required.
hover_select_object	Add label to hovered object if required.
hover_unselect_object	Remove all hover labels from the scene.

Properties

picked_dict	Return the dictionary of picked objects.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.picker import Picker
```

Property detail

property Picker.**picked_dict**: dict

Return the dictionary of picked objects.

Method detail

Picker.**pick_select_object**(*custom_object: ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot | ansys.tools.visualization_interface.types.edge_plot.EdgePlot*, *pt: numpy.ndarray*) → None

Add actor to picked list and add label if required.

Parameters

custom_object
[Union[MeshObjectPlot, EdgePlot]] The object to be selected.

pt
[np.ndarray] The point where the object was picked.

Picker.**pick_unselect_object**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot | ansys.tools.visualization_interface.types.edge_plot.EdgePlot) → None

Remove actor from picked list and remove label if required.

Parameters

custom_object
[Union[MeshObjectPlot, EdgePlot]] The object to be unselected.

Picker.**hover_select_object**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot | ansys.tools.visualization_interface.types.edge_plot.EdgePlot, *actor:* *pyvista.Actor*) → None

Add label to hovered object if required.

Parameters

custom_object
[Union[MeshObjectPlot, EdgePlot]] The object to be hovered over.

actor
[vtkActor] The actor corresponding to the hovered object.

Picker.**hover_unselect_object**()

Remove all hover labels from the scene.

Description

Module for managing picking and hovering of objects in a PyVista plotter.

The pyvista.py module

Summary

Classes

PyVistaBackendInterface	Provides the interface for the Visualization Interface Tool plotter.
PyVistaBackend	Provides the generic plotter implementation for PyAnsys libraries.

Constants

DARK_MODE_THRESHOLD

PyVistaBackendInterface

class ansys.tools.visualization_interface.backends.pyvista.pyvista.**PyVistaBackendInterface**(*use_trame: bool | None = None*, *allow_picking: bool | None = False*, *allow_hovering: bool | None = False*, *plot_picked_nar bool | None = False*, *show_plane: bool | None = False*, *use_qt: bool | None = False*, *show_qt: bool | None = True*, *custom_picker: ansys.tools.visuali = None*, *custom_picker_kwa Dict[str, Any] | None = None*, ***plotter_kwargs*)

Bases: ansys.tools.visualization_interface.backends._base.BaseBackend

Provides the interface for the Visualization Interface Tool plotter.

This class is intended to be used as a base class for the custom plotters in the different PyAnsys libraries. It provides the basic plotter functionalities, such as adding objects and enabling widgets and picking capabilities. It also provides the ability to show the plotter using the trame service.

You can override the `plot_iter()`, `plot()`, and `picked_operation()` methods. The `plot_iter()` method is intended to plot a list of objects to the plotter, while the `plot()` method is intended to plot a single object to the plotter. The `show()` method is intended to show the plotter. The `picked_operation()` method is intended to perform an operation on the picked objects.

Parameters

use_trame
[Optional[bool], default: None] Whether to activate the usage of the trame UI instead of the Python window.

allow_picking
[Optional[bool], default: False] Whether to allow picking capabilities in the window. Incompatible with hovering. Picking will take precedence over hovering.

allow_hovering
[Optional[bool], default: False] Whether to allow hovering capabilities in the window. Incompatible with picking. Picking will take precedence over hovering.

plot_picked_names
[Optional[bool], default: False] Whether to plot the names of the picked objects.

show_plane
[Optional[bool], default: False] Whether to show the plane in the plotter.

use_qt
[Optional[bool], default: False] Whether to use the Qt backend for the plotter.

show_qt
[Optional[bool], default: True] Whether to show the Qt window.

custom_picker
[AbstractPicker, default: None] Custom picker class that extends the AbstractPicker class.

custom_picker_kwargs
[Optional[Dict[str, Any]], default: None] Keyword arguments to pass to the custom picker class.

Overview

Abstract methods

plot_iter	Plot one or more compatible objects to the plotter.
plot	Plot a single object to the plotter.

Methods

enable_widgets	Enable the widgets for the plotter.
add_widget	Add one or more custom widgets to the plotter.
picker_callback	Define the callback for the element picker.
hover_callback	Define the callback for the element hover.
focus_point_selection	Focus the camera on a selected actor.
compute_edge_object_map	Compute the mapping between plotter actors and `EdgePlot` objects.
enable_picking	Enable picking capabilities in the plotter.
enable_set_focus_center	Enable setting the focus of the camera to the picked point.
enable_hover	Enable hover capabilities in the plotter.
disable_picking	Disable picking capabilities in the plotter.
disable_hover	Disable hover capabilities in the plotter.
disable_center_focus	Disable setting the focus of the camera to the picked point.
show	Plot and show any PyAnsys object.
show_plotter	Show the plotter or start the trame service.
picked_operation	Perform an operation on the picked objects.

Properties

pv_interface	PyVista interface.
scene	PyVista scene.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.pyvista import␣
↪PyVistaBackendInterface
```

Property detail

property PyVistaBackendInterface.**pv_interface:**
ansys.tools.visualization_interface.backends.pyvista.pyvista_interface.PyVistaInterface

PyVista interface.

property PyVistaBackendInterface.**scene:** pyvista.Plotter

PyVista scene.

Method detail

PyVistaBackendInterface.**enable_widgets**(*dark_mode: bool = False*) → None

Enable the widgets for the plotter.

Parameters

dark_mode
[bool, default: `False`] Whether to use dark mode for the widgets.

PyVistaBackendInterface.**add_widget**(*widget:* ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget | *List[*ansys.tools.visualization_interface.backends.pyvista.widgets.widget.PlotterWidget*]*)

Add one or more custom widgets to the plotter.

Parameters

widget
[Union[PlotterWidget, List[PlotterWidget]]] One or more custom widgets.

PyVistaBackendInterface.**picker_callback**(*actor: pyvista.Actor*) → None

Define the callback for the element picker.

Parameters

actor
[Actor] Actor to select for the picker.

PyVistaBackendInterface.**hover_callback**(*_widget*, *event_name*) → None

Define the callback for the element hover.

Parameters

actor
[Actor] Actor to hover for the picker.

PyVistaBackendInterface.**focus_point_selection**(*actor: pyvista.Actor*) → None

Focus the camera on a selected actor.

Parameters

actor
[Actor] Actor to focus the camera on.

PyVistaBackendInterface.**compute_edge_object_map**() → Dict[pyvista.Actor, *ansys.tools.visualization_interface.types.edge_plot.EdgePlot*]

Compute the mapping between plotter actors and EdgePlot objects.

Returns

Dict[Actor, EdgePlot]
Dictionary defining the mapping between plotter actors and EdgePlot objects.

PyVistaBackendInterface.**enable_picking**()

Enable picking capabilities in the plotter.

PyVistaBackendInterface.**enable_set_focus_center**()

Enable setting the focus of the camera to the picked point.

PyVistaBackendInterface.**enable_hover**()

Enable hover capabilities in the plotter.

PyVistaBackendInterface.**disable_picking**()

Disable picking capabilities in the plotter.

PyVistaBackendInterface.**disable_hover**()

Disable hover capabilities in the plotter.

PyVistaBackendInterface.**disable_center_focus**()

Disable setting the focus of the camera to the picked point.

PyVistaBackendInterface.**show**(*plottable_object: Any = None*, *screenshot: str | None = None*, *view_2d: Dict = None*, *dark_mode: bool = False*, ***kwargs: Dict[str, Any]*) → List[Any]

Plot and show any PyAnsys object.

The types of objects supported are MeshObjectPlot, pv.MultiBlock, and pv.PolyData.

Parameters

plottable_object
[Any, default: None] Object or list of objects to plot.

screenshot
[str, default: None] Path for saving a screenshot of the image that is being represented.

view_2d
[Dict, default: None] Dictionary with the plane and the viewup vectors of the 2D plane.

dark_mode
[bool, default: False] Whether to use dark mode for the widgets.

****kwargs**
[Any] Additional keyword arguments for the show or plot method.

Returns

List[Any]
List with the picked bodies in the picked order.

PyVistaBackendInterface.**show_plotter**(*screenshot: str | None = None*, ***kwargs*) → None

Show the plotter or start the trame service.

Parameters

plotter
[Plotter] Visualization Interface Tool plotter with the meshes added.

screenshot
[str, default: None] Path for saving a screenshot of the image that is being represented.

abstractmethod PyVistaBackendInterface.**plot_iter**(*plottable_object: Any*, ***plotting_options*)

Plot one or more compatible objects to the plotter.

Parameters

plottable_object
[Any] One or more objects to add.

****plotting_options**
[dict, default: None] Keyword arguments. For allowable keyword arguments, see the Plotter.add_mesh method.

abstractmethod PyVistaBackendInterface.**plot**(*plottable_object: Any*, ***plotting_options*)

Plot a single object to the plotter.

Parameters

plottable_object
[Any] Object to plot.

****plotting_options**
[dict, default: None] Keyword arguments. For allowable keyword arguments, see the Plotter.add_mesh method.

PyVistaBackendInterface.**picked_operation**() → None

Perform an operation on the picked objects.

PyVistaBackend

class ansys.tools.visualization_interface.backends.pyvista.pyvista.**PyVistaBackend**(*use_trame: bool | None = None*, *allow_picking: bool | None = False*, *allow_hovering: bool | None = False*, *plot_picked_names: bool | None = True*, *use_qt: bool | None = False*, *show_qt: bool | None = False*, *custom_picker: ansys.tools.visualization_inter = None*, ***plotter_kwargs*)

Bases: `PyVistaBackendInterface`

Provides the generic plotter implementation for PyAnsys libraries.

This class accepts `MeshObjectPlot`, `pv.MultiBlock` and `pv.PolyData` objects.

Parameters

use_trame
[bool, default: `None`] Whether to enable the use of trame. The default is `None`, in which case the `USE_TRAME` global setting is used.

allow_picking
[`Optional`[bool], default: `False`] Whether to allow picking capabilities in the window. Incompatible with hovering. Picking will take precedence over hovering.

allow_hovering
[`Optional`[bool], default: `False`] Whether to allow hovering capabilities in the window. Incompatible with picking. Picking will take precedence over hovering.

plot_picked_names
[bool, default: `True`] Whether to plot the names of the picked objects.

Overview

Methods

plot_iter	Plot the elements of an iterable of any type of object to the scene.
plot	Plot a pyansys or PyVista object to the plotter.
close	Close the plotter for PyVistaQT.
create_animation	Create an animation from a sequence of frames.
add_points	Add point markers to the scene.
add_lines	Add line segments to the scene.
add_planes	Add a plane to the scene.
add_text	Add text to the scene.
add_labels	Add labels at 3D point locations.
clear	Clear all actors from the scene and reset the plotter.

Properties

base_plotter	Return the base plotter object.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.pyvista import PyVistaBackend
```

Property detail

property PyVistaBackend.**base_plotter**

Return the base plotter object.

Method detail

PyVistaBackend.**plot_iter**(*plotting_list: List[Any]*, ***plotting_options*) → None

Plot the elements of an iterable of any type of object to the scene.

The types of objects supported are Body, Component, List[pv.PolyData], pv.MultiBlock, and Sketch.

Parameters

plotting_list
[List[Any]] List of objects to plot.

****plotting_options**
[dict, default: None] Keyword arguments. For allowable keyword arguments, see the Plotter.add_mesh method.

PyVistaBackend.**plot**(*plottable_object: Any*, ***plotting_options*)

Plot a pyansys or PyVista object to the plotter.

Parameters

plottable_object
[Any] Object to plot.

****plotting_options**
[dict, default: None] Keyword arguments. For allowable keyword arguments, see the Plotter.add_mesh method.

PyVistaBackend.**close**()

Close the plotter for PyVistaQT.

PyVistaBackend.**create_animation**(*frames: List[Any] | ansys.tools.visualization_interface.backends.pyvista.animation.FrameSequence*, *fps: int = 30*, *loop: bool = False*, *scalar_bar_args: dict | None = None*, ***plot_kwargs*) → *ansys.tools.visualization_interface.backends.pyvista.animation.Animation*

Create an animation from a sequence of frames.

This method creates an `Animation` object that can be used to visualize time-series simulation results, transient analyses, and dynamic phenomena.

Parameters

frames
[`List[Any]` or `FrameSequence`] Sequence of frame objects to animate. Can be a list of PyVista meshes, `MeshObjectPlot` objects, or a custom `FrameSequence` implementation for lazy loading.

fps
[`int`, `optional`] Frames per second for playback. Default is 30.

loop
[bool, `optional`] Whether to loop animation continuously. Default is False.

scalar_bar_args
[`dict`, `optional`] Scalar bar arguments to apply to all frames (e.g., `clim` for fixed color scale). If not provided, a global color scale is calculated automatically.

****plot_kwargs**
Additional keyword arguments passed to add_mesh for all frames (e.g., `cmap='viridis'`, `opacity=0.8`).

Returns

Animation
Animation controller object with playback controls.

See also

Animation
Animation controller class

Examples

Create and play an animation from transient simulation results:

```
>>> from ansys.tools.visualization_interface import Plotter
>>> plotter = Plotter(backend='pyvista')
>>> frames = [mesh1, mesh2, mesh3, mesh4]  # Time series data
>>> animation = plotter.backend.create_animation(frames, fps=30, loop=True)
>>> animation.play()
>>> animation.show()
```

Export animation to video:

```
>>> animation = plotter.backend.create_animation(frames)
>>> animation.save("output.mp4", quality=8)
```

Use fixed color scale for accurate comparison:

```
>>> animation = plotter.backend.create_animation(
...     frames,
...     scalar_bar_args={"clim": (0.0, 1.0), "title": "Displacement [m]"}
... )
```

PyVistaBackend.**add_points**(*points: List | Any*, *color: str = 'red'*, *size: float = 10.0*, ***kwargs*) → pyvista.Actor

Add point markers to the scene.

Parameters

points
[Union[List, Any]] Points to add. Can be a list of coordinates or array-like object. Expected format: [[x1, y1, z1], [x2, y2, z2], ...] or Nx3 array.

color
[str, default: "red"] Color of the points.

size
[float, default: 10.0] Size of the point markers.

****kwargs**
[dict] Additional keyword arguments passed to PyVista's add_mesh method.

Returns

pv.Actor
PyVista actor representing the added points.

PyVistaBackend.**add_lines**(*points: List | Any*, *connections: List | Any | None = None*, *color: str = 'white'*, *width: float = 1.0*, ***kwargs*) → pyvista.Actor

Add line segments to the scene.

Parameters

points
[Union[List, Any]] Points defining the lines. Can be a list of coordinates or array-like object. Expected format: [[x1, y1, z1], [x2, y2, z2], ...] or Nx3 array.

connections
[Optional[Union[List, Any]], default: None] Line connectivity. If None, connects points sequentially. Expected format: [[start_idx1, end_idx1], [start_idx2, end_idx2], ...] or Mx2 array where M is the number of lines.

color
[str, default: "white"] Color of the lines.

width
[float, default: 1.0] Width of the lines.

****kwargs**
[dict] Additional keyword arguments passed to PyVista's add_mesh method.

Returns

pv.Actor
PyVista actor representing the added lines.

PyVistaBackend.**add_planes**(*center: tuple[float, float, float] = (0.0, 0.0, 0.0)*, *normal: tuple[float, float, float] = (0.0, 0.0, 1.0)*, *i_size: float = 1.0*, *j_size: float = 1.0*, ***kwargs*) → pyvista.Actor

Add a plane to the scene.

Parameters

center
[Tuple[float, float, float], default: (0.0, 0.0, 0.0)] Center point of the plane (x, y, z).

normal
[Tuple[float, float, float], default: (0.0, 0.0, 1.0)] Normal vector of the plane (x, y, z).

i_size
[float, default: 1.0] Size of the plane in the i direction.

j_size
[float, default: 1.0] Size of the plane in the j direction.

****kwargs**
[dict] Additional keyword arguments passed to PyVista's add_mesh method (e.g., color, opacity).

Returns

pv.Actor
PyVista actor representing the added plane.

PyVistaBackend.**add_text**(*text: str*, *position: tuple[float, float] | str*, *font_size: int = 12*, *color: str = 'white'*, ***kwargs*) → pyvista.Actor

Add text to the scene.

Parameters

text
[str] Text string to display.

position
[Union[Tuple[float, float], str]] Position for the text. Can be:

- 2D tuple (x, y) for screen coordinates (pixels from bottom-left)
- String position like 'upper_left', 'upper_right', 'lower_left', 'lower_right', 'upper_edge', 'lower_edge' (PyVista-specific)

font_size
[int, default: 12] Font size for the text.

color
[str, default: "white"] Color of the text.

****kwargs**
[dict] Additional keyword arguments passed to PyVista's add_text method.

Returns

pv.Actor
PyVista actor representing the added text.

PyVistaBackend.**add_labels**(*points: List | Any*, *labels: List[str]*, *font_size: int = 12*, *point_size: float = 5.0*, ***kwargs*) → pyvista.Actor

Add labels at 3D point locations.

Parameters

points
[Union[List, Any]] Points where labels should be placed. Can be a list of coordinates or array-like object. Expected format: [[x1, y1, z1], ...] or Nx3 array.

labels
[List[str]] List of label strings to display at each point.

font_size
[int, default: 12] Font size for the labels.

point_size
[float, default: 5.0] Size of the point markers shown with labels.

****kwargs**
[dict] Additional keyword arguments passed to PyVista's add_point_labels method.

Returns

pv.Actor
PyVista actor representing the added labels.

PyVistaBackend.**clear**() → None

Clear all actors from the scene and reset the plotter.

This method removes all previously added objects (meshes, points, lines, text, etc.) from the visualization scene by fully reinitializing the plotter.

Description

Provides a wrapper to aid in plotting.

Module detail

pyvista.**DARK_MODE_THRESHOLD = 120**

The pyvista_interface.py module

Summary

Classes

PyVistaInterface	Provides the middle class between PyVista plotting operations and PyAnsys objects.

PyVistaInterface

class ansys.tools.visualization_interface.backends.pyvista.pyvista_interface.**PyVistaInterface**(*scene: pyvista.Plot | pyvis-taqt.Backgr | None = None, color_opts: Dict | None = None, num_points: int = 100, en-able_widget bool = True, show_plane bool = False, use_qt: bool = False, show_qt: bool = True, **plot-ter_kwargs*)

Provides the middle class between PyVista plotting operations and PyAnsys objects.

The main purpose of this class is to simplify interaction between PyVista and the PyVista backend provided. This class is responsible for creating the PyVista scene and adding the PyAnsys objects to it.

Parameters

scene
[Plotter, default: None] Scene for rendering the objects. If passed, off_screen needs to be set manually beforehand for documentation and testing.

color_opts
[dict, default: None] Dictionary containing the background and top colors.

num_points
[int, default: 100] Number of points to use to render the shapes.

enable_widgets

[bool, default: `True`] Whether to enable widget buttons in the plotter window. Widget buttons must be disabled when using trame for visualization.

show_plane
[bool, default: `False`] Whether to show the XY plane in the plotter window.

use_qt
[bool, default: `False`] Whether to use the Qt backend for the plotter window.

show_qt
[bool, default: `True`] Whether to show the Qt plotter window.

Overview

Methods

view_xy	View the scene from the XY plane.
view_xz	View the scene from the XZ plane.
view_yx	View the scene from the YX plane.
view_yz	View the scene from the YZ plane.
view_zx	View the scene from the ZX plane.
view_zy	View the scene from the ZY plane.
enable_parallel_projection	Enable parallel projection for the camera.
disable_parallel_projection	Disable parallel projection for the camera.
clip	Clip a given mesh with a plane.
plot_meshobject	Plot a generic `MeshObjectPlot` object to the scene.
plot_edges	Plot the outer edges of an object to the plot.
hide_children	Hide all the children of a given `MeshObjectPlot` object.
show_children	Show all the children of a given `MeshObjectPlot` object.
toggle_subtree_visibility	Toggle visibility of an object and its entire subtree.
plot	Plot any type of object to the scene.
plot_iter	Plot elements of an iterable of any type of objects to the scene.
show	Show the rendered scene on the screen.
set_add_mesh_defaults	Set the default values for the plotting options.

Properties

scene	Rendered scene object.
object_to_actors_map	Mapping between the PyVista actor and the PyAnsys objects.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.pyvista_interface import␣
↪PyVistaInterface
```

Property detail

property PyVistaInterface.**scene:** *pyvista.plotting.plotter.Plotter*

Rendered scene object.

Returns

Plotter
Rendered scene object.

property PyVistaInterface.object_to_actors_map: Dict[pyvista.Actor, *ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot*]

Mapping between the PyVista actor and the PyAnsys objects.

Method detail

PyVistaInterface.**view_xy**() → None

View the scene from the XY plane.

PyVistaInterface.**view_xz**() → None

View the scene from the XZ plane.

PyVistaInterface.**view_yx**() → None

View the scene from the YX plane.

PyVistaInterface.**view_yz**() → None

View the scene from the YZ plane.

PyVistaInterface.**view_zx**() → None

View the scene from the ZX plane.

PyVistaInterface.**view_zy**() → None

View the scene from the ZY plane.

PyVistaInterface.**enable_parallel_projection**() → None

Enable parallel projection for the camera.

PyVistaInterface.**disable_parallel_projection**() → None

Disable parallel projection for the camera.

PyVistaInterface.**clip**(*mesh: pyvista.PolyData | pyvista.MultiBlock | pyvista.UnstructuredGrid*, *plane:* ansys.tools.visualization_interface.utils.clip_plane.ClipPlane) → pyvista.PolyData | pyvista.MultiBlock

Clip a given mesh with a plane.

Parameters

mesh
[Union[pv.PolyData, pv.MultiBlock]] Mesh.

normal
[str, default: "x"] Plane to use for clipping. Options are "x", "-x", "y", "-y", "z", and "-z".

origin
[tuple, default: None] Origin point of the plane.

plane
[ClipPlane, default: None] Clipping plane to cut the mesh with.

Returns

Union[pv.PolyData,pv.MultiBlock]
Clipped mesh.

PyVistaInterface.**plot_meshobject**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot, *plot_children: bool = True*, ***plotting_options*)

Plot a generic `MeshObjectPlot` object to the scene.

Parameters

custom_object
[`MeshObjectPlot`] Object to add to the scene.

plot_children
[bool, default: `True`] Whether to plot the children of the object.

****plotting_options**
[`dict`, default: `None`] Keyword arguments. For allowable keyword arguments, see the `Plotter.add_mesh` method.

PyVistaInterface.**plot_edges**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot, ***plotting_options*) → None

Plot the outer edges of an object to the plot.

This method has the side effect of adding the edges to the `MeshObjectPlot` object that you pass through the parameters.

Parameters

custom_object
[`MeshObjectPlot`] Custom object with the edges to add.

****plotting_options**
[`dict`, default: `None`] Keyword arguments. For allowable keyword arguments, see the `Plotter.add_mesh` method.

PyVistaInterface.**hide_children**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot) → None

Hide all the children of a given `MeshObjectPlot` object.

Parameters

custom_object
[`MeshObjectPlot`] Custom object whose children will be hidden.

PyVistaInterface.**show_children**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot) → None

Show all the children of a given `MeshObjectPlot` object.

Parameters

custom_object
[`MeshObjectPlot`] Custom object whose children will be shown.

PyVistaInterface.**toggle_subtree_visibility**(*custom_object:* ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot, *include_root: bool = True*) → None

Toggle visibility of an object and its entire subtree.

This method toggles the visibility state of the given object and all its descendants. If the object is currently visible, it and all children will be hidden. If hidden, they will all be shown.

Parameters

custom_object
[MeshObjectPlot] Root object of the subtree to toggle.

include_root
[bool, default: True] Whether to toggle the root object's visibility. If False, only children are toggled.

Examples

Toggle visibility of picked object and its children:

```
>>> picked = plotter.pick()
>>> if picked:
...     backend.toggle_subtree_visibility(picked[0])
```

Toggle only children, keeping parent visible:

```
>>> backend.toggle_subtree_visibility(obj, include_root=False)
```

PyVistaInterface.**plot**(*plottable_object: pyvista.PolyData | pyvista.MultiBlock | ansys.tools.visualization_interface.types.mesh_object_plot.MeshObjectPlot | pyvista.UnstructuredGrid | pyvista.StructuredGrid*, *plot_children: bool = False*, ***plotting_options*) → None

Plot any type of object to the scene.

Supported object types are List[pv.PolyData], MeshObjectPlot, and pv.MultiBlock.

Parameters

plottable_object
[Union[pv.PolyData, pv.MultiBlock, MeshObjectPlot, pv.UnstructuredGrid, pv.StructuredGrid]] Object to plot.

****plotting_options**
[dict, default: None] Keyword arguments. For allowable keyword arguments, see the Plotter.add_mesh method.

PyVistaInterface.**plot_iter**(*plotting_list: List[Any]*, ***plotting_options*) → None

Plot elements of an iterable of any type of objects to the scene.

Parameters

plotting_list
[List[Any]] List of objects to plot.

****plotting_options**
[dict, default: None] Keyword arguments. For allowable keyword arguments, see the Plotter.add_mesh method.

PyVistaInterface.**show**(*show_plane: bool = False*, *jupyter_backend: str | None = None*, ***kwargs: Dict | None*) → None

Show the rendered scene on the screen.

Parameters

show_plane
[bool, default: True] Whether to show the XY plane.

jupyter_backend
[str, default: None] PyVista Jupyter backend.

****kwargs**
[dict, default: None] Plotting and show keyword arguments. For allowable keyword arguments, see the Plotter.show and Plotter.show methods.

Notes

For more information on supported Jupyter backends, see Jupyter Notebook Plotting in the PyVista documentation.

PyVistaInterface.**set_add_mesh_defaults**(*plotting_options: dict*) → None

Set the default values for the plotting options.

Parameters

plotting_options
[dict] Keyword arguments. For allowable keyword arguments, see the Plotter.add_mesh method.

Description

Provides plotting for various PyAnsys objects.

The trame_local.py module

Summary

Classes

TrameVisualizer	Defines the trame layout view.

Constants

CLIENT_TYPE

TrameVisualizer

class ansys.tools.visualization_interface.backends.pyvista.trame_local.**TrameVisualizer**

Defines the trame layout view.

Overview

Methods

set_scene	Set the trame layout view and the mesh to show through the PyVista plotter.
show	Start the trame server and show the mesh.

Attributes

server
plotter

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.trame_local import␣
↪TrameVisualizer
```

Attribute detail

TrameVisualizer.**server**

TrameVisualizer.**plotter = None**

Method detail

TrameVisualizer.**set_scene**(*plotter*)

Set the trame layout view and the mesh to show through the PyVista plotter.

Parameters

plotter
[Plotter] PyVista plotter with the rendered mesh.

TrameVisualizer.**show**()

Start the trame server and show the mesh.

Description

Provides trame visualizer interface for visualization.

Module detail

trame_local.**CLIENT_TYPE = 'vue2'**

The trame_remote.py module

Summary

Functions

send_pl	Send the plotter meshes to a remote trame service.
send_mesh	Send a mesh to a remote trame service.

Description

Module for trame websocket client functions.

Module detail

trame_remote.**send_pl**(*plotter: pyvista.Plotter*, *host: str = 'localhost'*, *port: int = 8765*)

Send the plotter meshes to a remote trame service.

Since plotter can't be pickled, we send the meshes list instead.

Parameters

plotter
[pv.Plotter] Plotter to send.

host
[str, optional] Websocket host to connect to, by default "localhost".

port
[int, optional] Websocket port to connect to, by default 8765.

trame_remote.**send_mesh**(*mesh: pyvista.PolyData | pyvista.MultiBlock*, *host: str = 'localhost'*, *port: int = 8765*)

Send a mesh to a remote trame service.

Parameters

mesh
[Union[pv.PolyData, pv.MultiBlock]] Mesh to send.

host
[str, optional] Websocket host to connect to, by default "localhost".

port
[int, optional] Websocket port to connect to, by default 8765.

The trame_service.py module

Summary

Classes

TrameService	Trame service class.

TrameService

class ansys.tools.visualization_interface.backends.pyvista.trame_service.**TrameService**(*websocket_host: str = 'localhost'*, *websocket_port: int = 8765*)

Trame service class.

Initializes a trame service where you can send meshes to plot in a trame webview plotter.

Parameters

websocket_host
[str, optional] Host where the webserver will listen for new plotters and meshes, by default "localhost".

websocket_port
[int, optional] Port where the webserver will listen for new plotters and meshes, by default 8765.

Overview

Methods

clear_plotter	Clears the web view in the service.
set_scene	Sets the web view scene for the trame service.
run	Start the trame web view and the websocket services.

Import detail

```
from ansys.tools.visualization_interface.backends.pyvista.trame_service import␣
↪TrameService
```

Method detail

TrameService.**clear_plotter**()
Clears the web view in the service.

TrameService.**set_scene**()
Sets the web view scene for the trame service.

TrameService.**run**()
Start the trame web view and the websocket services.

Description

Trame service module.

Description

Provides interfaces.

The usd package

Summary

Subpackages

web	USD-to-HTML web export pipeline (GLB + Three.js templates).

Submodules

html_export	Self-contained USD → Three.js HTML export.
usd_interface	USD backend interface for visualization using python-usd-viewer.
vtk_converter	Asset resolver for handling various file types and USD asset paths.

The web package

Summary

Submodules

glb	GLB conversion entrypoint for USD web viewer export.
glb_builder	Low-level helpers to build GLB buffers and base glTF scene state.
glb_lights	USD-to-glTF light conversion helpers.
glb_materials	Material and texture conversion helpers for GLB export.
glb_mesh	Mesh geometry conversion helpers for GLB export.
templates	HTML template rendering helpers for web viewer export.

The glb.py module

Summary

Functions

convert_usd_to_glb	Convert a USD stage to a self-contained GLB binary.

Description

GLB conversion entrypoint for USD web viewer export.

Module detail

glb.**convert_usd_to_glb**(*path: pathlib.Path*) → bytes

Convert a USD stage to a self-contained GLB binary.

Parameters

path
[Path] Path to the USD file to convert.

Returns

bytes
The GLB binary data.

The glb_builder.py module

Summary

Classes

GLBBuilder	Utility for writing GLTF buffers and accessors.

Functions

create_gltf_scene	Create an initialized glTF document and default scene.

GLBBuilder

class ansys.tools.visualization_interface.backends.usd.web.glb_builder.**GLBBuilder**(*gltf: Any*)

Utility for writing GLTF buffers and accessors.

This class manages the binary buffer and associated glTF structures when converting USD assets to GLB format. It provides methods to append data to the shared buffer and create corresponding buffer views and accessors.

Parameters

gltf
[Any] An initialized glTF document (e.g. pygltflib.GLTF2) to populate.

Overview

Methods

add_buffer_view	Append binary data to the shared buffer and return its buffer-view index.
add_accessor	Create an accessor for an existing buffer view and return its index.
finalize	Finalize the GLTF document and return the serialized GLB bytes.

Attributes

gltf
binary_blob

Import detail

```
from ansys.tools.visualization_interface.backends.usd.web.glb_builder import GLBBuilder
```

Attribute detail

GLBBuilder.**gltf**

GLBBuilder.**binary_blob**

Method detail

GLBBuilder.**add_buffer_view**(*data: bytes*, *target: int | None*) → int

Append binary data to the shared buffer and return its buffer-view index.

Parameters

data
[bytes] The binary data to append to the buffer.

target
[int | None] The target type for the buffer view (e.g., pygltflib.ARRAY_BUFFER or pygltflib.ELEMENT_ARRAY_BUFFER).

GLBBuilder.**add_accessor**(*buffer_view_idx: int*, *component_type: int*, *count: int*, *accessor_type: str*, *min_values: list[float] | None = None*, *max_values: list[float] | None = None*) → int

Create an accessor for an existing buffer view and return its index.

Parameters

buffer_view_idx
[int] The index of the buffer view this accessor references.

component_type
[int] The GLTF component type (e.g., pygltflib.FLOAT, pygltflib.UNSIGNED_SHORT).

count
[int] The number of elements in the accessor.

accessor_type
[str] The GLTF accessor type (e.g., "SCALAR", "VEC3", "MAT4").

min_values
[list[float] | None, optional] Minimum values for the accessor (used for bounding boxes), by default None.

max_values
[list[float] | None, optional] Maximum values for the accessor (used for bounding boxes), by default None.

GLBBuilder.**finalize**() → bytes

Finalize the GLTF document and return the serialized GLB bytes.

Returns

bytes
The complete GLB file as a byte string.

Description

Low-level helpers to build GLB buffers and base glTF scene state.

Module detail

glb_builder.**create_gltf_scene**() → tuple[Any, Any]

Create an initialized glTF document and default scene.

Returns

tuple[Any, Any]
A tuple containing the glTF document and the default scene object.

The glb_lights.py module

Summary

Functions

append_usd_lights_to_scene	Convert supported USD lights into glTF KHR_lights_punctual lights.

Description

USD-to-glTF light conversion helpers.

Module detail

glb_lights.**append_usd_lights_to_scene**(*stage: Any*, *gltf: Any*, *scene: Any*) → None

Convert supported USD lights into glTF KHR_lights_punctual lights.

Parameters

stage
[Any] The USD stage containing the lights.

gltf
[Any] The glTF document to modify.

scene
[Any] The glTF scene to which the lights will be added.

The glb_materials.py module

Summary

Functions

get_material_index	Resolve GLTF material index from USD bindings or displayColor fallback.

Description

Material and texture conversion helpers for GLB export.

Module detail

glb_materials.**get_material_index**(*prim: Any*, *display_colors: Any*, *color_interpolation: Any*, *gltf: Any*, *builder:* ansys.tools.visualization_interface.backends.usd.web.glb_builder.GLBBuilder, *texture_cache: dict[tuple[str, str, str], int]*, *image_cache: dict[str, int]*, *sampler_cache: dict[tuple[int, int], int]*) → int | None

Resolve GLTF material index from USD bindings or displayColor fallback.

Parameters

prim
[Any] The USD prim to get the material for.

display_colors
[Any] The displayColor values to use for the material.

color_interpolation
[Any] The interpolation method for the display colors.

gltf
[Any] The glTF document to modify with new textures, images, and samplers as needed.

builder
[GLBBuilder] The GLBBuilder instance managing the binary buffer.

texture_cache
[dict[tuple[str, str, str], int]] Cache mapping (file_path, wrap_s, wrap_t) to glTF texture indices.

image_cache
[dict[str, int]] Cache mapping file paths to glTF image indices.

sampler_cache
[dict[tuple[int, int], int]] Cache mapping (wrap_s, wrap_t) to glTF sampler indices.

Returns

int | None
The index of the resolved GLTF material, or None if no material is found.

The glb_mesh.py module

Summary

Functions

triangulate_faces	Convert polygon faces to triangles using fan triangulation.
iter_usd_mesh_prims	Collect mesh prims from a USD stage.
convert_mesh_prim_to_gltf	Convert one USD mesh prim into GLTF mesh/node entries.

Description

Mesh geometry conversion helpers for GLB export.

Module detail

glb_mesh.**triangulate_faces**(*indices: list[int]*, *counts: list[int]*) → list[int]

Convert polygon faces to triangles using fan triangulation.

Parameters

indices
[list[int]] The list of vertex indices for all faces, as defined by USD's faceVertexIndices.

counts
[list[int]] The list of vertex counts for each face, as defined by USD's faceVertexCounts.

Returns

list[int]
The list of triangle indices.

glb_mesh.**iter_usd_mesh_prims**(*stage: Any*) → list[Any]

Collect mesh prims from a USD stage.

Parameters

stage
[Any] The USD stage to traverse for mesh prims.

Returns

list[Any]
A list of USD mesh prims found in the stage.

glb_mesh.**convert_mesh_prim_to_gltf**(*prim: Any*, *gltf: Any*, *scene: Any*, *builder:* ansys.tools.visualization_interface.backends.usd.web.glb_builder.GLBBuilder, *texture_cache: dict[tuple[str, str, str], int]*, *image_cache: dict[str, int]*, *sampler_cache: dict[tuple[int, int], int]*) → None

Convert one USD mesh prim into GLTF mesh/node entries.

Parameters

prim
[Any] The USD mesh prim to convert.

gltf
[Any] The GLTF object to which the mesh will be added.

scene
[Any] The GLTF scene to which the mesh node will be added.

builder
[GLBBuilder] The GLBBuilder instance used for constructing the GLTF.

texture_cache
[dict[tuple[str, str, str], int]] A cache mapping texture identifiers to GLTF texture indices.

image_cache
[dict[str, int]] A cache mapping image file paths to GLTF image indices.

sampler_cache
[dict[tuple[int, int], int]] A cache mapping sampler settings to GLTF sampler indices.

The templates.py module

Summary

Functions

validate_template	Raise ValueError if any string in *required* is absent from *template*.
build_viewer_html_glb	Build a self-contained HTML viewer that renders a base64-encoded GLB.

Description

HTML template rendering helpers for web viewer export.

Module detail

templates.**validate_template**(*template: str*, *required: list[str]*) → None

Raise ValueError if any string in *required* is absent from *template*.

Parameters

template
[str] Template source text to validate.

required
[list[str]] Strings that must all appear verbatim in *template*.

Raises

ValueError
If one or more required strings are absent.

templates.**build_viewer_html_glb**(*glb_b64: str*, *model_name: str*, *template_path: pathlib.Path | None = None*) → str

Build a self-contained HTML viewer that renders a base64-encoded GLB.

The template is validated with *validate_template()* before substitution. Both `__GLB_B64_JSON__` and `__MODEL_NAME_JSON__` must appear verbatim in the template source; they are replaced with `json.dumps`-encoded values so that special characters are escaped automatically.

Parameters

glb_b64
[str] Base64-encoded GLB binary.

model_name
[str] Display name shown in the viewer (replaces `__MODEL_NAME_JSON__`).

template_path
[Path | None, default: None] Path to a custom HTML template. When None, the built-in `glb_template.html` is used. See the *Custom HTML templates* section of the user guide for the required placeholder contract.

Raises

FileNotFoundError
If *template_path* is given but the file does not exist.

ValueError
If the template is missing one or both required placeholders.

Description

USD-to-HTML web export pipeline (GLB + Three.js templates).

The `html_export.py` module

Summary

Functions

export_usd_to_html	Convert a USD asset to a self-contained Three.js HTML viewer.

Description

Self-contained USD → Three.js HTML export.

Converts a USD asset (file or in-memory `pxr.Usd.Stage`) into a self-contained HTML viewer page. The generated page embeds the geometry as a base64 GLB and requires only a CDN connection for Three.js at runtime.

Module detail

html_export.**export_usd_to_html**(*source: str | Path | object*, *output_path: str | Path | None = None*, ** (Keyword-only parameters separator (PEP 3102))*, *show_mesh_lines: bool = True*, *line_color: str = '#ffffff'*, *line_opacity: float = 0.9*, *template_path: str | Path | None = None*) → pathlib.Path

Convert a USD asset to a self-contained Three.js HTML viewer.

Parameters

source
[str | Path | pxr.Usd.Stage] USD file path (.usd / .usda / .usdc / .usdz) or an in-memory USD stage.

output_path
[str | Path | None, default: None] Destination for the HTML file. When None, placed alongside the source as {stem}_viewer.html.

show_mesh_lines
[bool, default: True] When True, injects Three.js LineSegments derived from polygon edges of all UsdGeomMesh prims in the stage for a wireframe overlay.

line_color
[str, default: "#ffffff"] CSS hex color for the mesh-edge overlay.

line_opacity
[float, default: 0.9] Opacity of the mesh-edge overlay (0.0 - 1.0).

template_path
[str | Path | None, default: None] Path to a custom HTML template. The template must contain __GLB_B64_JSON__ and __MODEL_NAME_JSON__ as literal placeholders that will be replaced with JSON-encoded values at render time. When show_mesh_lines=True, the template must also include the wireframe injection anchors (const binary = atob(glbBase64); and scene.add(gltf.scene);). When None, the built-in glb_template.html is used.

Returns

pathlib.Path
Absolute path to the written HTML file.

Raises

ImportError
If usd-core is not installed.

FileNotFoundError
If a file path is given but does not exist on disk, or if template_path points to a missing file.

ValueError
If line_opacity is outside [0.0, 1.0], line_color is not a valid CSS hex color, or the template is missing required placeholders or wireframe injection anchors.

The usd_interface.py module

Summary

Classes

USDInterface	USD backend interface using python-usd-viewer.

USDInterface

class ansys.tools.visualization_interface.backends.usd.usd_interface.**USDInterface**(*title: str = 'USD Viewer'*, *size: Tuple[int, int] = (750, 750)*)

Bases: ansys.tools.visualization_interface.backends._base.BaseBackend

USD backend interface using python-usd-viewer.

Parameters

title
[str, default: "USD Viewer"] Title of the viewer window.

size
[tuple[int, int], default: (750, 750)] Width and height of the viewer window in pixels.

Overview

Abstract methods

add_points	Not implemented for the USD backend.
add_lines	Not implemented for the USD backend.
add_planes	Not implemented for the USD backend.
add_text	Not implemented for the USD backend.
add_labels	Not implemented for the USD backend.
clear	Not implemented for the USD backend.

Methods

plot	Add a plottable object to the USD stage.
plot_iter	Add all objects in an iterable to the USD stage.
show	Display the current USD stage in the viewer window.

Import detail

```
from ansys.tools.visualization_interface.backends.usd.usd_interface import USDInterface
```

Method detail

USDInterface.**plot**(*plottable_object: Any*, ***plotting_options*) → None

Add a plottable object to the USD stage.

Parameters

plottable_object
[Any] Supported types:

- pxr.Usd.Stage — replaces the current stage.

- `str` / `pathlib.Path` — USD files replace the current stage; VTK-compatible files are converted and merged into it.
- *`MeshObjectPlot`* — the underlying PyVista mesh is converted and added.
- Any PyVista or VTK dataset — converted via *`VTKConverter`*.

****plotting_options**
[`dict`] Reserved for future backend-specific options.

`USDInterface.`**`plot_iter`**(*plotting_list: Iterable*) → None

Add all objects in an iterable to the USD stage.

`USDInterface.`**`show`**(*plottable_object=None*, *screenshot=None*, *name_filter=None*, ***kwargs*) → None

Display the current USD stage in the viewer window.

Opens the Qt-based USD viewer and starts the event loop. This call blocks until the window is closed.

Parameters

plottable_object
[`Any`, `optional`] Optional object to plot before showing.

screenshot
[`str`, `optional`] Unused — screenshots are not supported by the USD backend.

name_filter
[`str`, `optional`] Unused — name filtering is not supported by the USD backend.

****kwargs**
[`dict`] Additional keyword arguments (unused).

`abstractmethod` `USDInterface.`**`add_points`**(*points*, *color='red'*, *size=10.0*, ***kwargs*)

Not implemented for the USD backend.

`abstractmethod` `USDInterface.`**`add_lines`**(*points*, *connections=None*, *color='white'*, *width=1.0*, ***kwargs*)

Not implemented for the USD backend.

`abstractmethod` `USDInterface.`**`add_planes`**(*center=(0.0, 0.0, 0.0)*, *normal=(0.0, 0.0, 1.0)*, *i_size=1.0*, *j_size=1.0*, ***kwargs*)

Not implemented for the USD backend.

`abstractmethod` `USDInterface.`**`add_text`**(*text*, *position*, *font_size=12*, *color='white'*, ***kwargs*)

Not implemented for the USD backend.

`abstractmethod` `USDInterface.`**`add_labels`**(*points*, *labels*, *font_size=12*, *point_size=5.0*, ***kwargs*)

Not implemented for the USD backend.

`abstractmethod` `USDInterface.`**`clear`**() → None

Not implemented for the USD backend.

Description

USD backend interface for visualization using python-usd-viewer.

The `vtk_converter.py` module

Summary

Classes

VTKConverter	Convert VTK files to USD format for visualization.

VTKConverter

class ansys.tools.visualization_interface.backends.usd.vtk_converter.**VTKConverter**

Convert VTK files to USD format for visualization.

Overview

Methods

load_asset	Load a VTK asset into a given stage.

Static methods

convert_vtk_file_to_usd	Convert a VTK file to a USD file.
convert_vtk_to_usd	Convert a VTK file to a USD stage.
get_vtk_reader	Get the appropriate VTK reader based on the file extension.
convert_polydata_to_usd_mesh	Convert VTK polydata to USD mesh geometry.
convert_usd_to_vtk	Convert a USD mesh to VTK polydata.

Import detail

```
from ansys.tools.visualization_interface.backends.usd.vtk_converter import VTKConverter
```

Method detail

static VTKConverter.**convert_vtk_file_to_usd**(*vtk_file_path: str | pathlib.Path*, *stage: pxr.Usd.Stage = None*) → None

Convert a VTK file to a USD file.

Parameters

vtk_file_path
[Union[str, Path]] Path to the VTK file to convert.

stage
[Usd.Stage] Stage to add the converted USD data to.

static VTKConverter.**convert_vtk_to_usd**(*data: vtk.vtkDataSet*, *stage: pxr.Usd.Stage = None*, *mesh_name: str = 'VTKMesh'*) → pxr.Usd.Stage

Convert a VTK file to a USD stage.

Parameters

vtk_file_path
[Union[str, Path]] Path to the VTK file to convert.

stage
[Usd.Stage] Stage to add the VTK data to.

Returns

`Usd.Stage`
Stage containing the VTK data.

static VTKConverter.**get_vtk_reader**(*file_path: pathlib.Path*) → vtk.vtkAlgorithm

Get the appropriate VTK reader based on the file extension.

Parameters

file_path
[Path] Path to the VTK file.

static VTKConverter.**convert_polydata_to_usd_mesh**(*polydata: vtk.vtkPolyData*, *stage: pxr.Usd.Stage = None*, *mesh_name: str = 'VTKMesh'*) → None

Convert VTK polydata to USD mesh geometry.

Parameters

polydata
[vtk.vtkPolyData] VTK polydata to convert.

stage
[Usd.Stage] USD stage to add the mesh to.

mesh_name
[str, default: "VTKMesh"] Name of the mesh in USD.

static VTKConverter.**convert_usd_to_vtk**(*stage: pxr.Usd.Stage*, *mesh_path: str | None = None*) → vtk.vtkPolyData | None

Convert a USD mesh to VTK polydata.

Parameters

stage
[Usd.Stage] USD stage containing the mesh.

mesh_path
[str, default: None] Path to the mesh in USD. If None, the first mesh in the stage is used.

Returns

`Optional[vtk.vtkPolyData]`
Converted VTK polydata or None if conversion failed.

VTKConverter.**load_asset**(*asset_path: str*, *stage: pxr.Usd.Stage*) → pxr.Usd.Stage | None

Load a VTK asset into a given stage.

Parameters

asset_path
[str] Path to the asset file.

stage
[Usd.Stage] Stage to add the asset to.

Returns

`Optional[Usd.Stage]`
Stage with the loaded asset or None if loading failed.

Description

Asset resolver for handling various file types and USD asset paths.

Description

USD backend initialization.

Description

Provides interfaces.

The `types` package

Summary

Submodules

edge_plot	Provides the edge type for plotting.
mesh_object_plot	Provides the `MeshObjectPlot` class.

The `edge_plot.py` module

Summary

Classes

EdgePlot	Provides the mapper class for relating PyAnsys object edges with its PyVista actor.

`EdgePlot`

class ansys.tools.visualization_interface.types.edge_plot.**EdgePlot**(*actor: pyvista.Actor | plotly.graph_objects.Mesh3d*, *edge_object: Any*, *parent: Any = None*)

Provides the mapper class for relating PyAnsys object edges with its PyVista actor.

Parameters

actor
[`Union[Actor, Mesh3d]`] PyVista actor that represents the edge.

edge_object
[`Edge`] PyAnsys object edge that is represented by the PyVista actor.

parent
[`MeshObjectPlot`, default: `None`] Parent PyAnsys object of the edge.

Overview

Properties

actor	PyVista actor of the object.
edge_object	PyAnsys edge.
parent	Parent PyAnsys object of the edge.
name	Name of the edge.

Import detail

```
from ansys.tools.visualization_interface.types.edge_plot import EdgePlot
```

Property detail

property EdgePlot.**actor:** pyvista.Actor

PyVista actor of the object.

Returns

Actor
PyVista actor.

property EdgePlot.**edge_object:** Any

PyAnsys edge.

Returns

Any
PyAnsys edge.

property EdgePlot.**parent:** Any

Parent PyAnsys object of the edge.

Returns

Any
Parent PyAnsys object.

property EdgePlot.**name:** str

Name of the edge.

Returns

str
Name of the edge.

Description

Provides the edge type for plotting.

The mesh_object_plot.py module

Summary

Classes

MeshObjectPlot	Relates a custom object with a mesh, provided by the consumer library.

MeshObjectPlot

class ansys.tools.visualization_interface.types.mesh_object_plot.**MeshObjectPlot**(*custom_object: Any*, *mesh: pyvista.PolyData | pyvista.MultiBlock | plotly.graph_objects.Mesh3d*, *actor: pyvista.Actor = None*, *edges: List[ansys.tools.visualization_ = None*, *children: List[MeshObjectPlot] = None*, *parent: MeshObjectPlot = None*)

Relates a custom object with a mesh, provided by the consumer library.

Overview

Methods

add_child	Set a child MeshObjectPlot to the current object.
is_visible_in_tree	Check if this object is visible considering parent visibility.

Properties

parent	Get the parent MeshObjectPlot of the current object.
mesh	Mesh of the object in PyVista format.
custom_object	Custom object.
actor	PyVista actor of the object in the plotter.
edges	Edges of the object.
name	Name of the object.
mesh_type	Type of the mesh.
visible	Whether this object is currently visible.

Import detail

```
from ansys.tools.visualization_interface.types.mesh_object_plot import MeshObjectPlot
```

Property detail

property MeshObjectPlot.**parent:** *MeshObjectPlot*

Get the parent MeshObjectPlot of the current object.

This method is used to set a parent MeshObjectPlot to the current object. It is useful when the custom object has a hierarchical structure, and the consumer library wants to relate the parent objects with their meshes.

Parameters

parent
[MeshObjectPlot] Parent MeshObjectPlot to be set.

property MeshObjectPlot.**mesh:** pyvista.PolyData | pyvista.MultiBlock | plotly.graph_objects.Mesh3d

Mesh of the object in PyVista format.

Returns

Union[pv.PolyData, pv.MultiBlock]
Mesh of the object.

property MeshObjectPlot.**custom_object:** Any

Custom object.

Returns

Any
Custom object.

property MeshObjectPlot.**actor:** pyvista.Actor

PyVista actor of the object in the plotter.

Returns

pv.Actor
PyVista actor of the object.

property MeshObjectPlot.**edges:** List[*ansys.tools.visualization_interface.types.edge_plot.EdgePlot*]

Edges of the object.

Returns

List[EdgePlot]
Edges of the object.

property MeshObjectPlot.**name:** str

Name of the object.

Returns

str
Name of the object.

property MeshObjectPlot.**mesh_type: Type**

Type of the mesh.

Returns

type
Type of the mesh.

property MeshObjectPlot.**visible: bool**

Whether this object is currently visible.

This property reflects the visibility state of the object. If an actor is assigned, it reads the visibility from the actor to stay synchronized.

Returns

bool
True if the object is visible, False otherwise.

Method detail

MeshObjectPlot.**add_child**(*child:* MeshObjectPlot)

Set a child MeshObjectPlot to the current object.

This method is used to set a child MeshObjectPlot to the current object. It is useful when the custom object has a hierarchical structure, and the consumer library wants to relate the child objects with their meshes.

Parameters

child
[MeshObjectPlot] Child MeshObjectPlot to be set.

MeshObjectPlot.**is_visible_in_tree**() → bool

Check if this object is visible considering parent visibility.

An object is only truly visible if both itself and all its ancestors in the tree are visible.

Returns

bool
True if object and all ancestors are visible, False otherwise.

Description

Provides the MeshObjectPlot class.

Description

Provides custom types.

The utils package

Summary

Submodules

clip_plane	Provides the `ClipPlane` class.
color	Provides an enum with the color to use for the plotter actors.
logger	Provides the singleton helper class for the logger.
plotting_options	Provides the `PlottingOptions` dataclass for framework-specific plotting kwargs.
vtkhdf_converter	Utils module for VTKHDF management.

The `clip_plane.py` module

Summary

Classes

ClipPlane	Provides the clipping plane for clipping meshes in the plotter.

ClipPlane

class ansys.tools.visualization_interface.utils.clip_plane.**ClipPlane**(*normal: Tuple[float, float, float] = (1, 0, 0)*, *origin: Tuple[float, float, float] = (0, 0, 0)*)

Provides the clipping plane for clipping meshes in the plotter.

The clipping plane is defined by both normal and origin vectors.

Parameters

normal
[Tuple[float, float, float], default: (1, 0, 0)] Normal of the plane.

origin
[Tuple[float, float, float], default: (0, 0, 0)] Origin point of the plane.

Overview

Properties

normal	Normal of the plane.
origin	Origin of the plane.

Import detail

```
from ansys.tools.visualization_interface.utils.clip_plane import ClipPlane
```

Property detail

property ClipPlane.**normal: Tuple[float, float, float]**

Normal of the plane.

Returns

Tuple[float, float, float]
Normal of the plane.

property ClipPlane.**origin: Tuple**[float, float, float]
Origin of the plane.

Returns

Tuple[float, float, float]
Origin of the plane.

Description

Provides the `ClipPlane` class.

The `color.py` module

Summary

Enums

Color	Provides an enum with the color to use for the plotter actors.

Color

class ansys.tools.visualization_interface.utils.color.**Color**
Bases: enum.Enum

Provides an enum with the color to use for the plotter actors.

Overview

Attributes

DEFAULT	Default color for the plotter actors.
PICKED	Color for the actors that are currently picked.
EDGE	Default color for the edges.
PICKED_EDGE	Color for the edges that are currently picked.

Import detail

```
from ansys.tools.visualization_interface.utils.color import Color
```

Attribute detail

Color.**DEFAULT = '#D6F7D1'**
Default color for the plotter actors.

Color.**PICKED = '#BB6EEE'**
Color for the actors that are currently picked.

Color.**EDGE = '#000000'**
Default color for the edges.

Color.**PICKED_EDGE = '#9C9C9C'**

Color for the edges that are currently picked.

Description

Provides an enum with the color to use for the plotter actors.

The logger.py module

Summary

Classes

SingletonType	Provides the singleton helper class for the logger.
VizLogger	Provides the singleton logger for the visualizer.

Attributes

logger

SingletonType

class ansys.tools.visualization_interface.utils.logger.**SingletonType**

Bases: type

Provides the singleton helper class for the logger.

Overview

Special methods

__call__	Call to redirect new instances to the singleton instance.

Import detail

```
from ansys.tools.visualization_interface.utils.logger import SingletonType
```

Method detail

SingletonType.**__call__**(**args*, ***kwargs*)

Call to redirect new instances to the singleton instance.

VizLogger

class ansys.tools.visualization_interface.utils.logger.**VizLogger**(*level: int = logging.ERROR*, *logger_name: str = 'VizLogger'*)

Bases: object

Provides the singleton logger for the visualizer.

Parameters

to_file
[bool, default: `False`] Whether to include the logs in a file.

Overview

Methods

get_logger	Get the logger.
set_level	Set the logger output level.
enable_output	Enable logger output to a given stream.
add_file_handler	Save logs to a file in addition to printing them to the standard output.

Import detail

```
from ansys.tools.visualization_interface.utils.logger import VizLogger
```

Method detail

`VizLogger.`**`get_logger`**`()`

Get the logger.

Returns

Logger
Logger.

`VizLogger.`**`set_level`**`(`*level: int*`)`

Set the logger output level.

Parameters

level
[`int`] Output Level of the logger.

`VizLogger.`**`enable_output`**`(`*stream=None*`)`

Enable logger output to a given stream.

If a stream is not specified, `sys.stderr` is used.

Parameters

stream: TextIO, default: ``sys.stderr``
Stream to output the log output to.

`VizLogger.`**`add_file_handler`**`(`*logs_dir: str = './.log'*`)`

Save logs to a file in addition to printing them to the standard output.

Parameters

logs_dir
[`str`, default: `"./.log"`] Directory of the logs.

Description

Provides the singleton helper class for the logger.

Module detail

logger.**logger**

The plotting_options.py module

Summary

Classes

PlottingOptions	Structured representation of all plotting kwargs.

PlottingOptions

class ansys.tools.visualization_interface.utils.plotting_options.**PlottingOptions**

Structured representation of all plotting kwargs.

Calling `from_kwargs()` splits a raw `**kwargs` dictionary into:

- **Framework fields** (`name_filter`, `clipping_plane`) — consumed by the visualization framework and **not** forwarded to the renderer.
- **extra** — every remaining key, forwarded verbatim to the underlying rendering library (PyVista `add_mesh`, Plotly `Figure.show`, etc.).

Parameters

name_filter
[`str`, `optional`] Regular expression with the desired name or names to include in the plotter. Objects whose `name` does not match the expression are skipped.

clipping_plane
[`ClipPlane`, `optional`] Plane used to clip the mesh before adding it to the scene.

extra
[`dict`] Renderer-specific options forwarded verbatim to the underlying library.

Overview

Constructors

from_kwargs	Parse and consume **all** keys from *kwargs*.

Attributes

name_filter
clipping_plane
extra

Import detail

```
from ansys.tools.visualization_interface.utils.plotting_options import PlottingOptions
```

Attribute detail

PlottingOptions.**name_filter**: str | None = None

PlottingOptions.**clipping_plane**:
ansys.tools.visualization_interface.utils.clip_plane.ClipPlane | None = None

PlottingOptions.**extra**: Dict[str, Any]

Method detail

classmethod PlottingOptions.**from_kwargs**(*kwargs: Dict*) → *PlottingOptions*

Parse and consume **all** keys from *kwargs*.

Framework-specific keys (matching dataclass fields other than `extra`) are mapped to their typed fields. All remaining keys are collected into *extra*. The supplied *kwargs* dictionary is **cleared** in-place so that callers can safely inspect it afterwards and know it has been fully consumed.

Parameters

kwargs
[dict] Mutable keyword-argument dictionary to parse. All keys are removed in-place.

Returns

PlottingOptions
Instance populated from *kwargs*.

Description

Provides the `PlottingOptions` dataclass for framework-specific plotting kwargs.

The `vtkhdf_converter.py` module

Summary

Functions

pd_to_vtkhdf	Write the PyVista PolyData directly to a VTKHDF file.
vtkhdf_to_pd	Read a VTKHDF file and convert it to PyVista PolyData.
ug_to_vtkhdf	Write the PyVista UnstructuredGrid directly to a VTKHDF file.
vtkhdf_to_ug	Read a VTKHDF file and convert it to PyVista UnstructuredGrid.

Description

Utils module for VTKHDF management.

Module detail

vtkhdf_converter.**pd_to_vtkhdf**(*pd: pyvista.PolyData*, *output_vtkhdf_file: str | pathlib.Path*) → pathlib.Path

Write the PyVista PolyData directly to a VTKHDF file.

Parameters

pd
[pv.PolyData] The PyVista PolyData to be written.

output_vtkhdf_file
[Union[str, Path]] The output VTKHDF file path.

Returns

Path
The path to the saved VTKHDF file.

vtkhdf_converter.**vtkhdf_to_pd**(*input_vtkhdf_file: str | pathlib.Path*) → pyvista.PolyData

Read a VTKHDF file and convert it to PyVista PolyData.

Parameters

input_vtkhdf_file
[Union[str, Path]] The input VTKHDF file path.

Returns

pv.PolyData
The converted PyVista PolyData.

vtkhdf_converter.**ug_to_vtkhdf**(*ug: pyvista.UnstructuredGrid*, *output_vtkhdf_file: str | pathlib.Path*) → pathlib.Path

Write the PyVista UnstructuredGrid directly to a VTKHDF file.

Parameters

ug
[pv.UnstructuredGrid] The PyVista UnstructuredGrid to be written.

output_vtkhdf_file
[Union[str, Path]] The output VTKHDF file path.

Returns

Path
The path to the saved VTKHDF file.

vtkhdf_converter.**vtkhdf_to_ug**(*input_vtkhdf_file: str | pathlib.Path*) → pyvista.UnstructuredGrid

Read a VTKHDF file and convert it to PyVista UnstructuredGrid.

Parameters

input_vtkhdf_file
[Union[str, Path]] The input VTKHDF file path.

Returns

pv.UnstructuredGrid
The converted PyVista UnstructuredGrid.

Description

Provides the Utils package.

The `plotter.py` module

Summary

Classes

Plotter	Base plotting class containing common methods and attributes.

Plotter

class ansys.tools.visualization_interface.plotter.**Plotter**(*backend: ansys.tools.visualization_interface.backends._base.BaseBacke | None = None*)

Base plotting class containing common methods and attributes.

This class is responsible for plotting objects using the specified backend.

Parameters

backend
[`BaseBackend`, `optional`] Plotting backend to use, by default PyVistaBackend.

Overview

Methods

plot_iter	Plots multiple objects using the specified backend.
plot	Plots an object using the specified backend.
show	Show the plotted objects.
animate	Create an animation from a sequence of frames.
add_points	Add point markers to the scene.
add_lines	Add line segments to the scene.
add_planes	Add a plane to the scene.
add_text	Add text to the scene.
add_labels	Add labels at 3D point locations.
clear	Clear all actors from the scene.

Properties

backend	Return the base plotter object.

Import detail

```
from ansys.tools.visualization_interface.plotter import Plotter
```

Property detail

property Plotter.**backend**

Return the base plotter object.

Method detail

Plotter.**plot_iter**(*plotting_list: List*, ***plotting_options*)

Plots multiple objects using the specified backend.

Parameters

plotting_list
[List] List of objects to plot.

plotting_options
[dict] Additional plotting options.

Plotter.**plot**(*plottable_object: Any*, ***plotting_options*)

Plots an object using the specified backend.

Parameters

plottable_object
[Any] Object to plot.

plotting_options
[dict] Additional plotting options.

Plotter.**show**(*plottable_object: Any = None*, *screenshot: str = None*, ***kwargs*) → List

Show the plotted objects.

Parameters

plottable_object
[Any, optional] Object to show, by default None.

screenshot
[str, optional] Path to save a screenshot, by default None.

kwargs
[dict] Additional options the selected backend accepts.

Returns

List
List of picked objects.

Plotter.**animate**(*frames: List[Any]*, *fps: int = 30*, *loop: bool = False*, *scalar_bar_args: dict | None = None*, ***plot_kwargs*)

Create an animation from a sequence of frames.

This method provides a convenient way to create animations from time-series simulation results, transient analyses, and dynamic phenomena. It wraps the backend's animation functionality in a simple, consistent API.

Parameters

frames
[List[Any]] Sequence of frame objects to animate. Can be PyVista meshes, MeshObjectPlot objects, or any plottable objects.

fps
[int, optional] Frames per second for playback. Default is 30.

loop
[bool, optional] Whether to loop animation continuously. Default is False.

scalar_bar_args
[dict, optional] Scalar bar arguments to apply to all frames (e.g., clim for fixed color scale). If not provided, a global color scale is calculated automatically to ensure visual integrity across frames.

****plot_kwargs**
Additional keyword arguments passed to add_mesh for all frames (e.g., cmap='viridis', opacity=0.8).

Returns

Animation
Animation controller object with playback controls: - play(): Start animation - pause(): Pause animation - stop(): Stop and reset to first frame - step_forward(): Advance one frame - step_backward(): Rewind one frame - seek(frame_index): Jump to specific frame - save(filename): Export to video (MP4, GIF, AVI) - show(): Display with plotter

Raises

ValueError
If frames list is empty or fps is not positive.

NotImplementedError
If the backend does not support animations.

See also

Animation
Animation controller class with detailed playback controls

Notes

- Fixed color scales are recommended (and calculated by default) to ensure visual integrity and prevent misleading animations where color meanings change between frames.
- For large datasets (1000+ frames or >5M cells), consider implementing a custom FrameSequence with lazy loading capabilities.
- The animation uses the backend's native capabilities. Currently, only PyVista backend supports animations.

Examples

Create and play a simple animation from transient simulation results:

```
>>> from ansys.tools.visualization_interface import Plotter
>>> import pyvista as pv
>>> # Create example meshes representing time steps
>>> sphere = pv.Sphere()
>>> frames = []
>>> for i in range(20):
...     mesh = sphere.copy()
```

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```
...     mesh["displacement"] = np.random.rand(mesh.n_points) * i * 0.1
...     frames.append(mesh)
>>> plotter = Plotter()
>>> animation = plotter.animate(frames, fps=10, loop=True)
>>> animation.show()
```

Export animation to video:

```
>>> animation = plotter.animate(frames, fps=30)
>>> animation.save("simulation.mp4", quality=8)
```

Use fixed color scale for accurate comparison:

```
>>> animation = plotter.animate(
...     frames,
...     fps=30,
...     scalar_bar_args={"clim": (0.0, 1.0), "title": "Displacement [m]"}
... )
>>> animation.play()
>>> animation.show()
```

Control playback programmatically:

```
>>> animation = plotter.animate(frames)
>>> animation.play()  # Start animation
>>> # ... after some time ...
>>> animation.pause()  # Pause
>>> animation.step_forward()  # Advance one frame
>>> animation.seek(10)  # Jump to frame 10
>>> animation.stop()  # Reset to beginning
```

`Plotter.`**`add_points`**(*points: List | Any*, *color: str = 'red'*, *size: float = 10.0*, ***kwargs*) → Any

Add point markers to the scene.

This method provides a backend-agnostic way to add point markers to the visualization scene. The points will be rendered using the active backend's native point rendering capabilities.

Parameters

points
[`Union`[`List`, `Any`]] Points to add. Can be a list of coordinates or array-like object. Expected format: [[x1, y1, z1], [x2, y2, z2], ...] or Nx3 array.

color
[`str`, default: "red"] Color of the points. Can be a color name (e.g., 'red', 'blue') or hex color code (e.g., '#FF0000').

size
[`float`, default: 10.0] Size of the point markers in pixels or display units (interpretation depends on backend).

****kwargs**
[`dict`] Additional backend-specific keyword arguments for advanced customization.

Returns

Any
Backend-specific actor or object representing the added points. Can be used for further manipulation or removal.

Examples

Add simple point markers at three locations:

```
>>> from ansys.tools.visualization_interface import Plotter
>>> plotter = Plotter()
>>> points = [[0, 0, 0], [1, 0, 0], [0, 1, 0]]
>>> plotter.add_points(points, color='blue', size=15)
>>> plotter.show()
```

Add points with custom styling:

```
>>> import numpy as np
>>> points = np.random.rand(100, 3)
>>> plotter.add_points(points, color='yellow', size=8)
>>> plotter.show()
```

Plotter.**add_lines**(*points: List | Any*, *connections: List | Any | None = None*, *color: str = 'white'*, *width: float = 1.0*, ***kwargs*) → Any

Add line segments to the scene.

This method provides a backend-agnostic way to add lines to the visualization scene. Lines can connect points sequentially or based on explicit connectivity information.

Parameters

points
[Union[List, Any]] Points defining the lines. Can be a list of coordinates or array-like object. Expected format: [[x1, y1, z1], [x2, y2, z2], ...] or Nx3 array.

connections
[Optional[Union[List, Any]], default: None] Line connectivity. If None, connects points sequentially (0->1, 1->2, 2->3, ...). If provided, should define line segments as pairs of point indices: [[start_idx1, end_idx1], [start_idx2, end_idx2], ...] or Mx2 array where M is the number of line segments.

color
[str, default: "white"] Color of the lines. Can be a color name or hex color code.

width
[float, default: 1.0] Width of the lines in pixels or display units (interpretation depends on backend).

****kwargs**
[dict] Additional backend-specific keyword arguments for advanced customization.

Returns

Any
Backend-specific actor or object representing the added lines. Can be used for further manipulation or removal.

Examples

Add a line connecting points sequentially:

```
>>> from ansys.tools.visualization_interface import Plotter
>>> plotter = Plotter()
>>> points = [[0, 0, 0], [1, 0, 0], [1, 1, 0], [0, 1, 0]]
>>> plotter.add_lines(points, color='green', width=2.0)
>>> plotter.show()
```

Add specific line segments with explicit connectivity:

```
>>> points = [[0, 0, 0], [1, 0, 0], [0, 1, 0], [1, 1, 0]]
>>> connections = [[0, 1], [2, 3], [0, 2]]  # Connect specific pairs
>>> plotter.add_lines(points, connections=connections, color='red', width=3.0)
>>> plotter.show()
```

Plotter.**add_planes**(*center: Tuple[float, float, float] = (0.0, 0.0, 0.0)*, *normal: Tuple[float, float, float] = (0.0, 0.0, 1.0)*, *i_size: float = 1.0*, *j_size: float = 1.0*, ***kwargs*) → Any

Add a plane to the scene.

This method provides a backend-agnostic way to add plane objects to the visualization scene. Planes are useful for showing reference planes, symmetry planes, or cutting planes.

Parameters

center
[Tuple[float, float, float], default: (0.0, 0.0, 0.0)] Center point of the plane in 3D space (x, y, z).

normal
[Tuple[float, float, float], default: (0.0, 0.0, 1.0)] Normal vector of the plane (x, y, z). The vector will be normalized by the backend if needed.

i_size
[float, default: 1.0] Size of the plane in the i direction (local coordinate system).

j_size
[float, default: 1.0] Size of the plane in the j direction (local coordinate system).

****kwargs**
[dict] Additional backend-specific keyword arguments for advanced customization (e.g., color, opacity, resolution).

Returns

Any
Backend-specific actor or object representing the added plane. Can be used for further manipulation or removal.

Examples

Add a horizontal plane at z=0:

```
>>> from ansys.tools.visualization_interface import Plotter
>>> plotter = Plotter()
>>> plotter.add_planes(center=(0, 0, 0), normal=(0, 0, 1), i_size=2.0, j_size=2.0)
>>> plotter.show()
```

Add a vertical plane with custom styling:

```
>>> plotter.add_planes(
...     center=(1, 0, 0),
...     normal=(1, 0, 0),
...     i_size=3.0,
...     j_size=3.0,
...     color='lightblue',
...     opacity=0.5
... )
>>> plotter.show()
```

Plotter.**add_text**(*text: str*, *position: Tuple[float, float] | str*, *font_size: int = 12*, *color: str = 'white'*, ***kwargs*) → Any

Add text to the scene.

This method provides a backend-agnostic way to add text labels to the visualization scene. Text is positioned using 2D screen coordinates.

Parameters

text
[str] Text string to display.

position
[Union[Tuple[float, float], str]] Position for the text. Can be:

- 2D tuple (x, y) for screen/viewport coordinates (pixels from bottom-left)
- String position like 'upper_left', 'upper_right', 'lower_left', 'lower_right', 'upper_edge', 'lower_edge' (backend-dependent support)

font_size
[int, default: 12] Font size for the text in points.

color
[str, default: "white"] Color of the text. Can be a color name or hex color code.

****kwargs**
[dict] Additional backend-specific keyword arguments for advanced customization (e.g., font_family, bold, italic, shadow).

Returns

Any
Backend-specific actor or object representing the added text. Can be used for further manipulation or removal.

Examples

Add text at a screen position:

```
>>> from ansys.tools.visualization_interface import Plotter
>>> plotter = Plotter()
>>> plotter.add_text("Title", position=(10, 10), font_size=18, color='yellow')
>>> plotter.show()
```

Add text using a named position:

```
>>> plotter.add_text(
...     "Corner Label",
...     position='upper_right',
...     font_size=14,
...     color='red'
... )
>>> plotter.show()
```

Plotter.**add_labels**(*points: List | Any*, *labels: List[str]*, *font_size: int = 12*, *point_size: float = 5.0*, ***kwargs*) → Any

Add labels at 3D point locations.

This method provides a backend-agnostic way to add text labels at specific 3D coordinates in the visualization scene. Labels are displayed next to marker points.

Parameters

points
[Union[List, Any]] Points where labels should be placed. Can be a list of coordinates or array-like object. Expected format: [[x1, y1, z1], ...] or Nx3 array.

labels
[List[str]] List of label strings to display at each point. Must have the same length as points.

font_size
[int, default: 12] Font size for the labels.

point_size
[float, default: 5.0] Size of the point markers shown with labels.

****kwargs**
[dict] Additional backend-specific keyword arguments for advanced customization (e.g., text_color, shape, fill_shape).

Returns

Any
Backend-specific actor or object representing the added labels. Can be used for further manipulation or removal.

Examples

Add labels at specific locations:

```
>>> from ansys.tools.visualization_interface import Plotter
>>> plotter = Plotter()
>>> points = [[0, 0, 0], [1, 0, 0], [0, 1, 0]]
>>> labels = ['Origin', 'X-axis', 'Y-axis']
>>> plotter.add_labels(points, labels, font_size=14)
>>> plotter.show()
```

Add labels with custom styling:

```
>>> plotter.add_labels(
...     points,
...     labels,
```

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```
...     font_size=16,
...     point_size=10,
...     text_color='yellow',
...     shape='rounded_rect'
... )
>>> plotter.show()
```

Plotter.**clear**() → None

Clear all actors from the scene.

This method removes all previously added objects (meshes, points, lines, text, etc.) from the visualization scene, therefore allowing the plotter to be reused after `show()` has been called.

Examples

Clear before showing:

```
>>> from ansys.tools.visualization_interface import Plotter
>>> import pyvista as pv
>>> plotter = Plotter()
>>> plotter.plot(pv.Sphere())
>>> plotter.clear()  # Changed mind, remove sphere
>>> plotter.plot(pv.Cube())
>>> plotter.show()
```

Clear after showing

```
>>> plotter = Plotter()
>>> plotter.plot(pv.Sphere())
>>> plotter.show()
>>> plotter.clear()  # Reset the plotter to reuse it
>>> plotter.plot(pv.Cube())
>>> plotter.show()
```

Description

Module for the Plotter class.

4.1.2 Description

Visualization Interface Tool is a Python client library for visualizing the results of Ansys simulations.

4.1.3 Module detail

visualization_interface.**USE_TRAME: bool = False**

visualization_interface.**DOCUMENTATION_BUILD: bool**

Whether the documentation is being built or not.

visualization_interface.**TESTING_MODE: bool**

Whether the library is being built or not, used to avoid showing plots while testing.

visualization_interface.**USE_HTML_BACKEND: bool**

Whether the library is being built or not, used to avoid showing plots while testing.

visualization_interface.__version__

CHAPTER

FIVE

EXAMPLES

This section shows how to use the Visualization Interface Tool to perform many different types of operations.

CHAPTER

SIX

BASIC USAGE EXAMPLES

These examples show how to use the general plotter included in the Visualization Interface Tool.

CHAPTER

SEVEN

BASIC PLOTLY USAGE EXAMPLES

These examples show how to use the general plotter with Plotly backend included in the Visualization Interface Tool.

CHAPTER

EIGHT

BASIC USD USAGE EXAMPLES

These examples show how to use the general plotter with the USD backend included in the Visualization Interface Tool.

CHAPTER

NINE

ADVANCED USAGE EXAMPLES

These examples show how to use the Visualization Interface Tool to postprocess simulation data.

9.1 Basic usage examples

These examples show how to use the general plotter included in the Visualization Interface Tool.

9.1.1 Use trame as a remote service

This example shows how to launch a trame service and use it as a remote service.

First, we need to launch the trame service. We can do this by running the following code:

```
# import required libraries
from ansys.tools.visualization_interface.backends.pyvista.trame_service import (
    TrameService,
)

# create a trame service, in whatever port is available in your system
ts = TrameService(websocket_port=8765)

# run the service
ts.run()
```

Now, we can send meshes and plotter to the trame service. We can do this by running the following code in a separate terminal:

```
# import required libraries
import time

import pyvista as pv

from ansys.tools.visualization_interface.backends.pyvista.trame_remote import (
    send_mesh,
    send_pl,
)

# create an example plotter
plotter = pv.Plotter()
plotter.add_mesh(pv.Cube())

# send some example meshes
```

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```
send_mesh(pv.Sphere())
send_mesh(pv.Sphere(center=(3, 0, 0)))
time.sleep(4)

# if we send a plotter, the previous meshes will be deleted.
send_pl(plotter)
```

Total running time of the script: (0 minutes 0.000 seconds)

9.1.2 Use a PyVista Qt backend

PyVista Qt is a package that extends the PyVista functionality through the usage of Qt. Qt applications operate in a separate thread than VTK, you can simultaneously have an active VTK plot and a non-blocking Python session.

This example shows how to use the PyVista Qt backend to create a plotter

```
import pyvista as pv

from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.backends.pyvista import PyVistaBackend
```

Open a pyvistaqt window

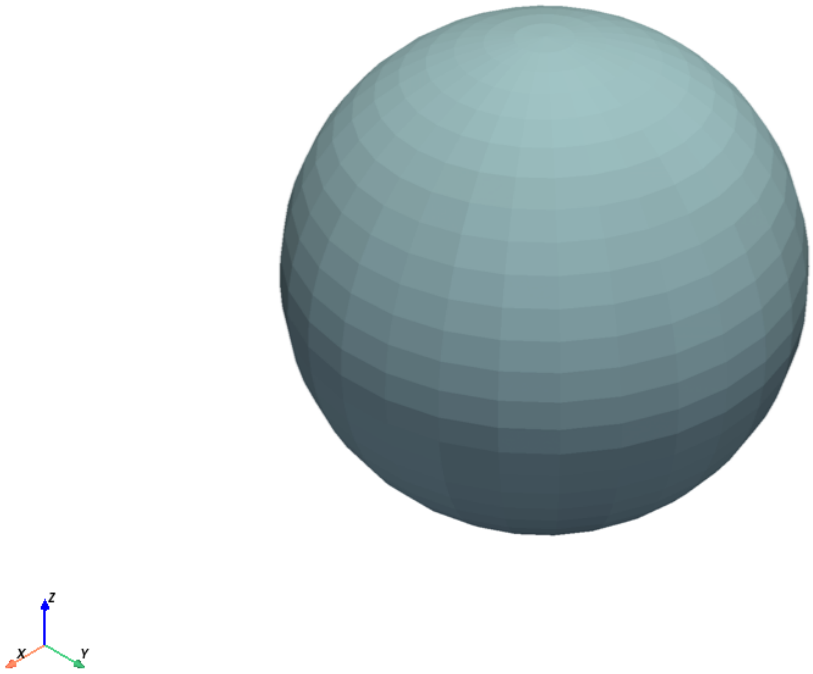
```
cube = pv.Cube()
pv_backend = PyVistaBackend(use_qt=True, show_qt=True)
pl = Plotter(backend=pv_backend)
pl.plot(cube)
pl.backend.enable_widgets()
pv_backend.scene.show()
```

Parallel VTK window

```
sphere = pv.Sphere()

pl_parallel = Plotter()
pl_parallel.plot(sphere)
pl_parallel.show()
```

Static Scene



Interactive Scene

```
[]
```

Close the pyvistaqt window

```
pv_backend.close()
```

Integrate the plotter in a Qt application

```
pv_backend = PyVistaBackend(use_qt=True, show_qt=False)
pv_backend.enable_widgets()

# You can use this plotter in a Qt application
pl = pv_backend.scene
```

Total running time of the script: (0 minutes 18.790 seconds)

9.1.3 Use a clipping plane

This example shows how to use a clipping plane in the Visualization Interface Tool to cut a mesh.

```
import pyvista as pv

from ansys.tools.visualization_interface import ClipPlane, Plotter

mesh = pv.Cylinder()
```

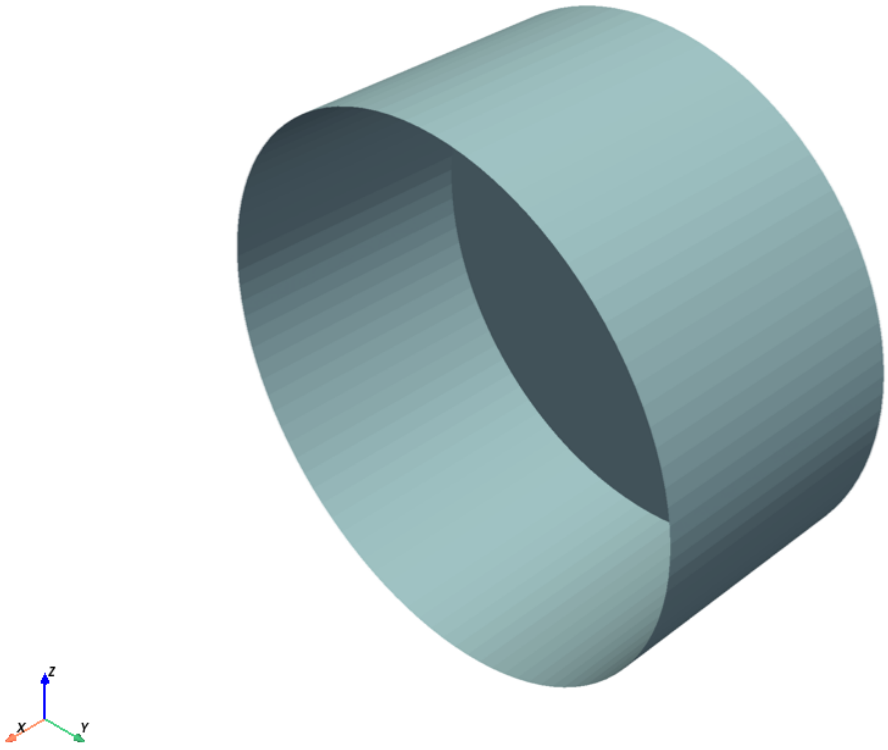
Create a plotter and clip the mesh

```
pl = Plotter()

# Create a clipping plane
clipping_plane = ClipPlane(normal=(1, 0, 0), origin=(0, 0, 0))

# Add the mesh to the plotter with the clipping plane
pl.plot(mesh, clipping_plane=clipping_plane)
pl.show()
```

Static Scene



Interactive Scene

```
[]
```

Total running time of the script: (0 minutes 0.166 seconds)

9.1.4 Use the `MeshObjectPlot` class

The Visualization Interface Tool provides the `MeshObject` helper class to relate a custom object with its mesh. With a custom object, you can take advantage of the full potential of the Visualization Interface Tool.

This example shows how to use the `MeshObjectPlot` class to plot your custom objects.

Relate `CustomObject` class with a PyVista mesh

```
import pyvista as pv

# Note that the ``CustomObject`` class must have a way to get the mesh
# and a name or ID.

class CustomObject:
    def __init__(self):
        self.name = "CustomObject"
        self.mesh = pv.Cube()

    def get_mesh(self):
        return self.mesh

    def name(self):
        return self.name

# Create a custom object
custom_object = CustomObject()
```

Create a `MeshObjectPlot` instance

```
from ansys.tools.visualization_interface import MeshObjectPlot

# Create an instance

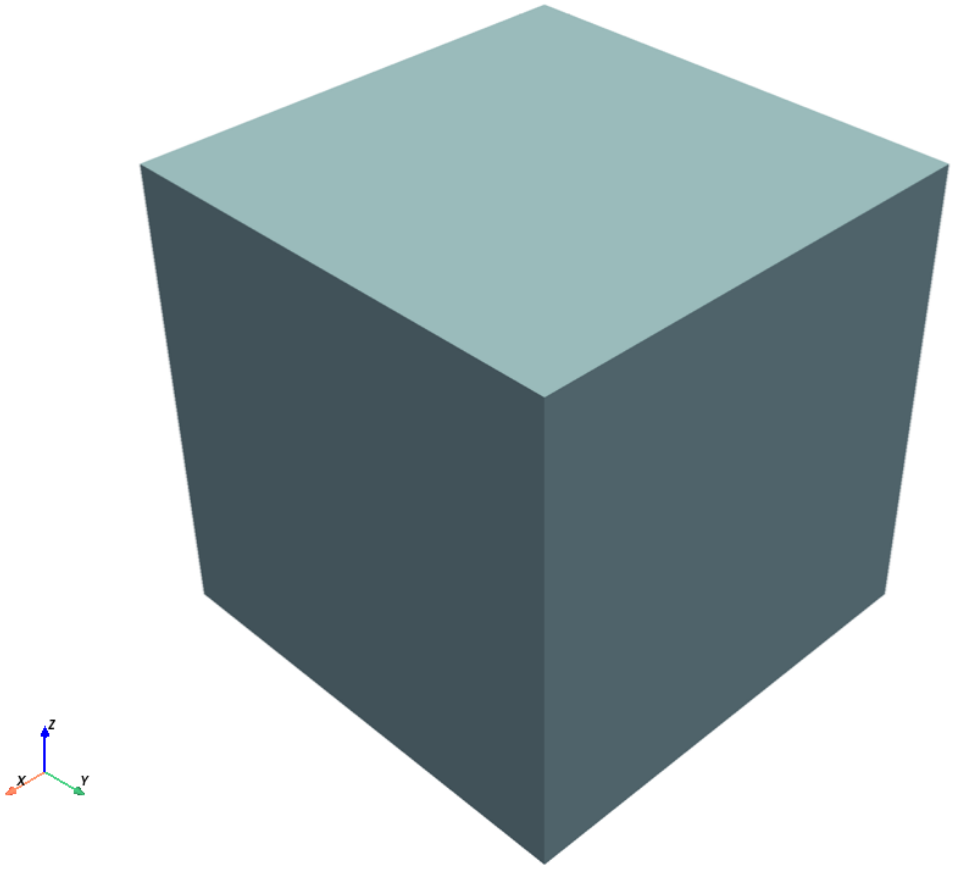
mesh_object = MeshObjectPlot(custom_object, custom_object.get_mesh())
```

Plot the `MeshObjectPlot` instance

```
from ansys.tools.visualization_interface import Plotter

pl = Plotter()
pl.plot(mesh_object)
pl.show()
```

Static Scene



Interactive Scene

```
[]
```

Total running time of the script: (0 minutes 0.156 seconds)

9.1.5 Use the plotter

This example shows how to add one or more meshes to the plotter.

Add a mesh to the plotter

This code shows how to add a single mesh to the plotter.

```
import pyvista as pv

from ansys.tools.visualization_interface import Plotter

mesh = pv.Cube()

# Create a plotter
pl = Plotter()
```

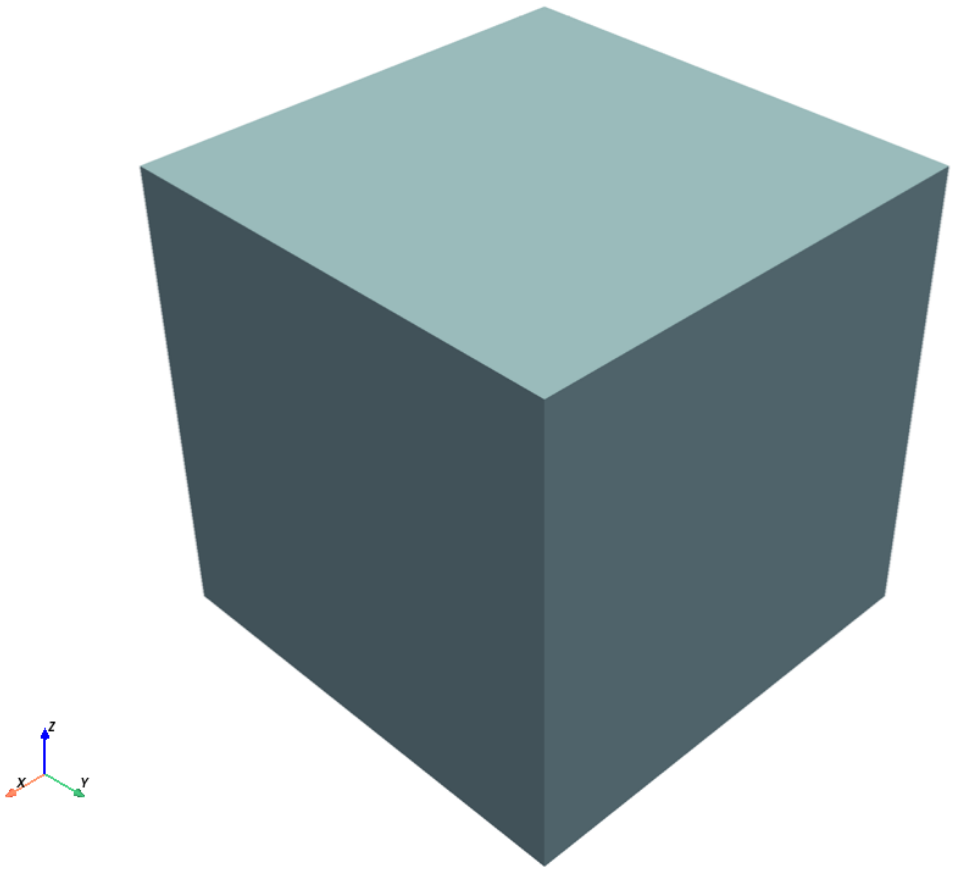
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```
# Add the mesh to the plotter
pl.plot(mesh)

# Show the plotter
pl.show()
```

Static Scene



Interactive Scene

```
[]
```

Getting a screenshot

Now we will check how to get a screenshot from our plotter.

```
import pyvista as pv

from ansys.tools.visualization_interface import Plotter

```

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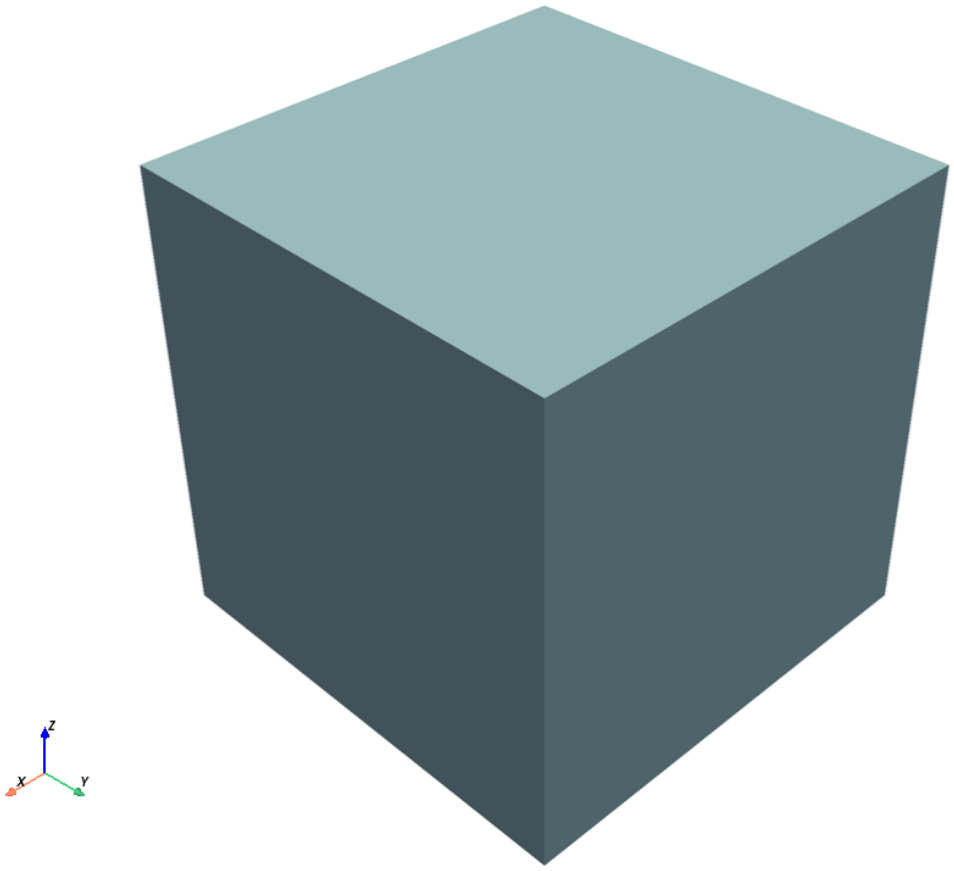
```
mesh = pv.Cube()

# Create a plotter
pl = Plotter()

# Add the mesh to the plotter
pl.plot(mesh)

# Show the plotter
pl.show()
```

Static Scene



Interactive Scene

```
[]
```

Add a list of meshes

This code shows how to add a list of meshes to the plotter.

```
import pyvista as pv

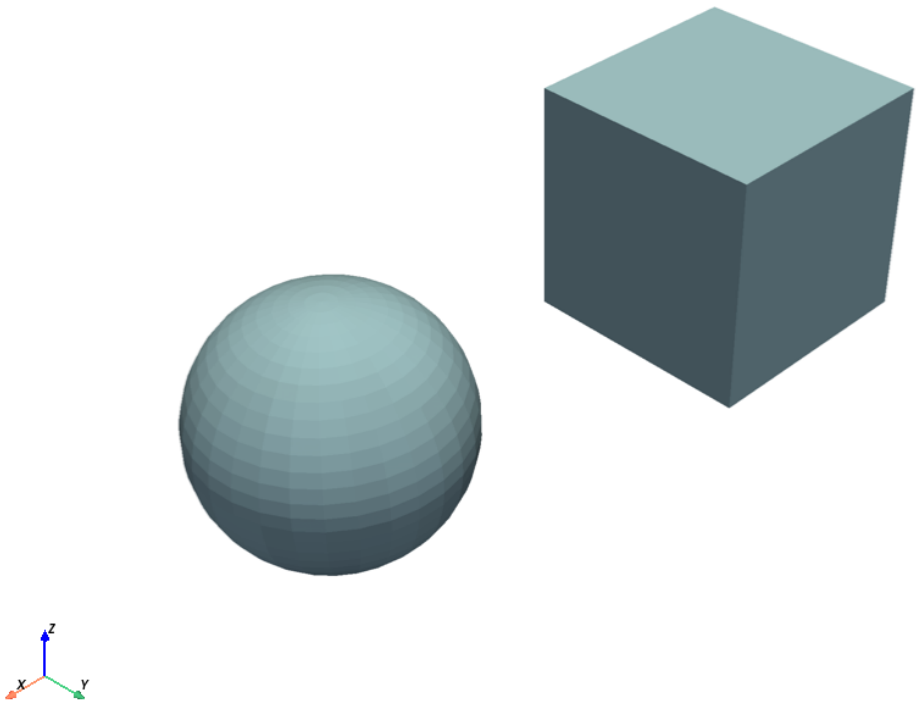
from ansys.tools.visualization_interface import Plotter

mesh1 = pv.Cube()
mesh2 = pv.Sphere(center=(2, 0, 0))
mesh_list = [mesh1, mesh2]
# Create a plotter
pl = Plotter()

# Add a list of meshes to the plotter
pl.plot(mesh_list)

# Show the plotter
pl.show()
```

Static Scene



Interactive Scene

```
[]
```

Total running time of the script: (0 minutes 0.462 seconds)

9.1.6 Animation Example

This example demonstrates how to create and display animations for time-series simulation data using the PyVista backend.

```
import numpy as np
import pyvista as pv

from ansys.tools.visualization_interface import Plotter
```

Create sample animation data

Generate a series of meshes representing a wave propagation over time.

```
def create_wave_mesh(time_step, n_points=50):
    """Create a mesh with a wave pattern for a given time step."""
    # Create grid
    x = np.linspace(-5, 5, n_points)
    y = np.linspace(-5, 5, n_points)
    x, y = np.meshgrid(x, y)

    # Create wave pattern that evolves over time
    t = time_step * 0.2
    z = np.sin(np.sqrt(x**2 + y**2) - t) * np.exp(-0.1 * np.sqrt(x**2 + y**2))

    # Create structured grid
    mesh = pv.StructuredGrid(x, y, z)
    mesh["displacement"] = np.abs(z).ravel()

    return mesh


# Create 30 frames
frames = [create_wave_mesh(i) for i in range(30)]
```

Display animation with interactive controls

Create and show an animation with play/pause, stop, and frame navigation.

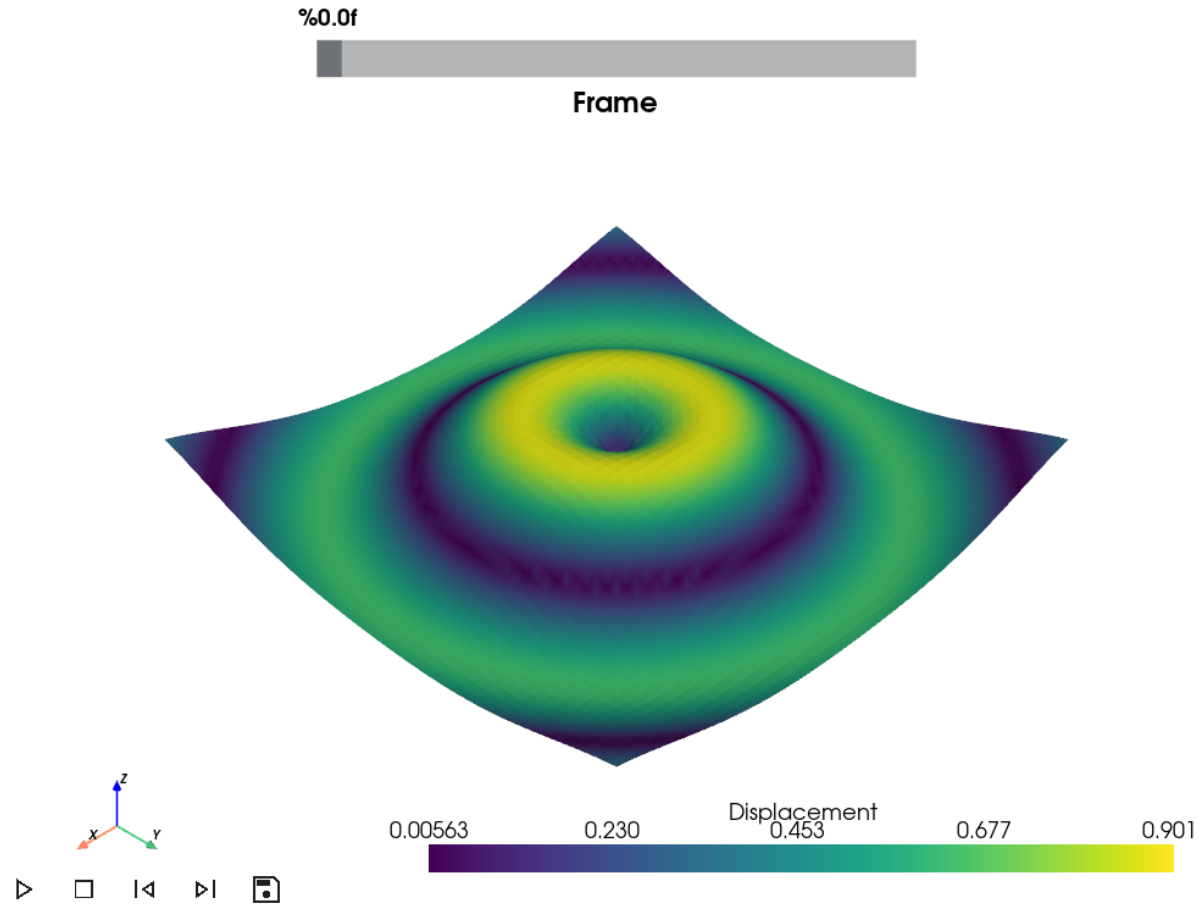
```
plotter = Plotter()
animation = plotter.animate(
    frames,
    fps=20,
    loop=True,
    scalar_bar_args={"title": "Displacement"}
)
```

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```
# Display with interactive controls
animation.show()
```

Static Scene



Interactive Scene

Interactive Controls

The animation window includes the following controls:

- **Play/Pause button** (green): Start/stop animation playback
- **Stop button** (red): Reset to first frame
- **Previous/Next buttons** (blue): Step through frames manually
- **Save GIF button** (orange): Export animation (requires imageio)
- **Frame slider** (top): Jump to any frame

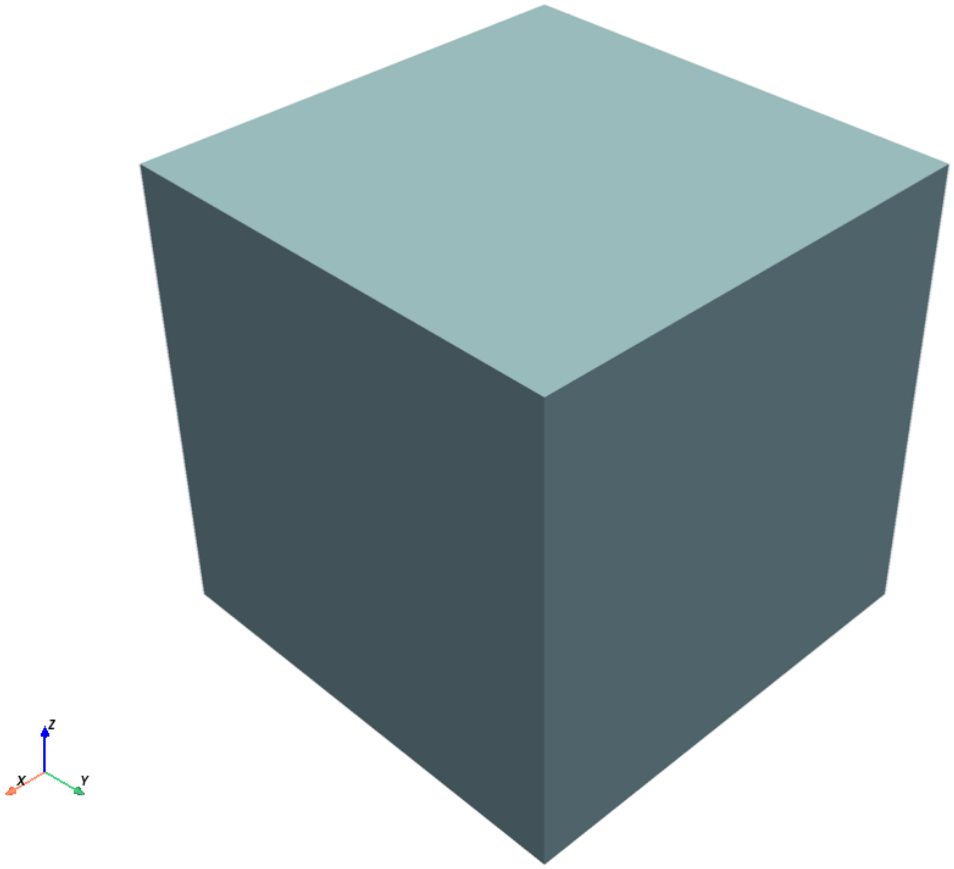
You can rotate, zoom, and pan the view while the animation plays.

Total running time of the script: (0 minutes 1.042 seconds)

9.1.7 MeshObjectPlot tree structure

This example shows how to add a tree structure of MeshObjectPlot to the plotter.

Static Scene



Interactive Scene

```
[]
```

```
import pyvista as pv
from ansys.tools.visualization_interface import Plotter

class CustomObject:
    def __init__(self):
        self.name = "CustomObject"
        self.mesh = pv.Cube(center=(1, 1, 0))
```

(continues on next page)

(continued from previous page)

```
    def get_mesh(self):
        return self.mesh

    def name(self):
        return self.name



# Create a custom objects
custom_cube = CustomObject()
custom_cube.name = "CustomCube"

custom_sphere = CustomObject()
custom_sphere.mesh = pv.Sphere(center=(0, 0, 5))
custom_sphere.name = "CustomSphere"

custom_sphere1 = CustomObject()
custom_sphere1.mesh = pv.Sphere(center=(5, 0, 5))
custom_sphere1.name = "CustomSphere"

from ansys.tools.visualization_interface import MeshObjectPlot

# Create an instance
mesh_object_cube = MeshObjectPlot(custom_cube, custom_cube.get_mesh())
mesh_object_sphere = MeshObjectPlot(custom_sphere, custom_sphere.get_mesh())
mesh_object_sphere1 = MeshObjectPlot(custom_sphere1, custom_sphere1.get_mesh())

mesh_object_cube.add_child(mesh_object_sphere)
mesh_object_sphere.add_child(mesh_object_sphere1)

pl = Plotter()
pl.plot(mesh_object_cube, plot_children=True)


pl.backend._pl.hide_children(mesh_object_cube)
pl.show()
```

Total running time of the script: (0 minutes 0.164 seconds)

9.1.8 Activate the picker

This example shows how to activate the picker, which is the tool that you use to select an object in the plotter and get its name.

Relate `CustomObject` class with a PyVista mesh

```
import pyvista as pv

# Note that the ``CustomObject`` class must have a way to get the mesh
# and a name or ID.

class CustomObject:
```

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```
    def __init__(self):
        self.name = "CustomObject"
        self.mesh = pv.Cube(center=(1, 1, 0))

    def get_mesh(self):
        return self.mesh

    def name(self):
        return self.name



# Create a custom object
custom_cube = CustomObject()
custom_cube.name = "CustomCube"
custom_sphere = CustomObject()
custom_sphere.mesh = pv.Sphere(center=(0, 0, 5))
custom_sphere.name = "CustomSphere"
```

Create two `MeshObjectPlot` instances

```
from ansys.tools.visualization_interface import MeshObjectPlot

# Create an instance
mesh_object_cube = MeshObjectPlot(custom_cube, custom_cube.get_mesh())
mesh_object_sphere = MeshObjectPlot(custom_sphere, custom_sphere.get_mesh())
```

Activate the picking capabilities

```
from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.backends.pyvista import PyVistaBackend

pv_backend = PyVistaBackend(allow_picking=True, plot_picked_names=True)
pl = Plotter(backend=pv_backend)
pl.plot(mesh_object_cube)
pl.plot(mesh_object_sphere)
pl.show()
```

Static Scene







Interactive Scene

```
[]
```

Activate the hover capabilities

```
from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.backends.pyvista import PyVistaBackend

pv_backend = PyVistaBackend(allow_hovering=True)
pl = Plotter(backend=pv_backend)
pl.plot(mesh_object_cube)
pl.plot(mesh_object_sphere)
pl.show()
```

Static Scene







Interactive Scene

```
[]
```

Using StructuredGrid mesh

```
import numpy as np


class CustomStructuredObject:
    def __init__(self):
        self.name = "CustomObject"
        xrng = np.arange(-10, 10, 2, dtype=np.float32)
        yrng = np.arange(-10, 10, 5, dtype=np.float32)
        zrng = np.arange(-10, 10, 1, dtype=np.float32)
        x, y, z = np.meshgrid(xrng, yrng, zrng, indexing='ij')
        grid = pv.StructuredGrid(x, y, z)
        self.mesh = grid

    def get_mesh(self):
        return self.mesh
```

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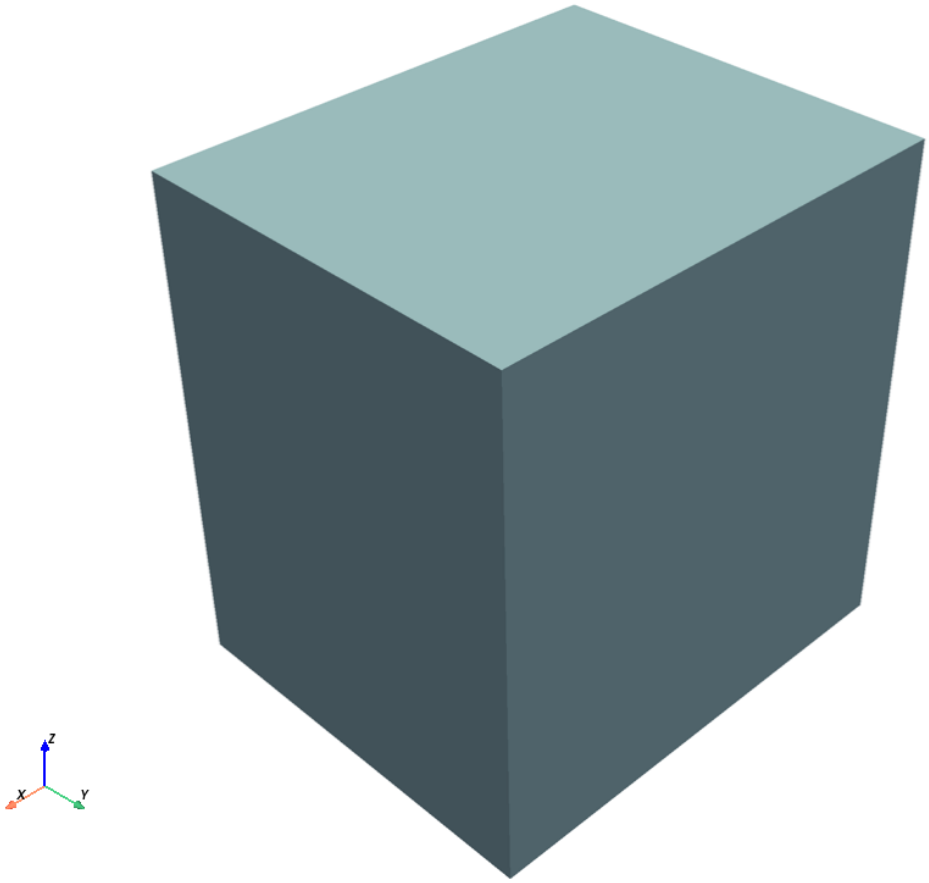
(continued from previous page)

```
    def name(self):
        return self.name


pv_backend = PyVistaBackend()
pl = Plotter(backend=pv_backend)

structured_object = CustomStructuredObject()
mo_plot = MeshObjectPlot(structured_object, structured_object.get_mesh())
pl.plot(mo_plot)
pl.show()
```

Static Scene



Interactive Scene

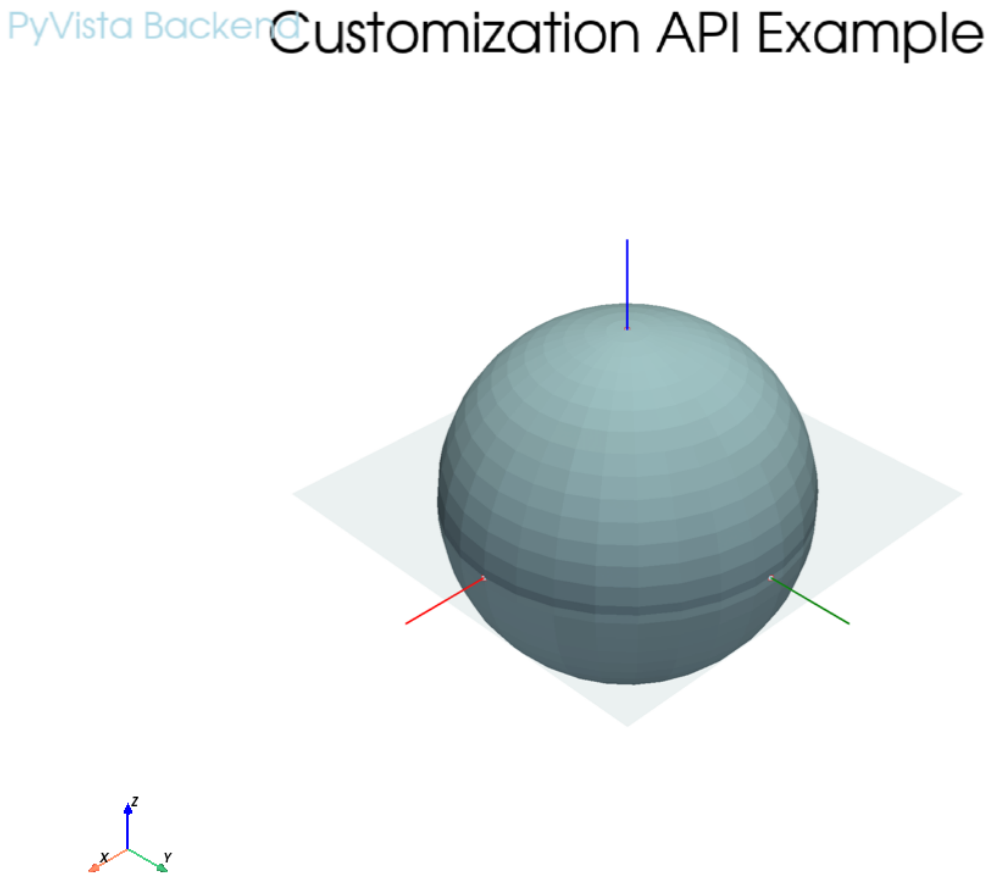
```
[]
```

Total running time of the script: (0 minutes 0.451 seconds)

9.1.9 Customization API example

This example demonstrates how to use the customization API of the visualization interface to add various elements to a PyVista scene, such as points, lines, planes, and text annotations. The example also shows how to plot a simple sphere mesh and customize its appearance.

Static Scene



Interactive Scene

```
/home/runner/work/ansys-tools-visualization-interface/ansys-tools-visualization-
↪interface/.venv/lib/python3.14/site-packages/ansys/tools/visualization_interface/
↪backends/pyvista/pyvista.py:808: UserWarning: Points is not a float type. This can␣
↪cause issues when transforming or applying filters. Casting to ``np.float32``. Disable␣
↪this by passing ``force_float=False``.
  point_cloud = pv.PolyData(points_array)
/home/runner/work/ansys-tools-visualization-interface/ansys-tools-visualization-
↪interface/.venv/lib/python3.14/site-packages/ansys/tools/visualization_interface/
↪backends/pyvista/pyvista.py:1024: UserWarning: Points is not a float type. This can␣
↪cause issues when transforming or applying filters. Casting to ``np.float32``. Disable␣
↪this by passing ``force_float=False``.
  point_cloud = pv.PolyData(points_array)

[]
```

```
from ansys.tools.visualization_interface import Plotter
import pyvista as pv

# Create a plotter using the Plotly backend and add basic geometry.

plotter = Plotter()

# Add a sphere - this works fine
sphere = pv.Sphere(radius=1.0, center=(0, 0, 0))
plotter.plot(sphere)


# Add point markers to highlight specific locations.

key_points = [
    [1, 0, 0],   # Point on X axis
    [0, 1, 0],   # Point on Y axis
    [0, 0, 1],   # Point on Z axis
]

plotter.add_points(key_points, color='red', size=10)

# Add line segments to show coordinate axes.

# X axis
x_axis = [[0, 0, 0], [1.5, 0, 0]]
plotter.add_lines(x_axis, color='red', width=4.0)

# Y axis
y_axis = [[0, 0, 0], [0, 1.5, 0]]
plotter.add_lines(y_axis, color='green', width=4.0)

# Z axis
z_axis = [[0, 0, 0], [0, 0, 1.5]]
plotter.add_lines(z_axis, color='blue', width=4.0)


# Add a plane to show a reference surface.

plotter.add_planes(
    center=(0, 0, 0),
    normal=(0, 0, 1),
    i_size=2.5,
    j_size=2.5,
    color='lightblue',
    opacity=0.2
)


# Scene title at the top center
```

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```
plotter.add_text("Customization API Example", position="upper_edge", font_size=18, color=
↪'black')

# Additional labels at the top left corner using a string for the position as before
plotter.add_text("PyVista Backend", position="upper_left", font_size=12, color='lightblue
↪')
# Additional labels at the bottom left corner using pixel coordinates
plotter.add_text("3D Visualization", position=(0.95, 0.95), font_size=12, color=
↪'lightgreen')


# Add labels at specific 3D points to annotate key locations in space.

label_points = [
    [1, 0, 0],   # X axis endpoint
    [0, 1, 0],   # Y axis endpoint
    [0, 0, 1],   # Z axis endpoint
]

labels = ['X-axis', 'Y-axis', 'Z-axis']

plotter.add_labels(label_points, labels, font_size=16, point_size=8.0)


# The clear() method resets the plotter and can be called even after show().
# This allows reusing the same plotter for multiple visualizations.

# Uncomment to clear everything added above and start fresh:
# plotter.show()
# plotter.clear()
# plotter.plot(pv.Cube())  # Would show only a cube instead


# Display the visualization with all customizations.

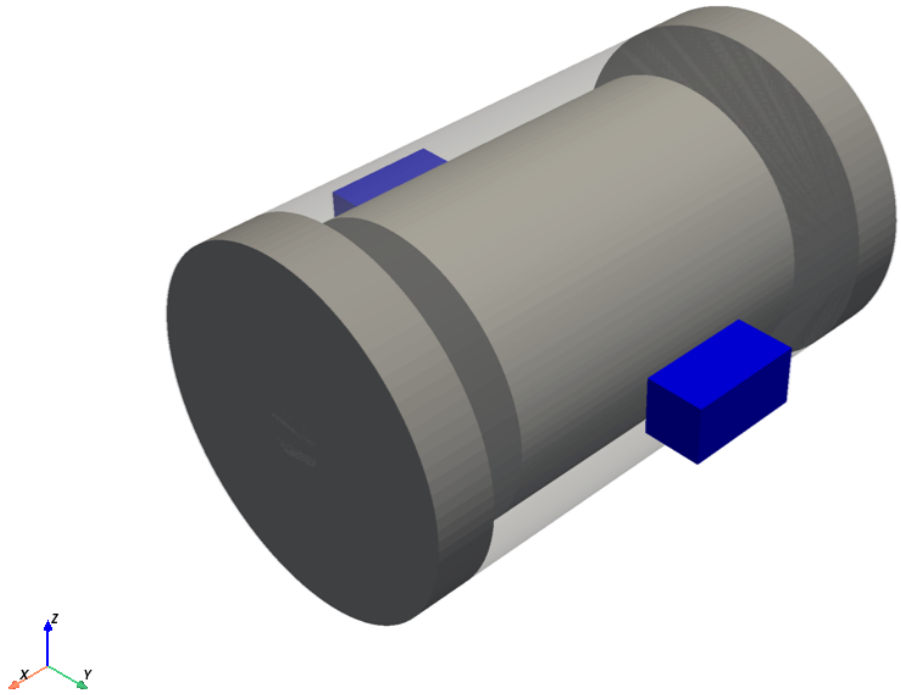
plotter.show()
```

Total running time of the script: (0 minutes 0.373 seconds)

9.1.10 Tree view menu

This example demonstrates the use of a tree view menu to manage the visibility of components in an encased motor block assembly. The tree menu allows users to interactively show or hide different parts of the assembly.

Static Scene



Interactive Scene

```
================================================================================
                          ENCASED MOTOR BLOCK ASSEMBLY
================================================================================

HIERARCHY:
  Motor Housing (semi-transparent)
    ├─ Stator
    │    └─ Rotor
    │         ├─ Shaft
    │         ├─ Front Bearing
    │         └─ Rear Bearing
    ├─ Front End Cap
    ├─ Rear End Cap
    ├─ Top Mounting Bracket
    └─ Bottom Mounting Bracket

TREE MENU (Right side):
  - Click eye icons to show/hide components
  - Hiding a parent hides all children
```

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```
  - Example: Hide 'Motor Housing' to hide entire assembly
  - Example: Hide 'Rotor' to hide rotor, shaft, and bearings

USE CASES:
  1. Hide housing to see internal components
  2. Hide stator to see only rotor assembly
  3. Hide brackets to focus on motor internals
  4. Hide individual bearings or shaft

CONTROLS:
  - Toggle button (bottom left) = Show/hide tree menu
  - Eye icons in menu = Toggle component visibility
================================================================================


[]
```

```python
import pyvista as pv
from ansys.tools.visualization_interface import Plotter, MeshObjectPlot
from ansys.tools.visualization_interface.backends.pyvista.pyvista import PyVistaBackend


# Simple helper class to hold component names
class MotorComponent:
    """Simple class to hold motor component name."""
    def __init__(self, name: str):
        self.name = name


# Create Motor Housing (outer casing)
housing = MeshObjectPlot(
    MotorComponent("Motor Housing"),
    pv.Cylinder(center=(0, 0, 0), direction=(1, 0, 0), radius=3, height=8)
)

# Create Stator (fixed part inside housing)
stator = MeshObjectPlot(
    MotorComponent("Stator"),
    pv.Cylinder(center=(0, 0, 0), direction=(1, 0, 0), radius=2.5, height=6)
)

# Create Rotor (rotating part)
rotor = MeshObjectPlot(
    MotorComponent("Rotor"),
    pv.Cylinder(center=(0, 0, 0), direction=(1, 0, 0), radius=1.8, height=5.5)
)
```

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```
# Create Shaft (central rotating shaft)
shaft = MeshObjectPlot(
    MotorComponent("Shaft"),
    pv.Cylinder(center=(0, 0, 0), direction=(1, 0, 0), radius=0.5, height=10)
)

# Create Bearings
bearing_front = MeshObjectPlot(
    MotorComponent("Front Bearing"),
    pv.Cylinder(center=(3.5, 0, 0), direction=(1, 0, 0), radius=0.8, height=0.5)
)

bearing_rear = MeshObjectPlot(
    MotorComponent("Rear Bearing"),
    pv.Cylinder(center=(-3.5, 0, 0), direction=(1, 0, 0), radius=0.8, height=0.5)
)

# Create End Caps
end_cap_front = MeshObjectPlot(
    MotorComponent("Front End Cap"),
    pv.Cylinder(center=(4.5, 0, 0), direction=(1, 0, 0), radius=3, height=1)
)

end_cap_rear = MeshObjectPlot(
    MotorComponent("Rear End Cap"),
    pv.Cylinder(center=(-4.5, 0, 0), direction=(1, 0, 0), radius=3, height=1)
)

# Create Mounting Brackets
bracket_top = MeshObjectPlot(
    MotorComponent("Top Mounting Bracket"),
    pv.Box(bounds=(-1, 1, 3, 4, -0.5, 0.5))
)

bracket_bottom = MeshObjectPlot(
    MotorComponent("Bottom Mounting Bracket"),
    pv.Box(bounds=(-1, 1, -4, -3, -0.5, 0.5))
)

# Build hierarchy - Housing contains all internal components
housing.add_child(stator)
housing.add_child(end_cap_front)
housing.add_child(end_cap_rear)
housing.add_child(bracket_top)
housing.add_child(bracket_bottom)

# Stator contains rotor
stator.add_child(rotor)

# Rotor contains shaft and bearings
rotor.add_child(shaft)
rotor.add_child(bearing_front)
```

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```
rotor.add_child(bearing_rear)

# Create plotter with picking enabled (automatically adds tree menu)
backend = PyVistaBackend()
pl = Plotter(backend=backend)

# Plot all components with different colors
pl.plot(housing, color="lightgray", opacity=0.3)  # Semi-transparent housing
pl.plot(stator, color="darkgray")
pl.plot(rotor, color="orange")
pl.plot(shaft, color="silver")
pl.plot(bearing_front, color="gold")
pl.plot(bearing_rear, color="gold")
pl.plot(end_cap_front, color="darkgray")
pl.plot(end_cap_rear, color="darkgray")
pl.plot(bracket_top, color="blue")
pl.plot(bracket_bottom, color="blue")

print("\n" + "="*80)
print(" " * 25 + "ENCASED MOTOR BLOCK ASSEMBLY")
print("="*80)
print("\nHIERARCHY:")
print("  Motor Housing (semi-transparent)")
print("    ├─ Stator")
print("    │   └─ Rotor")
print("    │       ├─ Shaft")
print("    │       ├─ Front Bearing")
print("    │       └─ Rear Bearing")
print("    ├─ Front End Cap")
print("    ├─ Rear End Cap")
print("    ├─ Top Mounting Bracket")
print("    └─ Bottom Mounting Bracket")
print("\nTREE MENU (Right side):")
print("  - Click eye icons to show/hide components")
print("  - Hiding a parent hides all children")
print("  - Example: Hide 'Motor Housing' to hide entire assembly")
print("  - Example: Hide 'Rotor' to hide rotor, shaft, and bearings")
print("\nUSE CASES:")
print("  1. Hide housing to see internal components")
print("  2. Hide stator to see only rotor assembly")
print("  3. Hide brackets to focus on motor internals")
print("  4. Hide individual bearings or shaft")
print("\nCONTROLS:")
print("  - Toggle button (bottom left) = Show/hide tree menu")
print("  - Eye icons in menu = Toggle component visibility")
print("="*80 + "\n")

# Show the plot
pl.show()
```

Total running time of the script: (0 minutes 0.260 seconds)

9.1.11 Create custom picker

This example shows how to create a custom picker. In this case we will show how the default picker is implemented through the `AbstractPicker` class.

Import the `AbstractPicker` class

```
# Import the abstract picker class
from ansys.tools.visualization_interface.backends.pyvista.picker import AbstractPicker

# Import custom object meshes
from ansys.tools.visualization_interface.types.mesh_object_plot import MeshObjectPlot

# Import plotter and color enum
from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.utils.color import Color
```

Create a custom picker class

```
class CustomPicker(AbstractPicker):
    """Custom picker class that extends the AbstractPicker.
    This custom picker changes the color of picked objects to red and adds a label with␣
↪the object's name.
    It also adds a label when hovering over an object.

    Parameters
    ----------
    plotter_backend : Plotter
        The plotter backend to use.
    plot_picked_names : bool, optional
        Whether to plot the names of picked objects, by default True.
    label : str, optional
        Extra parameter to exemplify the usage of custom parameters.
    """
    def __init__(self, plotter_backend: "Plotter", plot_picked_names: bool = True,␣
↪label: str = "This label: ") -> None:
        """Initialize the ``Picker`` class."""
        # Picking variables
        self._plotter_backend = plotter_backend
        self._plot_picked_names = plot_picked_names
        self._label = label

        # Map that relates PyVista actors with the added actors by the picker
        self._picker_added_actors_map = {}

        # Dictionary of picked objects in MeshObject format.
        self._picked_dict = {}

        # Map that saves original colors of the plotted objects.
        self._origin_colors = {}

        # Hovering variables
        self._added_hover_labels = []
```

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```
    def pick_select_object(self, custom_object: MeshObjectPlot, pt: "np.ndarray") ->␣
↪None:
        """Add actor to picked list and add label if required.

        Parameters
        ----------
        custom_object : MeshObjectPlot
            The object to be selected.
        pt : np.ndarray
            The point where the object was picked.
        """
        added_actors = []

        # Pick only custom objects
        if isinstance(custom_object, MeshObjectPlot):
            self._origin_colors[custom_object] = custom_object.actor.prop.color
            custom_object.actor.prop.color = Color.PICKED.value

        # Get the name for the text label
        text = custom_object.name

        # If picking names is enabled, add a label to the picked object
        if self._plot_picked_names:
            label_actor = self._plotter_backend.pv_interface.scene.add_point_labels(
                [pt],
                [self._label + text],
                always_visible=True,
                point_size=0,
                render_points_as_spheres=False,
                show_points=False,
            )
            # Add the label actor to the list of added actors
            added_actors.append(label_actor)

        # Add the picked object to the picked dictionary if not already present, to keep␣
↪track of it
        if custom_object.name not in self._picked_dict:
            self._picked_dict[custom_object.name] = custom_object
        # Add the picked object to the picked dictionary if not already present, to keep␣
↪track of it
        self._picker_added_actors_map[custom_object.actor.name] = added_actors

    def pick_unselect_object(self, custom_object: MeshObjectPlot) -> None:
        """Remove actor from picked list and remove label if required.

        Parameters
        ----------
        custom_object : MeshObjectPlot
            The object to be unselected.
        """
        # remove actor from picked list and from scene
```

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```
        if custom_object.name in self._picked_dict:
            self._picked_dict.pop(custom_object.name)

        # Restore original color if it was changed
        if isinstance(custom_object, MeshObjectPlot) and custom_object in self._origin_
↪colors:
            custom_object.actor.prop.color = self._origin_colors[custom_object]

        # Remove any added actors (like labels) associated with this picked object
        if custom_object.actor.name in self._picker_added_actors_map:
            self._plotter_backend._pl.scene.remove_actor(self._picker_added_actors_
↪map[custom_object.actor.name])
            self._picker_added_actors_map.pop(custom_object.actor.name)

    def hover_select_object(self, custom_object: MeshObjectPlot, actor: "Actor") -> None:
        """Add label to hovered object if required.

        Parameters
        ----------
        custom_object : MeshObjectPlot
            The object to be hovered over.
        actor : vtkActor
            The actor corresponding to the hovered object.
        """
        for label in self._added_hover_labels:
            self._plotter_backend._pl.scene.remove_actor(label)
        label_actor = self._plotter_backend._pl.scene.add_point_labels(
            [actor.GetCenter()],
            [custom_object.name],
            always_visible=True,
            point_size=0,
            render_points_as_spheres=False,
            show_points=False,
        )
        self._added_hover_labels.append(label_actor)

    def hover_unselect_object(self):
        """Remove all hover labels from the scene."""
        for label in self._added_hover_labels:
            self._plotter_backend._pl.scene.remove_actor(label)

    @property
    def picked_dict(self) -> dict:
        """Return the dictionary of picked objects.

        Returns
        -------
        dict
            Dictionary of picked objects.
        """
        return self._picked_dict
```

Initialize the plotter backend with the custom picker

```
from ansys.tools.visualization_interface.backends.pyvista import PyVistaBackend
pl_backend = PyVistaBackend(allow_picking=True, custom_picker=CustomPicker)
```



Create a custom object with a name to be picked

```
import pyvista as pv

class CustomObject:
    def __init__(self):
        self.name = "CustomObject"
        self.mesh = pv.Cube(center=(1, 1, 0))

    def get_mesh(self):
        return self.mesh

    def name(self):
        return self.name

# Create a custom object
custom_cube = CustomObject()
custom_cube.name = "CustomCube"
```

Create a `MeshObjectPlot` instance

```
from ansys.tools.visualization_interface import MeshObjectPlot
# Create an instance
mesh_object_cube = MeshObjectPlot(custom_cube, custom_cube.get_mesh())
```

Display the plotter and interact with the object

```
pl = Plotter(backend=pl_backend)
pl.plot(mesh_object_cube)
pl.show()
```

Total running time of the script: (0 minutes 0.113 seconds)

9.2 Basic Plotly usage examples

These examples show how to use the general plotter with Plotly backend included in the Visualization Interface Tool.

9.2.1 Plain usage of the plotly dash backend

This example shows the plain usage of the Plotly Dash backend in the Visualization Interface Tool to plot different objects, including PyVista meshes, custom objects, and Plotly-specific objects.

```
from ansys.tools.visualization_interface.backends.plotly.plotly_dash import␣
↪PlotlyDashBackend
from ansys.tools.visualization_interface.types.mesh_object_plot import MeshObjectPlot
from ansys.tools.visualization_interface import Plotter
import pyvista as pv
from plotly.graph_objects import Mesh3d


# Create a plotter with the Plotly backend
pl = Plotter(backend=PlotlyDashBackend())

# Create a PyVista mesh
mesh = pv.Sphere()
mesh2 = pv.Cube(center=(2,0,0))
# Plot the mesh
pl.plot(mesh, name="Sphere")
pl.plot(mesh2, name="Cube")

# ---------------------------------
# Start the server and show the plot
# ---------------------------------
#
# .. code-block:: python
#
#     pl.show()
```

Total running time of the script: (0 minutes 0.834 seconds)

9.2.2 Name filtering in the Plotly backend

The `name_filter` parameter accepts a regular-expression string and is available on *Plotter.plot()*, *Plotter.plot_iter()*, and *Plotter.show()*. Only `MeshObjectPlot` objects whose `name` matches the pattern are added to the figure; everything else is silently skipped.

```
import pyvista as pv

from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.backends.plotly.plotly_interface import␣
↪PlotlyBackend
from ansys.tools.visualization_interface.types.mesh_object_plot import MeshObjectPlot


class Part:
    """Minimal domain object with a name."""

    def __init__(self, name: str):
        self.name = name
```

Build a set of named parts

```
parts = [
    MeshObjectPlot(Part("Wheel_FL"), pv.Sphere(center=(-1,  1, 0), radius=0.3)),
    MeshObjectPlot(Part("Wheel_FR"), pv.Sphere(center=( 1,  1, 0), radius=0.3)),
    MeshObjectPlot(Part("Wheel_RL"), pv.Sphere(center=(-1, -1, 0), radius=0.3)),
    MeshObjectPlot(Part("Wheel_RR"), pv.Sphere(center=( 1, -1, 0), radius=0.3)),
    MeshObjectPlot(Part("Chassis"),  pv.Box(bounds=(-0.8, 0.8, -1.2, 1.2, 0, 0.4))),
]
```

Show with `name_filter` via `show()`

Pass `name_filter` directly to *Plotter.show()* together with the list of objects. The filter is applied while plotting, so only matching parts are added to the figure.

```
pl = Plotter(backend=PlotlyBackend())
pl.show(plottable_object=parts)

pl = Plotter(backend=PlotlyBackend())
pl.show(plottable_object=parts, name_filter="Wheel")
```

Total running time of the script: (0 minutes 0.134 seconds)

9.2.3 Plain usage of the plotly backend

This example shows the plain usage of the Plotly backend in the Visualization Interface Tool to plot different objects, including PyVista meshes, custom objects, and Plotly-specific objects.

```
from ansys.tools.visualization_interface.backends.plotly.plotly_interface import␣
↪PlotlyBackend
from ansys.tools.visualization_interface.types.mesh_object_plot import MeshObjectPlot
from ansys.tools.visualization_interface import Plotter
import pyvista as pv
from plotly.graph_objects import Mesh3d
```

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```
# Create a plotter with the Plotly backend
pl = Plotter(backend=PlotlyBackend())

# Create a PyVista mesh
mesh = pv.Sphere()

# Plot the mesh
pl.plot(mesh)


# Create a PyVista MultiBlock
multi_block = pv.MultiBlock()
multi_block.append(pv.Sphere(center=(-1, -1, 0)))
multi_block.append(pv.Cube(center=(-1, 1, 0)))

# Plot the MultiBlock
pl.plot(multi_block)

# Display the plotter

pl.show()
```

```
/home/runner/work/ansys-tools-visualization-interface/ansys-tools-visualization-
↪interface/.venv/lib/python3.14/site-packages/ansys/tools/visualization_interface/
↪backends/plotly/plotly_interface.py:73: PyVistaFutureWarning: The default value of␣
↪`algorithm` for the filter
`PolyData.extract_surface` will change in the future. It currently defaults to
``'dataset_surface'``, but will change to `None`. Explicitly set the `algorithm` keyword to
silence this warning.
  block = block.extract_surface()
```

Now create a custom object

```
class CustomObject:
    def __init__(self):
        self.name = "CustomObject"
        self.mesh = pv.Cube(center=(1, 1, 0))

    def get_mesh(self):
        return self.mesh

    def name(self):
        return self.name


# Create a custom object
custom_cube = CustomObject()
custom_cube.name = "CustomCube"
```

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```
# Create a MeshObjectPlot instance
mesh_object_cube = MeshObjectPlot(custom_cube, custom_cube.get_mesh())

# Plot the custom mesh object
pl.plot(mesh_object_cube)
```

Display the plotter again

Since Plotly is a web-based visualization, we can show the plot again to include the new object.

```
pl.show()
```

Add a Plotly Mesh3d object directly

```
custom_mesh3d = Mesh3d(
    x=[0, 1, 2],
    y=[0, 1, 0],
    z=[0, 0, 1],
    i=[0],
    j=[1],
    k=[2],
    color='lightblue',
    opacity=0.50
)
pl.plot(custom_mesh3d, name="CustomMesh3d")

# Show other plotly objects like Scatter3d
from plotly.graph_objects import Scatter3d

scatter = Scatter3d(
    x=[0, 1, 2],
    y=[0, 1, 0],
    z=[0, 0, 1],
    mode='markers',
    marker=dict(size=5, color='red')
)
pl.plot(scatter, name="CustomScatter3d")

pl.show()
```

Total running time of the script: (0 minutes 3.744 seconds)

9.2.4 Backend-Agnostic Customization APIs (Plotly)

This example demonstrates the backend-agnostic customization APIs with the Plotly backend.

The same API calls work with both PyVista and Plotly backends, demonstrating the true backend-agnostic nature of these methods.

```
import pyvista as pv
from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.backends.plotly.plotly_interface import␣
```

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```
↪PlotlyBackend

# Create a plotter using the Plotly backend and add basic geometry.

plotter = Plotter(backend=PlotlyBackend())

# Add a sphere - this works fine
sphere = pv.Sphere(radius=1.0, center=(0, 0, 0))
plotter.plot(sphere)

# Add point markers to highlight specific locations.

key_points = [
    [1, 0, 0],   # Point on X axis
    [0, 1, 0],   # Point on Y axis
    [0, 0, 1],   # Point on Z axis
]

plotter.add_points(key_points, color='red', size=10)

# Add line segments to show coordinate axes.

# X axis
x_axis = [[0, 0, 0], [1.5, 0, 0]]
plotter.add_lines(x_axis, color='red', width=4.0)

# Y axis
y_axis = [[0, 0, 0], [0, 1.5, 0]]
plotter.add_lines(y_axis, color='green', width=4.0)

# Z axis
z_axis = [[0, 0, 0], [0, 0, 1.5]]
plotter.add_lines(z_axis, color='blue', width=4.0)



# Add a plane to show a reference surface.

plotter.add_planes(
    center=(0, 0, 0),
    normal=(0, 0, 1),
    i_size=2.5,
    j_size=2.5,
    color='lightblue',
    opacity=0.2
)


# Add text annotations using 2D normalized coordinates (0-1 range).

# Scene title at the top center
plotter.add_text("Customization API Example", position=(0.5, 0.95), font_size=18, color=
```

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```
↪'black')

# Additional labels at the top corners
plotter.add_text("Plotly Backend", position=(0.05, 0.95), font_size=12, color='lightblue
↪')
plotter.add_text("3D Visualization", position=(0.95, 0.95), font_size=12, color=
↪'lightgreen')


# Add labels at specific 3D points to annotate key locations in space.

label_points = [
    [1, 0, 0],   # X axis endpoint
    [0, 1, 0],   # Y axis endpoint
    [0, 0, 1],   # Z axis endpoint
]

labels = ['X-axis', 'Y-axis', 'Z-axis']

plotter.add_labels(label_points, labels, font_size=16, point_size=8.0)


# The clear() method resets the plotter and can be called even after show().
# This allows reusing the same plotter for multiple visualizations.

# Uncomment to clear everything added above and start fresh:
# plotter.show()
# plotter.clear()
# plotter.plot(pv.Cube())  # Would show only a cube instead


# Display the visualization with all customizations.


# Uncomment to show in browser:
plotter.show()
```

Total running time of the script: (0 minutes 0.996 seconds)

9.3 Basic USD usage examples

These examples show how to use the general plotter with the USD backend included in the Visualization Interface Tool.

9.3.1 Plain usage of the USD backend

This example shows how to use the Python USD viewer backend in the Visualization Interface Tool to display meshes using the OpenUSD-based viewer from the `python-usd-viewer` package.

Plot a single mesh

This code shows how to plot a single PyVista mesh with the USD backend.

```
import pyvista as pv

from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.backends.usd.usd_interface import USDInterface

mesh = pv.Sphere()

# Create a plotter with the USD backend
pl = Plotter(backend=USDInterface())

# Add the mesh to the plotter
pl.plot(mesh)

# Show the plotter - opens a Qt window and blocks until it is closed
pl.show()
```

Plot multiple meshes

Use `plot_iter()` to add several meshes at once before showing the viewer.

```
import pyvista as pv

from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.backends.usd.usd_interface import USDInterface

meshes = [
    pv.Sphere(center=(0, 0, 0)),
    pv.Cube(center=(2, 0, 0)),
    pv.Cylinder(center=(-2, 0, 0)),
]

pl = Plotter(backend=USDInterface())
pl.plot_iter(meshes)
pl.show()
```

Load and display a USD file

If you already have a `.usd` / `.usda` / `.usdc` file on disk, pass its path directly to `plot()`. The file is opened as a `pxr.Usd.Stage` and displayed in the viewer.

```
import pyvista as pv
import tempfile, os

from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.backends.usd.usd_interface import USDInterface

# Write a simple USD file to a temporary location for demonstration purposes
usd_content = """\
#usda 1.0
```

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```
def Sphere "Ball"
{
    double radius = 1
}
"""
tmp = tempfile.NamedTemporaryFile(suffix=".usda", mode="w", delete=False)
tmp.write(usd_content)
tmp.close()

pl = Plotter(backend=USDInterface())
pl.plot(tmp.name)
pl.show()

os.unlink(tmp.name)
```

Total running time of the script: (0 minutes 0.000 seconds)

9.3.2 Export a USD file to HTML

This example shows how to convert a USD file (or an in-memory `pxr.Usd.Stage`) into a self-contained HTML viewer page. The page embeds all geometry as a base64-encoded GLB file and renders it with Three.js — only a CDN connection is required to open it in any modern browser.

Note

This feature requires the `[usd]` optional dependencies:

```
pip install ansys-tools-visualization-interface[usd]
```

Create a sample USD file

Build a minimal USD stage with one triangular mesh and save it to disk so the file-path examples below have something to read.

```
from pxr import Gf, Usd, UsdGeom

stage = Usd.Stage.CreateNew("my_model.usd")
mesh = UsdGeom.Mesh.Define(stage, "/Triangle")
stage.SetDefaultPrim(mesh.GetPrim())
mesh.GetPointsAttr().Set(
    [Gf.Vec3f(0, 0, 0), Gf.Vec3f(1, 0, 0), Gf.Vec3f(0, 1, 0)]
)
mesh.GetFaceVertexCountsAttr().Set([3])
mesh.GetFaceVertexIndicesAttr().Set([0, 1, 2])
stage.Save()
```

Export from a USD file on disk

Pass a file path string or `pathlib.Path` to `export_usd_to_html()`. The function returns the `pathlib.Path` to the generated HTML file.

```
from ansys.tools.visualization_interface import export_usd_to_html

html_path = export_usd_to_html("my_model.usd", "my_model_viewer.html")
print(f"Viewer written to: {html_path}")
```

```
Viewer written to: /home/runner/work/ansys-tools-visualization-interface/ansys-tools-
↪visualization-interface/examples/02-basic-usd-examples/my_model_viewer.html
```

Export from an in-memory stage

You can pass a `pxr.Usd.Stage` directly — no `.usd` file on disk is required. A temporary file is created automatically, used to generate the GLB, and removed when the function returns.

```
html_path = export_usd_to_html(stage, "triangle_viewer.html")
print(f"Viewer written to: {html_path}")
```

```
Viewer written to: /home/runner/work/ansys-tools-visualization-interface/ansys-tools-
↪visualization-interface/examples/02-basic-usd-examples/triangle_viewer.html
```

Add a wireframe overlay

Set `show_mesh_lines=True` to inject a Three.js `LineSegments` overlay that traces every polygon edge. Use `line_color` (CSS hex) and `line_opacity` (0–1) to style it.

```
html_path = export_usd_to_html(
    "my_model.usd",
    "my_model_viewer.html",
    show_mesh_lines=True,
    line_color="#00ffcc",
    line_opacity=0.7,
)
print(f"Viewer with wireframe written to: {html_path}")
```

```
Viewer with wireframe written to: /home/runner/work/ansys-tools-visualization-interface/
↪ansys-tools-visualization-interface/examples/02-basic-usd-examples/my_model_viewer.html
```

Total running time of the script: (0 minutes 0.250 seconds)

9.4 Advanced usage examples

These examples show how to use the Visualization Interface Tool to postprocess simulation data.

9.4.1 Postprocessing simulation results using the `MeshObjectPlot` class

The Visualization Interface Tool provides the `MeshObject` helper class to relate a custom object. With a custom object, you can take advantage of the full potential of the Visualization Interface Tool.

This example shows how to use the `MeshObjectPlot` class to plot your custom objects with scalar data on mesh.

Necessary imports

```
from ansys.fluent.core import examples
import pyvista as pv

from ansys.tools.visualization_interface.backends.pyvista import PyVistaBackend
from ansys.tools.visualization_interface import MeshObjectPlot, Plotter
```

Download the VTK file

A VTK dataset can be produced utilizing PyDPF for Ansys Flagship products simulations results file format.

```
mixing_elbow_file_src = examples.download_file("mixing_elbow.vtk", "result_files/fluent-
↪mixing_elbow_steady-state")
```

Define a custom object class

Note that the `CustomObject` class must have a way to get the mesh and a name or ID.

```
class CustomObject:
    def __init__(self):
        self.name = "CustomObject"
        self.mesh = pv.read(mixing_elbow_file_src)

    def get_mesh(self):
        return self.mesh

    def get_field_array_info(self):
        return self.mesh.array_names

    def name(self):
        return self.name


# Create a custom object
custom_vtk = CustomObject()
```

Create a `MeshObjectPlot` instance

```
mesh_object = MeshObjectPlot(custom_vtk, custom_vtk.get_mesh())

# Define the camera position
cpos = (
    (-0.3331763564757694, 0.08802797061044923, -1.055269197114142),
    (0.08813476356878325, -0.03975174212669032, -0.012819952697089087),
    (0.045604530283921085, 0.9935979348314435, 0.10336039239608838),
)
```

Get the available field data arrays

```
field_data_arrays = custom_vtk.get_field_array_info()
print(f"Field data arrays: {field_data_arrays}")
```

Plot the `MeshObjectPlot` instance with mesh object & field data (0)

```
pv_backend = PyVistaBackend()
pl = Plotter(backend=pv_backend)
pl.plot(
    mesh_object,
    scalars=field_data_arrays[0],
    show_edges=True,
    show_scalar_bar=True,
)
pl._backend.pv_interface.scene.camera_position = cpos
pl.show()
```

Plot the `MeshObjectPlot` instance with mesh object & other field data (1)

```
pv_backend = PyVistaBackend()
pl = Plotter(backend=pv_backend)
pl.plot(
    mesh_object,
    scalars=field_data_arrays[1],
    show_edges=True,
    show_scalar_bar=True,
)
pl._backend.pv_interface.scene.camera_position = cpos
pl.show()
```

CHAPTER

TEN

USD VIEWER USAGE

The Ansys Tools Visualization Interface provides a USD viewer that can be used to visualize USD stages. The viewer is built on top of the `python-usd-viewer` package, which provides a simple interface for visualizing USD stages.

10.1 USD viewer installation (live viewer)

The USD viewer provides an interactive 3D viewer for visualizing USD stages in real-time.

To use the live USD viewer, you need to install the optional dependencies with:

```
pip install 'ansys-tools-visualization-interface[usd-live-viewer]'
```

This installs the `python-usd-viewer` package along with required dependencies. For additional details, please refer to the USD viewer installation guide.

10.2 USD viewer usage

Once you have the USD viewer installed, you can use it to visualize USD stages. The Ansys Tools Visualization Interface provides a *USDInterface* class that serves as a bridge between the USD viewer and the Ansys Tools Visualization Interface plotter.

Here is a simple example of how to use the USD viewer through the Ansys Tools Visualization Interface:

```
from ansys.tools.visualization_interface import Plotter
from ansys.tools.visualization_interface.usd_interface import USDInterface

pl = Plotter(backend=USDInterface())

mesh = pv.Sphere()

pl.plot(mesh)

pl.show()
```

10.3 USD to HTML export

To convert a USD file to a self-contained HTML viewer page (backed by Three.js), use `export_usd_to_html()`.

The export pipeline works as follows:

1. The USD stage is traversed and its mesh primitives are converted to GLB format. GLB (GL Transmission Format Binary) is the binary container of the glTF standard—a compact, web-native 3D format understood natively by

Three.js without any additional plugins. It encodes mesh geometry (vertices, normals, indices), materials, and scene hierarchy in a single binary blob.

2. The GLB binary is base64-encoded so it can be stored as a plain string inside the HTML file, making the output completely self-contained—no separate model file is needed alongside the HTML.
3. The viewer decodes the string with the browser's built-in `atob()` function and passes the resulting binary buffer to `GLTFLoader`, which renders it with WebGL.

The generated file requires only a CDN connection (for Three.js) to open in any modern browser.

To use this feature, install the optional dependencies with:

```
pip install 'ansys-tools-visualization-interface[usd]'
```

```
from ansys.tools.visualization_interface import export_usd_to_html

# From a file path
html_path = export_usd_to_html("my_model.usd", "my_model_viewer.html")

# From an in-memory pxr.Usd.Stage (no .usd file needed)
from pxr import Gf, Usd, UsdGeom

stage = Usd.Stage.CreateInMemory()
mesh = UsdGeom.Mesh.Define(stage, "/Box")
stage.SetDefaultPrim(mesh.GetPrim())
mesh.GetPointsAttr().Set([Gf.Vec3f(0, 0, 0), Gf.Vec3f(1, 0, 0), Gf.Vec3f(0, 1, 0)])
mesh.GetFaceVertexCountsAttr().Set([3])
mesh.GetFaceVertexIndicesAttr().Set([0, 1, 2])

html_path = export_usd_to_html(stage, "triangle_viewer.html")
```

The optional `show_mesh_lines`, `line_color`, and `line_opacity` parameters control a wireframe edge overlay injected directly into the HTML:

```
html_path = export_usd_to_html(
    "my_model.usd",
    show_mesh_lines=True,
    line_color="#00ffcc",
    line_opacity=0.7,
)
```

10.4 Custom HTML templates

By default `export_usd_to_html()` uses the built-in `glb_template.html` viewer. Pass `template_path` to supply your own HTML file instead—for example to match a house style, add UI controls, or embed the viewer inside a larger page.

10.4.1 Template variables

Two placeholders are **always required**. They are replaced with JSON-encoded values at render time so that special characters are escaped automatically.

Placeholder	Replaced with
`__MODEL_NAME_JSON__`	A JSON string containing the source filename, for example `"my_model.usd"`. Use it wherever you want to display the model name.
`__GLB_B64_JSON__`	A JSON string containing the base64-encoded GLB binary. GLB is the binary form of the glTF standard: it packs all mesh geometry (vertices, normals, indices), materials, and scene hierarchy into a single blob, which Three.js can parse directly with `GLTFLoader`. Base64 encoding lets the binary data travel inside a plain HTML file without corruption. In JavaScript, assign it to a variable, decode it with `atob()`, copy the result into a `Uint8Array`, and pass the buffer to `GLTFLoader.parse()`.

When `show_mesh_lines=True`, two additional **anchor strings** must appear verbatim in the template. The exporter locates them by exact text match and inserts the wireframe code around them:

Anchor string	What gets injected
`const binary = atob(glbBase64);`	The edge-segment `Float32Array` and color/opacity constants are inserted immediately **before** this line.
`scene.add(gltf.scene);`	A `THREE.LineSegments` object is added to the scene immediately **after** this line.

10.4.2 Using a custom template

Pass the path to your template file via the `template_path` parameter:

```
from ansys.tools.visualization_interface import export_usd_to_html

html_path = export_usd_to_html(
    "my_model.usd",
    "my_model_viewer.html",
    template_path="my_custom_template.html",
)
```

To use a custom template **with** the wireframe overlay, the template must contain both anchor strings listed in the preceding table:

```
html_path = export_usd_to_html(
    "my_model.usd",
    "my_model_viewer.html",
    template_path="my_custom_template.html",
    show_mesh_lines=True,
    line_color="#00ffcc",
    line_opacity=0.7,
)
```

10.4.3 Validation errors

`export_usd_to_html()` validates templates before writing any output. A `ValueError` is raised listing every missing placeholder or anchor so you can fix all problems in one pass:

```
ValueError: Template is missing required placeholders: ['__GLB_B64_JSON__']

ValueError: Template is missing required placeholders:
    ['const binary = atob(glbBase64);', 'scene.add(gltf.scene);']
```

CHAPTER

ELEVEN

CONTRIBUTE

Overall guidance on contributing to a PyAnsys library appears in the Contributing topic in the *PyAnsys developer's guide*. Ensure that you are thoroughly familiar with this guide before attempting to contribute to the Visualization Interface Tool.

The following contribution information is specific to the Visualization Interface Tool.

11.1 Install in developer mode

Installing the Visualization Interface Tool in developer mode allows you to modify and enhance the source.

To clone and install the latest Visualization Interface Tool release in development mode, run these commands:

```
git clone https://github.com/ansys/ansys-tools-visualization-interface
cd ansys-tools-visualization-interface
python -m pip install --upgrade pip
pip install -e .
```

11.2 Run tests

The Visualization Interface Tool uses pytest for testing.

1. Prior to running tests, you must run this command to install test dependencies:

   ```
   pip install -e .[tests]
   ```

2. To then run the tests, navigate to the root directory of the repository and run this command:

   ```
   pytest
   ```

11.3 Adhere to code style

The Visualization Interface Tool follows the PEP8 standard as outlined in PEP 8 in the *PyAnsys developer's guide* and implements style checking using pre-commit.

To ensure your code meets minimum code styling standards, run these commands:

```
pip install pre-commit
pre-commit run --all-files
```

You can also install this as a pre-commit hook by running this command:

```
pre-commit install
```

This way, it's not possible for you to push code that fails the style checks:

```
$ pre-commit install
$ git commit -am "added my cool feature"
black....................................................................Passed
blacken-docs.............................................................Passed
isort....................................................................Passed
flake8...................................................................Passed
docformatter.............................................................Passed
codespell................................................................Passed
pydocstyle...............................................................Passed
check for merge conflicts................................................Passed
debug statements (python)................................................Passed
check yaml...............................................................Passed
trim trailing whitespace.................................................Passed
Add License Headers......................................................Passed
Validate GitHub Workflows................................................Passed
```

11.4 Build the documentation

You can build the Visualization Interface Tool documentation locally.

1. Prior to building the documentation, you must run this command to install documentation dependencies:

   ```
   pip install -e .[doc]
   ```

2. To then build the documentation, navigate to the `docs` directory and run this command:

   ```
   # On Linux or macOS
   make html

   # On Windows
   ./make.bat html
   ```

The documentation is built in the `docs/_build/html` directory.

You can clean the documentation build by running this command:

```
# On Linux or macOS
make clean

# On Windows
./make.bat clean
```

11.5 Post issues

Use the Visualization Interface Tool Issues page to report bugs and request new features. When possible, use the issue templates provided. If your issue does not fit into one of these templates, click the link for opening a blank issue.

If you have general questions about the PyAnsys ecosystem, email pyansys.core@ansys.com. If your question is specific to the Visualization Interface Tool, ask your question in an issue as described in the previous paragraph.

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